

Current MoE Reasoning Experiments

1 Scope

This report tracks the three active workflows: `gepaLLMAsJudge`, `probeTest`, and `correlation_pipeline`. The first two were called Experiments 0a and 0b in the previous combined manuscript. Their full protocols, intermediate results, tables, limitations, and external-validation results are retained here under their current names.

The previous combined manuscript for numbered Experiments 1–5 is preserved at `old/report/legacy_full_report.tex` together with its compiled PDF. Those archived results remain available for provenance but are not mixed into the current experimental plan.

2 Experimental Setup

Current experimental scope. The active repository contains three complementary Qwen workflows. `gepaLLMAsJudge` is an annotation-capability study that optimizes Qwen3.6-27B and Qwen3.8-27B prompts for sentence-level Schoenfeld episode classification. It does not provide evidence about Qwen router dynamics. `probeTest` is a representation-probe replication that teacher-forces the 38 released DeepSeek-R1 Schoenfeld traces through the official GPTQ 4-bit Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 checkpoint and tests whether the next sentence’s label is linearly decodable at its causal pre-sentence boundary. The separate `correlation_pipeline` generates benchmark solutions with Qwen3.5-35B-A3B, replays them through the matching checkpoint, and relates router and hidden-state summaries to correctness and reasoning-event labels.

GEPA judge design. The GEPA workflow starts from 3,125 annotated units in 38 released DeepSeek-R1 SAT responses. Complete responses are assigned to partitions before their units are flattened, preventing neighboring units from crossing splits. A local `llama.cpp` server supplies Qwen predictions. Each request contains the SAT problem and the previous, current, and next response units. The original context-aware seed instruction contains 21 audited synthetic contrastive demonstrations. The Qwen3.8 alignment instead imports the 40-example prompt selected by the earlier final fit as its external seed, so it does not dynamically sample new demonstrations. The implementation supports inverse-frequency weighted exact match and an informative same-model LLM-as-a-judge reward. Prompt-selection rules are evaluated with nested response-grouped cross-validation; the final-fit runs use a response-grouped 26/6/6 train/validation/test split.

Schoenfeld boundary-probe design. The probe workflow excludes the 38 synthetic `</think>` closing units, which are all labelled Monitor, leaving the paper-matched 3,087 units. Qwen receives each SAT instruction as the user message and the released response as a teacher-forced assistant continuation. For a target unit, the feature is the hidden state immediately before its first token, so the probe sees the problem and preceding response but cannot read the target sentence. Hidden-state index 0 is the embedding output and indices 1–40 are the successive language-model layers.

At each index, seven separate target-versus-rest logistic regressions use raw 2,048-dimensional activations, balanced class weights, L2 regularization with $C = 1$, and no standardization or PCA. The paper-matched

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protocol uses a target-stratified 80/20 sentence split with seed 42. This is a replication choice rather than a strong generalization design: all 38 response IDs occur in both partitions. The reported layer for each target maximizes accuracy on that same test split. For qualitative transfer inspection, the selected probes are applied to 20 examples each from GSM8K, MATH, PRM800K, and ProcessBench. The seven binary scores are independently trained and uncalibrated; their argmax is an inspection label, not a multiclass probability or new gold annotation. Only boundary vectors are stored; Qwen does not regenerate the released or benchmark solutions.

Correlation-pipeline design. The current report uses the six-benchmark run completed September 17, 2026: 4,647 attempts from 1,547 source problems, including all 32 attempts per AIME24, AIME25, and AMC23 problem and one per MATH-500, OlympiadBench, and Minerva problem. GPQA-Diamond and MMLU-Pro remain absent. Generation uses Qwen/Qwen3.5-35B-A3B-GPTQ-Int4; replay uses unsloth/Qwen3.5-35B-A3B with runtime 4-bit quantization at decoder indices 27, 34, 35, 36, 38, and 39. Compact features and top-eight expert IDs are retained. No raw activation tensors are saved.

All 412 token-limit completions remain included; 3,568/4,647 answers are correct (0.768). Full-reasoning and absolute-position views pool all reasoning tokens independently of tagging; whole-continuation metrics also include final-answer tokens. The position reference mean is 12,498.062 tokens, with ten disjoint windows and overflow. Class views use available sentence labels from sample zero, selected with error-rate-weighted quotas and a 100,000-unit cap (seed 42). Missing class/window features are not zero. The judge is unsloth/Qwen3.8-27B-NVFP4 with the frozen GEPA program and low reasoning effort; its accuracy on these traces is unmeasured.

All requested stages are complete. The 100 complete avg@32 groups support problem-level and within-problem analyses in whole-continuation and full-reasoning views; classes still have one tagged attempt per problem. Length, termination, sparse class outcomes, and changing window cohorts limit interpretation. Historical runs remain separate artifacts.

Table model and metric conventions. Every table caption identifies the evaluated model or models and the metrics shown. For classification, accuracy is the fraction of exact label matches, balanced accuracy is the unweighted mean of class recalls, precision and recall have their standard true-positive definitions, macro-F1 is the unweighted mean of per-class F1, and AUROC measures score ranking across the positive and negative classes. Cohen's κ is chance-corrected nominal agreement, Kendall's τ_b is rank agreement under the fixed episode-label ordering, and the GEPA composite averages their equal-weight $[0, 1]$ rescalings. Routing entropy is Shannon entropy of the softmax router distribution; the correlation pipeline's switch rate is the fraction of adjacent tokens whose top-1 expert changes; its top- k overlap is the adjacent-set intersection divided by k (not Jaccard overlap); and router margin is the top-1 minus top-2 router probability. Top- k mass sums the selected probabilities, while the boundary margin is $p_{(k)} - p_{(k+1)}$. Unless a caption states otherwise, N and support are observation counts, r is a Pearson-family correlation coefficient, and confidence intervals are percentile bootstrap intervals.

Artifact provenance. The active packages are located at `src/moe_exp/gepaLLMAsJudge`, `src/moe_exp/probeTest`, and `src/moe_exp/correlation_pipeline`. GEPA runs through `run_gepaLLMAsJudge.sh` and writes to `results/gepaLLMAsJudge`; boundary-probe outputs remain under `results/probeTest`. The correlation workflow is launched by its local `run_all.sh` and uses its documented output directory.

The Qwen judge artifacts formerly called Experiment 0a, including the final Qwen3.8 model-alignment run, and the Qwen3.5 boundary-probe artifacts formerly called Experiment 0b are complete. The six-benchmark Qwen correlation run is complete for all requested views; class labels remain sampled and cross-model tagging is ongoing. Earlier untagged and GGUF outputs are not pooled with the current estimates. Historical OLMoE and numbered-experiment outputs remain archived.

Table 1. Preliminary gepaLLMAsJudge agreement. The composite rescales Cohen’s κ and Kendall’s τ_b from $[-1, 1]$ to $[0, 1]$ and averages them. Coverage was 100% in every row. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: exact-match accuracy, Cohen’s κ , Kendall’s τ_b , their rescaled composite, and valid-output coverage.

Evaluation	Accuracy	κ	τ_b	Composite
Seed validation	0.6388	0.5581	0.4810	0.7598
Optimized validation	0.6855	0.6074	0.5847	0.7980
Optimized test	0.6607	0.5710	0.6066	0.7944
Few-shot validation	0.6806	0.6078	0.5830	0.7977
Few-shot test	0.6994	0.6178	0.6667	0.8211

3 gepaLLMAsJudge

The gepaLLMAsJudge workflow evaluates whether prompt optimization improves a local judge for the seven adapted Schoenfeld episode labels: Read, Analyze, Plan, Implement, Explore, Verify, and Monitor. The model was the Q4_K_XL quantization of Qwen3.6-27B served by llama.cpp on one RTX 3090. All splits preserve complete responses, preventing neighboring units from crossing partitions.

Preliminary single-split runs. With seed 42, the 38 responses were divided into 26 training, 6 validation, and 6 test documents (2,382/407/336 units). These first runs classified the current unit without neighboring context and optimized per-unit exact match.

The matched few-shot run used the same seed, document split, model, and GEPA budget, adding seven gold training examples to the instruction. It performs better than the optimized base condition on the held-out test set: accuracy rises from 66.07% to 69.94% (+3.87 percentage points), Cohen’s κ from 0.5710 to 0.6178, Kendall’s τ_b from 0.6066 to 0.6667, and composite agreement from 0.7944 to 0.8211. GEPA itself does not improve the few-shot validation result: the seed and selected optimized prompts both score 68.06% accuracy. The current comparison therefore favors few-shot prompting for held-out generalization, whereas the measurable GEPA optimization gain is confined to the base-prompt condition.

Context-aware nested cross-validation. The revised protocol excludes the same six responses from model selection and runs five outer folds over the remaining 32 responses. Within each fold, five responses form the inner validation set and all others outside the outer fold form the GEPA training set. Each request includes the problem statement, previous unit, current unit, and next unit. The seed instruction includes 21 audited synthetic contrastive demonstrations. GEPA receives inverse-frequency weighted exact match, whose split mean is balanced accuracy; the safety selector compares either balanced accuracy or macro-F1 and rejects any candidate whose class recall falls by more than the configured threshold.

Informative same-model reward. The current implementation can replace the static GEPA reward with an LLM-as-a-judge call to the same Qwen3.6-27B model. For each candidate label, the judge returns a short free-text explanation plus four diagnostic scores from 1 to 5: gold agreement, functional fit, contextual coherence, and boundary precision. Their normalized weighted mean is used only as GEPA’s optimization signal, with respective weights 0.40, 0.25, 0.15, and 0.20. Gold-label accuracy, balanced accuracy, macro-F1, and per-class recall remain the independent selection and reporting criteria. The judge runs deterministically with model reasoning disabled. Every judge call is persisted as JSONL with its component scores, normalized score, and free-text explanation, permitting a direct audit of feedback quality. Tables 2 and 3 remain static-reward cross-validation baselines; the completed LLM-judge final fit is reported separately below because its reused validation set is a prompt-development set rather than an outer-fold estimate.

Table 2. Aggregate selected-prompt outer-fold results over 2,789 units. “Balanced light” is the earlier 0.10 recall-gate run; the macro-F1 runs use a 0.05 gate. The six excluded responses were not evaluated. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, and the rescaled κ/τ_b composite.

Selection / budget	Accuracy	Balanced acc.	Macro-F1	Composite
Balanced / light	0.6701	0.6753	0.6480	0.8031
Macro-F1 / light	0.6734	0.6796	0.6505	0.8063
Macro-F1 / heavy	0.6669	0.6737	0.6437	0.8030

Table 3. Per-class outer-fold performance for the best current configuration, macro-F1 selection with the light GEPA budget. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: class support, precision, recall, and F1.

Class	Support	Precision	Recall	F1
Read	318	0.798	0.770	0.784
Analyze	685	0.757	0.632	0.689
Plan	205	0.524	0.590	0.555
Implement	650	0.880	0.722	0.793
Explore	193	0.679	0.492	0.571
Verify	538	0.669	0.606	0.636
Monitor	200	0.364	0.945	0.526

Macro-F1 selection with the light budget is the strongest cross-validated configuration, but its advantage over balanced-accuracy selection is small: +0.32 accuracy points, +0.43 balanced-accuracy points, and +0.24 macro-F1 points. At the response level it wins on 15 documents, loses on 13, and ties on 4, so the observed gain is not stable across documents. The light run selects a GEPA-optimized prompt in three of five folds; the recall gate retains the seed prompt in the other two because Plan or Monitor recall falls too far.

Increasing the budget does not help. The heavy run selects an optimized prompt in only two folds and is 0.65 accuracy points and 0.68 macro-F1 points below the light run. Paired predictions show 38 units correct only under light versus 20 correct only under heavy. This difference is concentrated in fold 5, where light discovers and selects a strong candidate whereas heavy retains the seed after its candidate violates the Analyze recall gate. Independent heavy search is therefore not a superset of light search and can discard a better prompt found by a smaller run.

The revised objective improves overall balance only marginally and does not resolve Monitor overprediction. The model emits Monitor 519 times for 200 gold Monitor units; the leading errors are Verify→Monitor (127), Plan→Monitor (65), and Explore→Monitor (42). The annotation audit flags 180/3,125 units (5.76%), including 145 multiline units and 49 structural/control markers such as `</think>` and “Final Answer.” These artifacts plausibly inflate Monitor recall while depressing precision and should be reviewed before further optimization.

All runs retain 100% final-output coverage and complete without OOM failures, but the light and heavy jobs log 31 and 28 truncated LM responses during optimization. Moreover, the outer folds currently evaluate only the safety-selected prompt, not both seed and optimized prompts, so they do not directly estimate GEPA’s paired outer-fold gain. Finally, the six responses excluded from current CV were inspected in the preliminary experiments; they are locked from the current code path but no longer constitute a pristine research holdout. The judge should therefore not replace validated annotation until structural artifacts are adjudicated, decoding is constrained, paired outer evaluation is added, and a genuinely unseen external test set is designated.

Table 4. Context-aware LLM-judge final-fit results on the 407-unit prompt-development validation set. Coverage is 100% in both rows. The locked test is not evaluated. *Model(s)*: Qwen3.6-27B Q4_K_XL, used both for episode-label prediction and the same-model GEPA reward. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, Cohen’s κ , Kendall’s τ_b , absolute change, and valid-output coverage.

Prompt	Accuracy	Balanced acc.	Macro-F1	κ	τ_b
21-example seed	0.6683	0.6765	0.6132	0.5988	0.5904
GEPA optimized	0.7543	0.7769	0.7270	0.6968	0.6832
Absolute change	+0.0860	+0.1003	+0.1138	+0.0980	+0.0929

Table 5. Per-class recall and optimized F1 for the completed LLM-judge final fit. *Model(s)*: Qwen3.6-27B Q4_K_XL, used both for episode-label prediction and the same-model GEPA reward. *Metrics*: class support, seed and optimized recall, and optimized per-class F1.

Class	Support	Seed recall	Opt. recall	Opt. F1
Read	45	0.778	0.756	0.791
Analyze	122	0.598	0.680	0.744
Plan	32	0.531	0.813	0.722
Implement	103	0.786	0.825	0.825
Explore	17	0.471	0.706	0.686
Verify	70	0.571	0.714	0.763
Monitor	18	1.000	0.944	0.557

Completed LLM-judge final fit. The August 11, 2026 run uses seed 42, the 21-example context-aware few-shot prompt, and the same-model informative reward on the original response-grouped 26/6/6 split (2,382/407/336 units). Seed and optimized prompts both return a valid label for all 407 validation units. Table 4 reports exact gold-label metrics rather than the LLM judge’s diagnostic score.

The optimized prompt fixes 49 seed errors and regresses on 14 previously correct units, a net gain of 35. All six validation responses improve, with response-level accuracy gains from 2.0 to 13.8 percentage points. Its primary effect is to curb the seed prompt’s overuse of Monitor: Monitor predictions fall from 74 to 43, including nine corrections from Monitor to true Plan and 15 from Monitor to true Verify. Recall rises most for Plan (+28.13 points), Explore (+23.53), Verify (+14.29), and Analyze (+8.20). Monitor recall falls slightly from 1.000 to 0.944, but precision rises from 0.243 to 0.395. The selected prompt still has weak Monitor precision and Analyze recall, and the remaining errors concentrate at the Analyze/Implement/Verify boundaries and Verify→Monitor. Implement is the only class whose F1 decreases, from 0.839 to 0.825, because its additional predictions include more false positives.

The internal same-model reward rises from 0.7264 for the seed to 0.8196 for the selected candidate. The optimization trace contains 145 candidates, 144 full validation evaluations, and 60,767 persisted judge calls, of which 148 are judge-output errors. A text audit finds 37 validation units for which the judge explicitly disputes the corpus gold label in at least one evaluation, showing that the reward itself detects annotation-sensitive boundaries rather than consistently treating gold as semantically authoritative. The structural audit flags 29 validation units; their accuracy remains 72.4% under both prompts, so the gain comes entirely from otherwise unflagged units.

GEPA expands the instruction from 21 to 40 demonstrations and from 7,648 to 32,717 characters. The extra rules and examples target observed corpus edge cases, raising inference cost and the risk of learning annotation

Table 6. Final Qwen3.8 model-alignment results on the 407-unit development validation set. Both prompts have 100% valid-output coverage; the 336-unit locked test is not evaluated. *Model(s)*: Qwen3.8-27B UD-Q4_K_XL, used for both classification and the same-model GEPA reward. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, Cohen’s κ , and Kendall’s τ_b .

Prompt	Accuracy	Balanced acc.	Macro-F1	κ	τ_b
Imported 40-example seed	0.7174	0.7400	0.6938	0.6523	0.6122
GEPA optimized	0.7125	0.7288	0.6743	0.6475	0.6009
Absolute change	−0.0049	−0.0112	−0.0196	−0.0049	−0.0113

Table 7. Per-class recall in the final Qwen3.8 alignment. The selection gate rejects any candidate whose recall drops by more than 0.10 for any class. *Model(s)*: Qwen3.8-27B UD-Q4_K_XL. *Metrics*: class support and seed versus GEPA-optimized recall.

Class	Support	Seed recall	Opt. recall
Read	45	0.778	0.756
Analyze	122	0.549	0.533
Plan	32	0.750	0.563
Implement	103	0.874	0.874
Explore	17	0.529	0.647
Verify	70	0.700	0.786
Monitor	18	1.000	0.944

conventions rather than a portable episode taxonomy. More importantly, GEPA evaluates 145 candidates against the same six-response validation set used for final selection. The nominal 407 units are also correlated within responses, and only one optimization seed is available. The result files explicitly record `locked_test_evaluated: false` and `test: null`. The observed gain is therefore strong evidence of successful prompt optimization on the development distribution, but not a held-out generalization estimate. The prompt should be frozen and compressed or ablated before a one-time test on genuinely unseen responses.

Final Qwen3.8 model-alignment pass. The August 27–28, 2026 alignment run uses Qwen3.8-27B UD-Q4_K_XL, seed 42, the heavy GEPA budget, and medium reasoning effort for the same-model judge. It keeps the same response-grouped 2,382/407/336 train/validation/locked-test split. Rather than drawing new few-shot examples, it imports the previous 40-example Qwen3.6 optimized prompt as its external seed. Gold-label agreement supplies 40% of the internal judge reward; functional fit, contextual coherence, and boundary precision supply the remaining 60%. Final selection uses exact balanced accuracy with a maximum allowed per-class recall drop of 0.10.

The optimized candidate changes 49 labels: it fixes 17 seed errors, breaks 19 seed-correct cases, and changes 13 errors to other errors. Its largest failure is Plan recall, which falls by 0.188 and therefore breaches the recall gate; the selected prompt is byte-identical to the imported seed. An unclustered exact McNemar test on the 17 versus 19 discordant correctness outcomes gives $p = 0.868$, offering no evidence of an aggregate improvement, although this unit-level calculation understates response clustering.

This negative exact-label result contrasts with the internal GEPA objective, whose best candidate reward rises from 0.7684 to 0.7762. Because only 40% of that reward is gold agreement, the discrepancy is consistent with optimizing the judge’s broader functional criteria without improving corpus-label classification. The optimized instruction is shorter (2,591 versus 5,201 words) but internally inconsistent: an early rule treats “Let me compute

Table 8. Accuracy-selected Schoenfeld boundary probes on the sentence-level test split. Prevalence is over all 3,087 units. Δ Maj. subtracts the always-negative accuracy. Balanced accuracy is derived from the saved test confusion matrix. Layer selection and evaluation use the same test split and are therefore selection-optimistic. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen hidden-state encoder, with one-versus-rest logistic probes. *Metrics*: target prevalence, selected hidden-state layer, accuracy, accuracy minus the majority baseline, balanced accuracy, F1, and AUROC.

Target	Prevalence	Layer	Accuracy	Δ Maj.	Balanced acc.	F1	AUROC
Read	12.21%	34	0.9239	+0.0461	0.8476	0.7044	0.9338
Analyze	25.62%	36	0.8107	+0.0669	0.7669	0.6465	0.8348
Plan	7.29%	39	0.9175	-0.0096	0.7405	0.4848	0.8551
Implement	21.87%	38	0.8964	+0.1151	0.8617	0.7714	0.9305
Explore	6.74%	40	0.9223	-0.0103	0.7486	0.4894	0.8884
Verify	20.38%	27	0.8754	+0.0792	0.8509	0.7260	0.9122
Monitor	5.90%	35	0.9304	-0.0106	0.7546	0.4819	0.9031

that” as Monitor while a later rule treats “Let me calculate” as Plan. Plan-to-Monitor errors rise from 6 to 10, consistent with that boundary confusion. The best internal candidate appears by iteration 3 and remains the best through iteration 19, so the remaining heavy search adds roughly 14.5 hours without a better candidate. The appropriate conclusion is that the final alignment completed correctly and the safety gate worked: GEPA found no safe improvement over the existing seed. The complete artifact is stored at `results/gepaLLMAsJudge/qwen3.8-27b-medium-final-s42-v3`.

4 probeTest

The probeTest workflow tests a different annotation mechanism from the generative judge in gepaLLMAsJudge. It asks whether Qwen’s representation of the prefix immediately before a sentence contains linearly accessible information about the upcoming Schoenfeld episode. It therefore measures prospective prefix-state decodability, not semantic classification after reading the sentence and not the latent dynamics of a Qwen-generated trace.

Within-corpus results. Table 8 reports the accuracy-selected layer for each one-versus-rest target. AUROC is 0.835–0.934 and balanced accuracy is 0.740–0.862, demonstrating substantial within-split ranking signal. Implement is strongest at the natural threshold (F1 0.771), followed by Verify (0.726) and Read (0.704). Analyze is intermediate. Plan, Explore, and Monitor have F1 below 0.50. Their nominal accuracies are also 0.96–1.06 percentage points below an always-negative majority classifier, so accuracy alone gives a misleading impression for these rare targets.

The signal emerges rapidly and is not maximal at the model output. Mean one-versus-rest AUROC across targets rises from 0.624 at the embedding output to 0.873 at layer 10 and 0.892 at layer 20, peaks at 0.899 at layer 30, and falls to 0.867 at layer 40. Mean F1 peaks at layer 28. Accuracy-based target selection sometimes obscures this profile: for Plan, the selected layer 39 has AUROC 0.855 whereas layer 21 reaches 0.922. AUROC or balanced accuracy, chosen inside a validation fold, better matches the scientific question of decodability.

The fits are numerically stable—none of the 287 regressions reports a convergence warning—but statistical overfitting remains visible. At the selected layers, train–test accuracy gaps are 16.7 points for Analyze, 10.1 for Implement, approximately 7.8 for Plan and Explore, and 4.5–5.5 for the other targets. Explore reaches 100% training accuracy. With 2,048 features, 2,469 training units, strong within-response dependence, and only 38

Table 9. Argmax inspection-label counts on external reasoning units. These are raw prediction frequencies, not accuracies. Table 10 compares them with the subsequent provisional pragmatic review. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: unit counts and seven-way argmax prediction counts.

Dataset	Units	Read	Analyze	Plan	Implement	Explore	Verify	Monitor
GSM8K	96	0	2	3	53	0	0	38
MATH	88	23	7	1	45	0	1	11
PRM800K	139	34	18	0	71	0	1	15
ProcessBench	258	69	18	12	117	0	17	25
Total	581	126	45	16	286	0	19	89

distinct responses, the effective independent sample size is much closer to the number of responses than to the number of sentences. Test-based layer selection adds further optimism.

External benchmark inspection. All 80 requested examples complete without omitted units, yielding 581 sentence boundaries. Table 9 shows that the seven-way inspection behavior collapses mainly to Implement, Read, and Monitor. Implement alone receives 49.2% of argmax assignments, whereas Explore is never selected. The transfer distribution also differs sharply from training: Analyze falls from 25.6% of gold units to 7.7% of predictions and Verify from 20.4% to 3.3%.

The independent binary scores do not form a clean multiclass decision rule. Seventy-four units (12.7%) have no score above 0.5 but are still assigned an argmax label, while 180 (31.0%) activate more than one probe and 99 (17.0%) have a top-two margin below 0.1. A conservative display rule requiring a top score of at least 0.8 and a margin of at least 0.2 retains only 318/581 units (54.7%), mostly Implement, Read, and Monitor. This threshold is a useful abstention heuristic, not a calibrated guarantee of correctness.

Provisional pragmatic adjudication. To make the qualitative inspection measurable, every displayed unit was subsequently reviewed using the problem, the current unit, its neighbors, and the adapted Schoenfeld precedence rule. The annotation is a hybrid rule-assisted Codex pragmatic review, not human consensus or authoritative benchmark gold. It records one primary label for conventional single-label statistics and, where two functions are genuinely defensible, an accepted set containing both labels. This occurs for 187/581 units (32.2%), mostly at boundaries where a unit both announces and executes an operation or where a structural marker also carries a direct result. Of the annotations, 394 have high, 150 medium, and 37 low confidence; the low-confidence cases remain priority items for human review.

The probe agrees with the primary pragmatic choice on 292/581 units (50.3%) and with at least one accepted label on 353/581 (60.8%). Thus recognizing genuine taxonomy overlap recovers 61 additional compatible decisions, but it does not make the reference authoritative. ProcessBench has the strongest compatible agreement (68.6%), followed by GSM8K (62.5%); MATH has the lowest strict agreement (40.9%). Table 10 reports the complete dataset-level results.

The class-level comparison in Table 11 sharpens the interpretation. Implement is the only operationally strong external label (precision 0.804, recall 0.667, F1 0.729). Read is moderately useful (F1 0.524). Analyze and Plan are severely under-produced, while Verify and Monitor have low recall. Explore has 11 accepted reference memberships but is never predicted, hence zero recall. Because multi-label units contribute to two reference classes, accepted supports overlap and do not sum to 581.

Table 10. Agreement with the provisional pragmatic review. Primary accuracy requires the preferred label; compatible accuracy accepts either label on a multi-label unit. Macro-F1 and macro recall treat each accepted label as a positive reference membership. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: unit count, primary accuracy, compatible accuracy, multi-label count, accepted-membership macro-F1, and macro recall.

Dataset	Units	Primary acc.	Compatible acc.	Multi-label	Macro-F1	Macro recall
GSM8K	96	49.0%	62.5%	20	0.136	0.129
MATH	88	40.9%	53.4%	36	0.211	0.228
PRM800K	139	46.0%	49.6%	44	0.241	0.302
ProcessBench	258	56.2%	68.6%	87	0.396	0.350
Total	581	50.3%	60.8%	187	0.330	0.306

Table 11. Per-label comparison with accepted pragmatic memberships. Support counts every unit for which the label is accepted, including secondary labels on ambiguous units. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: accepted-label support, argmax prediction count, precision, recall, and F1.

Label	Accepted support	Probe predictions	Precision	Recall	F1
Read	84	126	0.437	0.655	0.524
Analyze	84	45	0.356	0.190	0.248
Plan	90	16	0.375	0.067	0.113
Implement	345	286	0.804	0.667	0.729
Explore	11	0	0.000	0.000	0.000
Verify	51	19	0.632	0.235	0.343
Monitor	103	89	0.382	0.330	0.354

Pragmatically, Implement is the most coherent transferred label: it marks long calculation, substitution, and algebra runs. Read often marks givens or restatements, particularly near the start of MATH and PRM800K solutions. Monitor behaves primarily as a solution-boundary or opening-state detector: all 20 GSM8K first units and 18/20 ProcessBench first units receive Monitor, whereas 19/20 MATH and 17/20 PRM800K first units receive Read. Because the target sentence is unseen at this point, this source dependence must arise from the problem, prefix position, and formatting rather than the displayed unit’s semantics. Analyze and Plan appear plausibly on some explanatory or phase-change boundaries but have low coverage. Explore is unusable for external tagging in the present model.

Verify has a narrower but potentially useful role as a review or reconsideration flag. All 17 ProcessBench Verify predictions occur at or after the benchmark’s annotated first error, including six within one trace that repeatedly reinterprets the question before ending with the wrong sign. This is not an error detector: errors generally occur early, Verify may merely mark late position or reflective language, and no PRM800K post-error unit is labelled Verify. It can prioritize passages for inspection but cannot judge correctness.

The qualitative stage also exposes an input problem independent of the classifier. Thirteen units are split immediately after abbreviations such as “Mr.”, producing fragments including “Therefore, Mr.” and inconsistent scores. External use should first adopt abbreviation- and LaTeX-aware segmentation. With the current outputs, the defensible practical representation is a coarse three-phase view—setup (Read and opening Monitor), execution

Table 12. Prespecified measures for the expanded correlation analysis.

Analysis	Unit	Measure	Reason
Continuous feature versus correctness or a binary reasoning-event indicator	Generation trace	Point-biserial r_{pb}	The predictor is continuous and the outcome is binary; the signed coefficient is exactly Pearson's r after coding the outcome as 0/1.
Mean feature versus avg@ N correctness	Source problem	Spearman's ρ	Avg@ N is bounded, often tied, and not assumed to have a linear interval relationship with a feature. Ranking tests monotone association and one row per problem avoids treating repeated attempts as independent.
Continuous feature–feature and continuous expert-rate associations	Trace or source problem	Spearman's ρ	These exploratory scans ask whether quantities vary monotonically without assuming comparable scales or a linear relationship.
Correct-versus-incorrect feature difference within a repeated problem	Mixed-outcome source problem	Mean paired contrast	This directly controls for the problem by forming one correct-minus-incorrect contrast per problem; uncertainty is obtained by resampling problems.

(Analyze, Plan, and Implement), and review (Verify)—plus an explicit unknown/ambiguous state. The seven raw argmax labels should not be used as automatic cognitive annotations.

5 Correlation Pipeline

5.1 Statistical measures and literature basis

The analysis selects the association measure from the outcome scale and unit of analysis, as summarized in Table 12. The detailed Qwen results below use the six-benchmark run completed September 17, 2026: 4,647 attempts, including token-limit completions, with full-reasoning, class, and position views. The 100 complete avg@32 groups now support repeated-attempt analysis. Classes use sampled labelled attempts; full and position views use all reasoning tokens. Historical snapshot coefficients are superseded here and remain separate archival artifacts. Section 6 adds the completed Gemma and GPT-OSS runs, with model-specific class coverage and annotation protocols.

For a continuous feature X and binary target $Y \in \{0, 1\}$, the first coefficient is

$$r_{pb} = \text{Corr}(X, Y). \quad (1)$$

Thus positive values mean that the feature tends to be higher for correct (or event-present) traces, and negative values mean that it tends to be lower. Liu et al. directly use point-biserial correlation for continuous language-model quantities against binary answer correctness, making it the closest precedent for this case [3].

For repeated-attempt datasets, correctness and each predictor are first aggregated once per source problem. Only groups containing exactly the declared avg@ N unique sample identifiers, no duplicate generations detected by the saved fingerprints, and a complete set of binary outcomes enter the primary Spearman analysis. Spearman correlation is appropriate here because its target is monotone rank association rather than a specifically linear change in avg@ N . Hong et al. correlate a scalar projection of residual-stream states with graded reasoning scores using Spearman correlation [2]; Xu et al. use Spearman correlation between language-model-derived conceptual quantities and human ratings [4]; and Hewitt and Manning use it to compare geometry derived from contextual representations with gold syntactic structure [1]. These studies are analogous precedents for rank

correlation with representation-derived quantities; they do not by themselves establish Spearman as a unique choice for `avg@32`. Zhang et al. additionally report Pearson and Spearman correlations together when relating model uncertainty, including entropy-based measures, to graded human hallucination judgments [5].

Uncertainty is reported with 95% percentile bootstrap intervals that resample whole source problems, keeping attempts from the same problem together. Xu et al. provide relevant precedent for bootstrap intervals and use 10,000 resamples in one category–feature analysis [4]; the resample count in this pipeline remains explicit and configurable rather than being justified by that number alone. Benjamini–Hochberg q -values are computed separately within each declared dataset/scope, target, aggregation, or cross-feature family. Naive trace-level p -values are retained only as diagnostics when a dataset contains repeated attempts, and layerwise and expert-identity screens remain exploratory.

5.2 Completed six-benchmark run and analysis population

The Qwen numerical results use the run completed on September 17, 2026, under `results/correlation_pipeline/reasoning-vllm-v1`. Forward replay and the saved aggregate analyses are complete for 4,647 attempts from 1,547 source problems: 500 MATH-500, 960 AIME24, 960 AIME25, 675 OlympiadBench, 1,280 AMC23, and 272 Minerva attempts. AIME and AMC retain 32 attempts per problem; the other benchmarks retain one. GPQA-Diamond and MMLU-Pro are absent.

This supersedes the September 15 tagged snapshot in the numerical report. That historical snapshot had 738 inputs and 699 attempts after termination filtering. The new run changes the population, termination policy, and token scope together: it is not a controlled before/after comparison or an annotation-only ablation. Historical snapshot outputs remain archived.

Generation, replay, and annotation. Generation serves Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 through vLLM with temperature 0.6 and top- p 0.95. Replay uses `unsloth/Qwen3.5-35B-A3B` with runtime 4-bit quantization at the probe-selected decoder indices 27, 34, 35, 36, 38, and 39. Hidden-state features describe block inputs, not exact post-attention router inputs. Router confidence is $1 - H(p)/\log 256$. The continuation analysis has 83 scalar features; across-layer means are emphasized and layerwise screens remain exploratory. Saved bootstrap intervals use 500 replicates and seed 42, resampling whole source problems.

The sentence judge is `unsloth/Qwen3.8-27B-NVFP4`, temperature zero, thinking enabled with low reasoning effort, and a 4,096-token completion budget. Inputs contain the question and previous/current/next sentences. The frozen program SHA-256 is `69706a02c0484f345cf57775729ce7551feb5932954b9ad0b8f5bb63585c6fd9`. Its accuracy on these generated math traces has not been independently measured; the earlier GGUF alignment evaluation uses a different deployment.

Population and token scopes. All supplied generations enter whole-continuation, full-reasoning, position, and expert analyses, including token-limit completions. Whole-continuation metrics include final-answer tokens. The `full/reasoning` view now uses every reasoning token, independently of annotation coverage. Position views also use all reasoning tokens within their absolute windows. Class views alone use available validated sentence labels: the sampling protocol selects sample zero, with error-rate-weighted benchmark quotas, seed 42, and a 100,000-sentence cap. Missing labels never exclude a generation from the other views. Missing class/window features are pairwise missing, not zero.

Table 13. Completed-run coverage and accuracy, Qwen3.5-35B-A3B. All supplied attempts are retained, including limit hits. AIME/AMC use 32 attempts per problem; other benchmarks use one. Unparseable answers with known gold are incorrect. Pooled accuracy is descriptive.

Dataset	Input	Limit hits	Analyzed	Correct	Errors	Input acc.	Analyzed acc.
MATH-500	500	17	500	345	155	0.690	0.690
AIME24	960	105	960	817	143	0.851	0.851
AIME25	960	165	960	773	187	0.805	0.805
Olympiad	675	89	675	322	353	0.477	0.477
AMC23	1280	17	1280	1238	42	0.967	0.967
Minerva	272	19	272	73	199	0.268	0.268
Pooled	4647	412	4647	3568	1079	0.768	0.768

Table 14. Whole-continuation aggregate features versus correctness: point-biserial r (finite traces), Qwen3.5-35B-A3B. Token-limit completions are included. A dash denotes an undefined coefficient.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	-0.291 (500)	-0.726 (960)	-0.727 (960)	-0.387 (675)	-0.238 (1,280)	-0.293 (272)
Router confidence	0.199 (500)	0.533 (960)	0.493 (960)	0.243 (675)	0.175 (1,280)	0.112 (272)
Leading-expert margin	0.125 (500)	0.562 (960)	0.479 (960)	0.218 (675)	0.295 (1,280)	0.075 (272)
Selected probability mass	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)
Selection-boundary margin	0.054 (500)	-0.035 (960)	0.222 (960)	0.118 (675)	-0.010 (1,280)	0.039 (272)
Selected/rest mean gap	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)
Hidden-state norm	0.203 (500)	-0.018 (960)	0.056 (960)	0.085 (675)	0.550 (1,280)	0.083 (272)
Hidden-state step distance	0.157 (500)	0.382 (960)	0.404 (960)	0.111 (675)	0.557 (1,280)	0.100 (272)
Hidden/router geometry	-0.152 (500)	-0.058 (960)	-0.077 (960)	-0.102 (675)	-0.229 (1,280)	-0.096 (272)
Top-1 switch rate	-0.071 (500)	-0.297 (960)	0.141 (960)	0.005 (675)	-0.160 (1,280)	-0.006 (272)
Adjacent selected-set overlap	0.047 (500)	0.022 (960)	-0.155 (960)	0.016 (675)	-0.062 (1,280)	-0.010 (272)

Termination and accuracy denominators. There are 3,568 correct answers among 4,647 attempts (0.768). All 412 completion-limit hits (8.9%) remain in the analysis; only two are correct. Among the 4,235 non-limited attempts, 3,566 are correct (0.842). There are 388 unparseable answers counted as incorrect under the known-gold scoring contract. These denominators differ from accuracy computed only among parseable answers in generation summaries. Termination and length are strong potential confounders: these results do not estimate natural-length performance or a length-adjusted routing effect.

5.3 Correctness associations and uncertainty

Whole-continuation router confidence has positive point-biserial coefficients on all six benchmarks: MATH-500 $r = 0.199$, AIME24 0.533, AIME25 0.493, Olympiad 0.243, AMC23 0.175, and Minerva 0.112. Saved pointwise 95% problem-bootstrap intervals exclude zero except for AMC23 $[-0.116, 0.678]$. MATH-500's interval is $[0.099, 0.276]$; AIME24's is $[0.382, 0.678]$ and AIME25's is $[0.302, 0.608]$. These are marginal associations, without length or dataset adjustment.

Leading-expert margin ($M = p_{(1)} - p_{(2)}$) is positive throughout; Minerva's interval includes zero. Hidden-state step distance is also positive throughout ($r = 0.100$ – 0.557), while token count is negative (-0.727 – -0.238). The change from the historical snapshot's signs cannot be attributed to adding labels alone: replay and inclusion semantics also changed. Attempts within AIME/AMC problems are dependent; naive trace-level p -values and their BH adjustments are diagnostic only.

Full-reasoning confidence gives similar coefficients: 0.191, 0.527, 0.486, 0.238, 0.164, and 0.124 in the same dataset order. All 4,647 traces have reasoning tokens. Similarity between these two scopes is descriptive; it does not remove length or termination effects.

5.4 Class-conditioned routing and hidden states

Table 15. Class coverage in the completed run. Qwen3.8-27B labels on Qwen3.5-35B-A3B generations. Units are tagged sentences; dataset columns count traces with selected class tokens. Classes overlap across traces; missing classes are not zero-valued features.

Class	Units	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva	Total traces
Read	6,358	378	30	26	514	27	270	1245
Analyze	22,378	424	30	30	619	38	271	1412
Plan	5,654	407	28	28	507	28	271	1269
Implement	23,844	477	30	30	667	39	271	1514
Explore	7,151	260	28	27	474	22	251	1062
Verify	10,745	400	29	30	617	38	270	1384
Monitor	23,870	451	30	29	583	29	272	1394

The 100,000 sampled sentences have complete labels across 1,546 nonempty attempt samples; one of the 1,547 sample-zero attempts has no labelled sentence. All seven class analyses are complete, with coverage reported separately from the full population. Explore has tokens in 1,062 traces; Verify in 1,384. AIME/AMC class views contain at most the 30/30/40 sample-zero attempts rather than all 32 attempts per problem. Annotation counts describe sampled sentences, not class prevalence in the complete corpus. Class transition metrics require adjacent original tokens within the same sentence.

Class conditioning remains heterogeneous. Verify confidence is 0.066 on MATH-500, 0.123 on Olympiad, and -0.028 on Minerva; Plan confidence is -0.067 , 0.039, and -0.121 , respectively. These separately selected cohorts do not test a difference between classes. The smaller AIME/AMC class cohorts remain especially sensitive to individual errors. Annotation errors, length, difficulty, and coverage are uncontrolled, so these screens do not identify beneficial or harmful reasoning states.

5.5 Class counts, transitions, and outcome associations

The call distinguishes class-conditioned router features from the proportion of reasoning assigned to each class (24:41–29:57). The following plots use *sampled sentence shares*: the number of labelled sentences of a class divided by the number of labelled sentences in that attempt. An observed zero share is retained; unlabelled sentences are never assigned a class. These are estimates from the sampling protocol, not complete-sequence class fractions. Each source problem contributes only its sample-zero attempt; benchmark comparisons retain their separate denominators. Attempts with no labelled sentences are excluded.

Table 16. Mean sampled sentence share (percent), with equal weight per labelled attempt. Empty-label attempts are excluded; absent classes in an observed sample have zero share. These are not full-corpus class proportions.

Dataset	Attempts	Read	Analyze	Plan	Implement	Explore	Verify	Monitor
MATH-500	499	8.9	19.5	9.3	29.3	4.1	11.8	17.1
AIME24	30	6.7	33.9	2.9	28.3	7.0	14.4	6.8
AIME25	30	5.7	33.2	4.8	27.9	6.7	13.4	8.3
Olympiad	675	6.9	26.8	6.0	29.7	6.4	13.2	11.0
AMC23	40	6.6	25.9	5.8	32.0	3.8	15.2	10.6
Minerva	272	9.4	19.8	8.4	21.8	7.7	10.3	22.6

Class-state dynamics remain descriptive. Relating complete transition paths to correctness and interventions conditioned on those paths is future work, as agreed at 29:32–30:20 and 45:23–48:10. The currently sampled labels do not justify reconstructing missing transitions.

5.6 Position-dependent estimates and coverage

The position reference uses all 4,647 supplied generations and mean reasoning length 12,498.062 tokens. Ten non-cumulative windows have boundaries $[i\bar{L}/10]$, approximately 1,249.8 tokens apart; overflow begins after the final boundary at 12,499. Every reasoning token contributes to its window regardless of tagging. Token count is omitted as a position predictor. Coverage falls from 4,647 traces in the first window to 1,789 in overflow, so later-window coefficients describe different cohorts.

Table 17. Reasoning length in tokens, all attempts including limit hits. This replaces the position-coverage count plot; lengths are observed under the generation budget.

Dataset	Attempts	Mean	Median	90th percentile
MATH-500	500	6,226.5	4,250	12,772
AIME24	960	15,908.4	14,011	32,766
AIME25	960	17,874.9	16,620	32,768
Olympiad	675	13,402.5	9,150	32,768
AMC23	1280	8,452.8	7,822	14,929
Minerva	272	9,805.0	7,465	19,533

5.7 Metric-pair dependence

Token count and confidence are negatively associated on all six datasets: $\rho = -0.681, -0.627, -0.836, -0.870, -0.845$, and -0.280 in MATH/AIME24/AIME25/Olympiad/AMC/Minerva order. This dependence is one reason confidence–correctness correlations need length-aware follow-up. Top-eight mass S and the selected/unselected mean probability gap G are algebraically redundant for $k = 8$ and $E = 256$:

$$G = \frac{S}{k} - \frac{1-S}{E-k} = \frac{E}{k(E-k)}S - \frac{1}{E-k}. \quad (2)$$

The affine relation survives token/layer averaging. These metrics should not count as independent corroboration or enter an unregularized model twice. The appendix presents correlation heatmaps for all 20 scopes, including class-conditioned metrics versus correctness and token count. The numeric tables remain available as a separate generated artifact.

The legacy regex flags (backtracking, contradiction, and self-correction) are removed from the report’s outcome panels, as requested at 19:08–19:39. They are not the seven predicted reasoning classes. Historical detector outputs remain in the source artifacts.

5.8 Repeated-attempt findings

There are now 100 eligible complete avg@32 groups: 30 AIME24, 30 AIME25, and 40 AMC23 problems. Problem means use all declared attempts, including limit hits, and binary outcomes are complete. Unlike the historical sample-zero snapshot, the current run supports both between-problem rank associations and within-problem correct/incorrect contrasts.

Table 18. Whole-continuation mean feature versus avg@32 correctness, Spearman ρ (source problems). All 32 attempts, including token-limit completions, enter each eligible group.

Feature	AIME24	AIME25	AMC23
Token count	−0.861 (30)	−0.909 (30)	−0.057 (40)
Router confidence	0.632 (30)	0.705 (30)	−0.200 (40)
Leading-expert margin	0.583 (30)	0.638 (30)	0.190 (40)
Selected probability mass	−0.112 (30)	0.388 (30)	−0.215 (40)
Selection-boundary margin	−0.367 (30)	0.135 (30)	−0.145 (40)

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Feature	AIME24	AIME25	AMC23
Selected/rest mean gap	−0.112 (30)	0.388 (30)	−0.215 (40)
Hidden-state norm	−0.384 (30)	−0.186 (30)	0.484 (40)
Hidden-state step distance	0.238 (30)	0.384 (30)	0.504 (40)
Hidden/router geometry	0.273 (30)	0.046 (30)	−0.331 (40)
Top-1 switch rate	−0.381 (30)	0.156 (30)	0.121 (40)
Adjacent selected-set overlap	−0.047 (30)	−0.196 (30)	−0.327 (40)

Table 19. Within-problem confidence differences, Qwen3.5: mean correct minus incorrect, equally weighted over mixed-outcome problems; saved 95% whole-problem bootstrap intervals.

Dataset	Mixed problems	Difference	95% interval
AIME24	14	0.00268	[0.00152, 0.00412]
AIME25	15	0.00123	[0.00065, 0.00176]
AMC23	6	0.01102	[0.00455, 0.01709]

Mean whole-continuation confidence correlates positively with avg@32 on AIME24 ($\rho = 0.632$, CI [0.383, 0.818]) and AIME25 (0.705, CI [0.460, 0.854]). AMC23 is negative but uncertain (−0.200, CI [−0.565, 0.243]), despite its positive attempt-level coefficient. Thus attempt-level and problem-level associations answer different questions. AIME24 and AIME25 mean token count have strong negative avg@32 associations (−0.861 and −0.909).

Within-problem confidence differences are positive over 14, 15, and 6 mixed-outcome problems, respectively; Table 19 reports magnitudes and bootstrap intervals. Conditioning on a shared question reduces between-question variation but does not control attempt length or termination. Class views still lack repeated labelled attempts, so these repeated-attempt findings must not be generalized to class-specific effects.

5.9 Expert screen, storage, and implications

Table 20. Post-hoc largest absolute correctness coefficient per dataset in the expert-rate screen, Qwen3.5-35B-A3B. Point-biserial r and BH q within dataset/target; no expert bootstrap intervals. Numeric IDs pool six distinct layer-local modules. Repeated-attempt trace-level q -values are diagnostic, not cluster-adjusted inference.

Dataset	Pooled-ID rate	Traces	Errors	r	q
MATH-500	Top-1 61 / top-2 7	500	155	0.328	$3.36e - 11$
AIME24	Top-1 ID 123	960	143	−0.568	$1.87e - 80$
AIME25	Top-1 133 / top-2 85	960	187	0.556	$4.72e - 76$
Olympiad	Top-1 70 / top-2 181	675	353	0.312	$6.25e - 14$
AMC23	Top-8 ID 217	1280	42	−0.665	$2.83e - 161$
Minerva	Top-1 162 / top-2 165	272	199	0.409	$1.31e - 09$

The expert screen includes top-1 rates, any-top-eight rates, and up to 128 ordered top-1/top-2 pairs selected by routing support (minimum four traces). The largest absolute correctness coefficient per dataset is selected post hoc. Figure 11 shows all selected-pool IDs against correctness and total token count, without selecting a maximum. Numeric IDs pool six different layer-local modules; they do not identify a portable expert or establish an intervention target. Trace-level BH values on repeated-attempt datasets do not account for clustering.

Metric values associated with accuracy. Correlation signs alone do not specify the observed metric ranges associated with higher accuracy (42:06–45:17). Figure 12 therefore reports accuracy directly against router confidence, the selection-boundary margin, and the selected/rest mean probability gap. Bins are computed separately per benchmark; this is an exploratory whole-continuation analysis, not an online threshold-selection experiment.

Common router notation and unavailable extensions. Let $p_{(j)}$ denote the j th largest router probability, E the number of experts, and k the selected count ($E = 256$, $k = 8$ here). Router confidence is $C = 1 - H(p)/\log E$; leading-expert margin is $M = p_{(1)} - p_{(2)}$; selected probability mass is $S = \sum_{j \leq k} p_{(j)}$; selection-boundary margin is $B = p_{(k)} - p_{(k+1)}$; and selected/rest mean gap is $G = S/k - (1 - S)/(E - k)$. Thus M is not a selected-versus-rest statistic. These model-independent names replace ambiguous top-8/top-9 plot labels without changing the saved metric definitions.

The requested rank-prefix curves (top 2, 4, 6, then all selected versus the rest; 20:11–21:07 and 37:29–39:20) require additional probability summaries. Only the all-selected endpoint G is saved. Separate entropy of the selected and unselected distributions also cannot be recovered from global entropy, selected mass, or expert IDs. These extensions require new feature extraction/replay; no missing curves are fabricated.

Replay covers 62,088,209 continuation tokens. Compact tensor artifacts occupy 5.970 GB versus a projected 1,907.350 GB of raw selected-layer payload; dataset files total 9.259 GB (decimal units). Storage and source-fingerprint checks are recorded in `report/correlation_run_audit.json`. No raw router/hidden tensors or expert weights are saved. The completed run supplies six-benchmark, full-reasoning, class, position, lexical, and repeated-attempt analyses. Next steps are independent label assessment, length-/termination-aware models, common-cohort class/window comparisons, and validation holding out whole problems before any routing intervention.

6 Three-model correlation results

As of September 19, 2026, forward replay and saved correlation analyses are complete for Qwen3.5-35B-A3B, Gemma-4-26B-A4B, and GPT-OSS-20B. Each includes 4,647 attempts from 1,547 source problems on the same six benchmarks, with 100 complete avg@32 groups. All token-limit completions are retained. Completion of class analysis means analysis of available labels, not full annotation coverage. GPT-OSS class tokens occur only in MATH-500 and part of AIME24; the other four benchmarks have no class estimates. Qwen uses the local deterministic NVFP4 judge described above. Gemma and GPT-OSS use imported September 18 Qwen3.8-27B labels, with temperature 1, top- p 0.95, top- k 20, low reasoning effort and a 16,384-token judge budget. Consequently class comparisons also differ in annotation protocol and cohort. Generation uses Qwen GPTQ-Int4, Gemma NVFP4, and GPT-OSS checkpoints respectively; replay settings and selected decoder indices are below. Token counts and absolute position windows are model-specific. Correlations are marginal descriptive associations, not controlled model effects.

Table 21. Replay configuration and mean reasoning length used for position windows.

Model	Replay	Decoder indices	Mean tokens
Qwen3.5-35B-A3B	unsloth-4bit	27, 34, 35, 36, 38, 39	12,498.1
Gemma-4-26B-A4B	bnb-4bit	16, 19, 21, 22, 25, 27, 29	10,512.6
GPT-OSS-20B	mxfp4-bf16	15, 19, 21, 23	5,977.6

Table 22. All-attempt accuracy and termination by model. Invalid known-gold answers count as incorrect; repeated attempts are not independent problems.

Model	Dataset	Attempts	Correct	Accuracy	Limit hits
Qwen3.5-35B-A3B	MATH-500	500	345	0.690	17
Qwen3.5-35B-A3B	AIME24	960	817	0.851	105
Qwen3.5-35B-A3B	AIME25	960	773	0.805	165
Qwen3.5-35B-A3B	Olympiad	675	322	0.477	89
Qwen3.5-35B-A3B	AMC23	1280	1238	0.967	17
Qwen3.5-35B-A3B	Minerva	272	73	0.268	19
Gemma-4-26B-A4B	MATH-500	500	353	0.706	6
Gemma-4-26B-A4B	AIME24	960	852	0.887	54
Gemma-4-26B-A4B	AIME25	960	791	0.824	103
Gemma-4-26B-A4B	Olympiad	675	349	0.517	36
Gemma-4-26B-A4B	AMC23	1280	1249	0.976	3
Gemma-4-26B-A4B	Minerva	272	82	0.301	2
GPT-OSS-20B	MATH-500	500	332	0.664	4
GPT-OSS-20B	AIME24	960	681	0.709	99
GPT-OSS-20B	AIME25	960	668	0.696	118
GPT-OSS-20B	Olympiad	675	285	0.422	34
GPT-OSS-20B	AMC23	1280	1152	0.900	6
GPT-OSS-20B	Minerva	272	31	0.114	0

Table 23. Class token-bearing attempts; classes overlap. Zero coverage implies unavailable estimates.

Model	Class	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Qwen3.5-35B-A3B	Read	378	30	26	514	27	270
Qwen3.5-35B-A3B	Analyze	424	30	30	619	38	271
Qwen3.5-35B-A3B	Plan	407	28	28	507	28	271
Qwen3.5-35B-A3B	Implement	477	30	30	667	39	271
Qwen3.5-35B-A3B	Explore	260	28	27	474	22	251
Qwen3.5-35B-A3B	Verify	400	29	30	617	38	270
Qwen3.5-35B-A3B	Monitor	451	30	29	583	29	272
Gemma-4-26B-A4B	Read	393	28	29	528	34	271
Gemma-4-26B-A4B	Analyze	424	30	30	615	40	269
Gemma-4-26B-A4B	Plan	294	27	28	397	20	244
Gemma-4-26B-A4B	Implement	478	30	30	656	39	271
Gemma-4-26B-A4B	Explore	202	26	25	387	20	235
Gemma-4-26B-A4B	Verify	408	30	30	597	37	261
Gemma-4-26B-A4B	Monitor	280	30	29	461	28	251
GPT-OSS-20B	Read	474	68	0	0	0	0
GPT-OSS-20B	Analyze	433	68	0	0	0	0
GPT-OSS-20B	Plan	324	68	0	0	0	0
GPT-OSS-20B	Implement	500	68	0	0	0	0
GPT-OSS-20B	Explore	291	44	0	0	0	0
GPT-OSS-20B	Verify	435	68	0	0	0	0
GPT-OSS-20B	Monitor	496	68	0	0	0	0

Across models, confidence–correctness associations do not have a uniform sign. Qwen is positive on all six benchmarks; Gemma is negative on AMC23 and Minerva, and GPT-OSS is negative on MATH-500, AIME25, AMC23, and Minerva. The pointwise intervals in the comparison figures show substantial uncertainty for several of these estimates. Token-count associations are negative except for Gemma on AMC23. These differences motivate model-specific validation rather than assuming that a Qwen-derived routing rule transfers to other models.

6.1 Qwen3.5-35B-A3B

Table 24. Qwen3.5-35B-A3B: whole-continuation correctness r (finite attempts).

Metric	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	−0.291 (500)	−0.726 (960)	−0.727 (960)	−0.387 (675)	−0.238 (1,280)	−0.293 (272)
Router confidence	0.199 (500)	0.533 (960)	0.493 (960)	0.243 (675)	0.175 (1,280)	0.112 (272)
Leading-expert margin	0.125 (500)	0.562 (960)	0.479 (960)	0.218 (675)	0.295 (1,280)	0.075 (272)
Selected probability mass	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)
Selection-boundary margin	0.054 (500)	−0.035 (960)	0.222 (960)	0.118 (675)	−0.010 (1,280)	0.039 (272)
Selected/rest mean gap	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)
Hidden-state norm	0.203 (500)	−0.018 (960)	0.056 (960)	0.085 (675)	0.550 (1,280)	0.083 (272)
Hidden-state step distance	0.157 (500)	0.382 (960)	0.404 (960)	0.111 (675)	0.557 (1,280)	0.100 (272)
Hidden/router geometry	−0.152 (500)	−0.058 (960)	−0.077 (960)	−0.102 (675)	−0.229 (1,280)	−0.096 (272)
Top-1 switch rate	−0.071 (500)	−0.297 (960)	0.141 (960)	0.005 (675)	−0.160 (1,280)	−0.006 (272)
Adjacent selected-set overlap	0.047 (500)	0.022 (960)	−0.155 (960)	0.016 (675)	−0.062 (1,280)	−0.010 (272)

Confidence coefficients in benchmark order are 0.199, 0.533, 0.493, 0.243, 0.175, 0.112.

Table 25. Qwen3.5-35B-A3B: problem mean versus avg@32, Spearman ρ (problems).

Metric	AIME24	AIME25	AMC23
Token count	−0.861 (30)	−0.909 (30)	−0.057 (40)
Router confidence	0.632 (30)	0.705 (30)	−0.200 (40)
Leading-expert margin	0.583 (30)	0.638 (30)	0.190 (40)
Selected probability mass	−0.112 (30)	0.388 (30)	−0.215 (40)
Selection-boundary margin	−0.367 (30)	0.135 (30)	−0.145 (40)
Selected/rest mean gap	−0.112 (30)	0.388 (30)	−0.215 (40)
Hidden-state norm	−0.384 (30)	−0.186 (30)	0.484 (40)
Hidden-state step distance	0.238 (30)	0.384 (30)	0.504 (40)
Hidden/router geometry	0.273 (30)	0.046 (30)	−0.331 (40)
Top-1 switch rate	−0.381 (30)	0.156 (30)	0.121 (40)
Adjacent selected-set overlap	−0.047 (30)	−0.196 (30)	−0.327 (40)

6.2 Gemma-4-26B-A4B

Table 26. Gemma-4-26B-A4B: whole-continuation correctness r (finite attempts).

Metric	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	−0.155 (500)	−0.672 (960)	−0.709 (960)	−0.240 (675)	0.080 (1,280)	−0.205 (272)
Router confidence	0.080 (500)	0.392 (960)	0.313 (960)	0.027 (675)	−0.256 (1,280)	−0.037 (272)
Leading-expert margin	0.108 (500)	0.297 (960)	0.456 (960)	0.036 (675)	−0.243 (1,280)	0.010 (272)
Selected probability mass	0.072 (500)	0.375 (960)	0.259 (960)	0.009 (675)	−0.217 (1,280)	−0.061 (272)
Selection-boundary margin	−0.027 (500)	0.261 (960)	0.068 (960)	−0.043 (675)	−0.128 (1,280)	−0.028 (272)
Selected/rest mean gap	0.072 (500)	0.375 (960)	0.259 (960)	0.009 (675)	−0.217 (1,280)	−0.061 (272)
Hidden-state norm	0.036 (500)	0.161 (960)	0.007 (960)	−0.005 (675)	0.104 (1,280)	0.003 (272)
Hidden-state step distance	0.010 (500)	0.272 (960)	0.440 (960)	0.001 (675)	−0.141 (1,280)	0.059 (272)
Hidden/router geometry	−0.068 (500)	0.020 (960)	0.000 (960)	−0.019 (675)	−0.043 (1,280)	−0.076 (272)
Top-1 switch rate	0.052 (500)	−0.133 (960)	0.222 (960)	0.048 (675)	−0.116 (1,280)	0.100 (272)
Adjacent selected-set overlap	−0.021 (500)	0.129 (960)	−0.244 (960)	−0.049 (675)	0.038 (1,280)	−0.117 (272)

Confidence coefficients in benchmark order are 0.080, 0.392, 0.313, 0.027, −0.256, −0.037.

Table 27. Gemma-4-26B-A4B: problem mean versus avg@32, Spearman ρ (problems).

Metric	AIME24	AIME25	AMC23
Token count	−0.744 (30)	−0.917 (30)	−0.099 (40)
Router confidence	0.473 (30)	0.468 (30)	0.071 (40)
Leading-expert margin	0.586 (30)	0.777 (30)	0.191 (40)
Selected probability mass	0.416 (30)	0.403 (30)	0.027 (40)

Continued from previous page

Metric	AIME24	AIME25	AMC23
Selection-boundary margin	0.345 (30)	0.182 (30)	-0.063 (40)
Selected/rest mean gap	0.416 (30)	0.403 (30)	0.027 (40)
Hidden-state norm	0.026 (30)	-0.042 (30)	-0.060 (40)
Hidden-state step distance	0.410 (30)	0.547 (30)	-0.063 (40)
Hidden/router geometry	0.300 (30)	0.042 (30)	-0.004 (40)
Top-1 switch rate	-0.367 (30)	0.285 (30)	-0.215 (40)
Adjacent selected-set overlap	0.213 (30)	-0.321 (30)	0.059 (40)

6.3 GPT-OSS-20B

Table 28. GPT-OSS-20B: whole-continuation correctness r (finite attempts).

Metric	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	-0.221 (500)	-0.606 (960)	-0.646 (960)	-0.362 (675)	-0.169 (1,280)	-0.094 (272)
Router confidence	-0.061 (500)	0.210 (960)	-0.075 (960)	0.051 (675)	-0.077 (1,280)	-0.160 (272)
Leading-expert margin	0.041 (500)	0.137 (960)	-0.176 (960)	0.154 (675)	-0.018 (1,280)	-0.082 (272)
Selected probability mass	-0.052 (500)	0.128 (960)	-0.102 (960)	0.089 (675)	-0.033 (1,280)	-0.166 (272)
Selection-boundary margin	-0.036 (500)	-0.218 (960)	-0.091 (960)	-0.064 (675)	-0.005 (1,280)	-0.032 (272)
Selected/rest mean gap	-0.052 (500)	0.128 (960)	-0.102 (960)	0.089 (675)	-0.033 (1,280)	-0.166 (272)
Hidden-state norm	0.192 (500)	0.494 (960)	0.366 (960)	0.329 (675)	0.049 (1,280)	-0.006 (272)
Hidden-state step distance	-0.150 (500)	-0.186 (960)	-0.404 (960)	-0.210 (675)	-0.094 (1,280)	0.045 (272)
Hidden/router geometry	0.002 (500)	-0.239 (960)	-0.311 (960)	-0.076 (675)	-0.027 (1,280)	0.053 (272)
Top-1 switch rate	0.181 (500)	0.149 (960)	0.376 (960)	0.110 (675)	-0.006 (1,280)	0.331 (272)
Adjacent selected-set overlap	-0.198 (500)	-0.068 (960)	-0.278 (960)	-0.143 (675)	-0.052 (1,280)	-0.317 (272)

Confidence coefficients in benchmark order are -0.061 , 0.210 , -0.075 , 0.051 , -0.077 , -0.160 .

Table 29. GPT-OSS-20B: problem mean versus avg@32, Spearman ρ (problems).

Metric	AIME24	AIME25	AMC23
Token count	-0.889 (30)	-0.908 (30)	-0.338 (40)
Router confidence	0.337 (30)	0.104 (30)	0.186 (40)
Leading-expert margin	0.248 (30)	-0.028 (30)	0.262 (40)
Selected probability mass	0.186 (30)	0.046 (30)	0.272 (40)
Selection-boundary margin	-0.356 (30)	-0.273 (30)	0.060 (40)
Selected/rest mean gap	0.186 (30)	0.046 (30)	0.272 (40)
Hidden-state norm	0.733 (30)	0.655 (30)	0.338 (40)
Hidden-state step distance	-0.341 (30)	-0.655 (30)	-0.289 (40)
Hidden/router geometry	-0.658 (30)	-0.474 (30)	-0.177 (40)
Top-1 switch rate	0.162 (30)	0.546 (30)	-0.151 (40)
Adjacent selected-set overlap	-0.029 (30)	-0.340 (30)	-0.089 (40)

Aggregate correctness and metric-pair coefficients were independently recomputed from saved CSVs in all 60 model/scope combinations; repeated-problem mean correlations and within-problem contrasts were also checked. This validates numerical consistency, not judge accuracy or replay fidelity. Audit counts and numerical errors are saved in `report/multimodel_correlation_audit.json`; input SHA-256 hashes are in `report/multimodel_correlation_sources.json`. The appended atlases include every Gemma and GPT-OSS scope, including empty class panels. Full correctness tables are available in `report/multimodel_correlation_tables.tex`.

7 Analysis

7.1 Conclusions supported so far

- (1) The static-reward, context-aware GEPA study reaches 0.650 cross-validated macro-F1 with the light budget. The advantage over balanced-accuracy selection is small and the heavy search is worse. In the separate

completed same-model LLM-judge final fit, development-set balanced accuracy rises from 0.677 to 0.777 and macro-F1 from 0.613 to 0.727, primarily through reduced Monitor overprediction. This establishes successful prompt optimization on the development distribution, not held-out generalization. The final Qwen3.8-27B alignment does not reproduce that gain: balanced accuracy falls from 0.740 for the imported seed to 0.729 for the GEPA candidate, and the prespecified recall gate correctly retains the seed.

- (2) The boundary probes show substantial sentence-split decodability of upcoming Schoenfeld episodes, with selected-layer AUROC of 0.835–0.934 and the strongest thresholded performance for Implement, Verify, and Read. This establishes accessible prefix-state signal within familiar responses, not generalization to unseen reasoning traces. External application is best interpreted as coarse phase inspection: a provisional pragmatic review gives 50.3% strict and 60.8% compatible agreement, with 187/581 multi-label units. Implement is strongest (F1 0.729), while the remaining labels are incomplete external tags. ProcessBench Verify predictions identify review-heavy post-error passages but do not identify correctness.
- (3) All three model correlation analyses are complete (Qwen, Gemma, and GPT-OSS), each with 4,647 attempts and 100 complete avg@32 groups. GPT-OSS class coverage is limited to MATH-500 and part of AIME24; class comparisons also differ in annotation protocol. The September 17 Qwen correlation run covers 4,647 attempts across six benchmarks, with all token-limit completions retained. Whole-continuation confidence coefficients are positive ($r = 0.112$ – 0.533), although AMC23’s problem-bootstrap interval includes zero. Mean confidence also correlates positively with avg@32 on AIME24 ($\rho = 0.632$) and AIME25 (0.705), while AMC23 is uncertain (-0.200). Full-reasoning and position metrics now use all reasoning tokens; classes use sampled sentence labels. The 412 limit hits, of which only two are correct, make termination and length essential follow-up controls. Expert maxima remain exploratory pooled-ID candidates.

7.2 What has not been established

Neither GEPA final fit provides an unbiased test estimate. The earlier run selects among 145 candidates on the same six-response validation set, has only one optimization seed, and leaves the locked 336-unit test unevaluated. The result does not establish that the expanded 40-demonstration prompt transfers to new responses or that the same-model reward is an independent evaluator. The Qwen3.8 alignment reuses that selected prompt as its seed, optimizes and selects on the same validation partition, and also leaves the locked test untouched; retaining the seed is evidence that this search found no safe development-set improvement, not evidence of generalization.

The boundary-probe run does not establish response-grouped generalization, calibrated seven-way episode classification, or the cognitive states of Qwen’s own generations. It uses Qwen to encode teacher-forced DeepSeek-R1 text, shares all 38 response IDs between train and test, and selects layers on the reported test set. Its external argmax outputs have no authoritative human or benchmark episode ground truth. Agreement with the provisional hybrid review is diagnostic and should not be called validated transfer accuracy.

The correlation results do not establish a dataset-invariant predictor or a causal mechanism. No length-conditioned or dataset-adjusted model has been fit. All attempts are retained, but completion limits censor natural response length. Class views condition on error-rate-weighted sentence sampling and available labels. GPQA-Diamond and MMLU-Pro remain absent. AIME/AMC have only 30/30/40 independent source problems, and their class cohorts have at most one labelled attempt per problem. Undefined bootstrap resamples are discarded, so pointwise intervals need particular care for sparse outcomes.

Seven class views are now available, but their labels have no independent accuracy evaluation on the generated math traces. Class views pool sampled labelled tokens; position views pool all reasoning tokens in each window. Their finite cohorts differ. Per-family BH adjustment does not correct selection across views or provide held-out validation. Numeric expert IDs pool distinct layer-local modules. Across all three workflows, correlation and linear decodability should not be interpreted as causal evidence.

8 Discussion

The completed GEPA final fit demonstrates that same-model informative feedback can guide a large prompt toward the development labels: it fixes 49 seed errors while introducing 14 regressions and improves every validation response. Its optimization trace also shows the central risk. The prompt expands from 21 to 40 demonstrations and is selected after 145 candidate evaluations on those same six responses. The next defensible test is therefore to freeze and ablate or compress the prompt, quantify response-level uncertainty, and evaluate it once on genuinely unseen responses.

The Qwen3.8 alignment is an important negative result. Although its internal same-model reward increases, exact corpus-label metrics decline and Plan recall breaches the prespecified 0.10 safety threshold. Its shorter optimized instruction also contains conflicting guidance for phrases such as “Let me compute that,” plausibly contributing to more Plan-to-Monitor errors. The safety gate therefore performs its intended role: optimization reward alone is insufficient evidence for deployment, and the byte-identical selected seed should now be frozen rather than optimized further on this validation set.

The episode probes add a complementary representation-level result: the prefix before a sentence often encodes the phase likely to follow, and the signal is strongest in middle-to-late layers rather than at the final model output. The current split cannot determine whether this reflects a portable episode representation, trace-specific transition regularities, position, or formatting. A response-grouped nested evaluation should select both layer and regularization on inner folds, evaluate once on unseen responses, repeat over seeds, and report response-bootstrap intervals. Standardized or norm-matched features and simple position, previous-unit, and formatting baselines are needed to separate hidden-state information from layer scale and trace structure.

For external use, the probe outputs should be treated as abstaining phase suggestions rather than annotations. Abbreviation- and LaTeX-aware segmentation should precede evaluation; independent one-versus-rest scores should either be calibrated on a common multiclass validation set or retain multiple labels. The pragmatic review confirms substantial ambiguity and only 60.8% compatible agreement. Independent human annotators should resolve the 37 low-confidence units and assess inter-annotator agreement before any transfer-accuracy claim. Cross-model recognition, as studied here, should also be separated from Qwen’s own generation by running the protocol on Qwen-generated traces.

The completed replay expands the evidence to all six benchmarks and complete repeated-attempt groups. Positive marginal confidence associations on AIME also appear in problem-level avg@32 analyses, while AMC23 illustrates that attempt-level and problem-level signs can differ. These observations need length- and termination-aware models, validation holding out whole problems, and common-cohort class/window comparisons. The changed population and replay semantics preclude interpreting differences from the September 15 snapshot as an effect of tagging. Layer-qualified expert rates and independent label assessment remain necessary before routing interventions.

9 Conclusion

The three active workflows support deliberately limited claims. GEPA improves the earlier same-model-judge development result substantially, but the final Qwen3.8 alignment finds no safe improvement over that imported prompt and neither run has been tested on an untouched response set. The boundary probes reveal strong within-split linear access to upcoming episode labels, especially for Implement, Verify, and Read, but their sentence split leaks every response across train and test and their external outputs collapse toward a coarse setup, execution, and review taxonomy. The completed six-benchmark run adds full-reasoning, class, position, lexical, metric-pair, and repeated-attempt analyses. Positive confidence associations on both AIME sets persist at the avg@32 problem level, but length, termination, cohort selection, and sparse class outcomes remain uncontrolled. These screens supply hypotheses for held-out evaluation without establishing a causal routing mechanism.

The immediate evidential requirements are a frozen-prompt GEPA evaluation on unseen responses, response-grouped nested boundary probes with independently validated external labels, a prespecified response to remaining generation censoring, extension of the current run to GPQA-Diamond and MMLU-Pro, and length-/termination-aware and common-cohort position analyses with problem-respecting validation before routing interventions are evaluated.

10 Immediate Work

The agreed order of work is:

- (1) freeze and archive the seed prompt retained by the completed final GEPA alignment, and reserve the locked test for a prespecified evaluation;
- (2) independently assess sentence labels, document token-limit sensitivity, and extend coverage to the missing benchmarks;
- (3) fit a prespecified length-/termination-aware, dataset-aware correctness model with problem-respecting validation and exploratory layerwise screening;
- (4) validate the completed repeated-problem analyses, test class/position contrasts on common cohorts, and qualify pooled expert-ID candidates by layer; and
- (5) extend the completed Qwen, Gemma, and GPT-OSS comparison with matched annotation coverage and held-out validation.

The detailed acceptance criteria and transcript ambiguities are maintained in NEXT_STEPS.md at the repository root.

A Reproducibility Checklist

Table 30. Living inventory for the three active workflows. *Model(s)*: Qwen3.6-27B and Qwen3.8-27B Q4_K_XL, Qwen/Qwen3.5-35B-A3B-GPTQ-Int4, and Qwen3.5-35B-A3B correlation-run variants. *Metrics*: workflow status, sample counts, headline metrics, and next actions; this is an inventory rather than a common-metric evaluation.

Item	Status	Note / next action
gepaLLMAsJudge	Cross-validation and two final-fit validations complete; test pending	Static-reward grouped cross-validation over 2,789 units gives 67.34% accuracy and 65.05% macro-F1 for light-budget GEPA; heavy is worse. The same-model LLM-judge final fit improves validation accuracy from 66.83% to 75.43% and macro-F1 from 0.613 to 0.727, but reuses six responses across 145 candidate evaluations. In the final Qwen3.8 alignment, GEPA instead reduces balanced accuracy from 0.740 to 0.729 and Plan recall from 0.750 to 0.563, so the safety gate correctly retains the imported 40-example seed. Neither run evaluates the 336-unit locked test. Freeze the selected seed and evaluate it only under a prespecified held-out protocol.
probeTest	Complete; grouped and human validation pending	Qwen3.5 GPTQ probes cover 3,087 units, seven targets, and 41 hidden-state indices; selected-layer AUROC is 0.835–0.934. A provisional hybrid review of 581 external units gives 50.3% primary and 60.8% compatible agreement, with 187 multi-label and 37 low-confidence units; Explore is never predicted. Grouped cross-validation, robust segmentation, multiclass calibration, and independent human adjudication remain pending.
correlation_pipeline	Three six-benchmark model runs complete	Qwen, Gemma, and GPT-OSS each analyze 4,647 attempts from 1,547 problems and 100 complete avg@32 groups. GPT-OSS class labels cover only MATH-500 and part of AIME24. The September 17 Qwen run analyzes all 4,647 attempts from 1,547 problems, including 412 token-limit completions. There are 3,568 correct answers (0.768). Full-reasoning, seven class, and eleven position scopes are complete. All 100 AIME/AMC avg@32 groups are eligible. Class labels remain sampled, and the added repeated attempts are not all tagged. Validate length-/termination-aware effects on held-out problems, assess label accuracy, compare common cohorts, and qualify expert IDs by layer.

A.1 Completed run and independent checks

The numerical source root is `results/correlation_pipeline/reasoning-vllm-v1`. Under it, `analysis/unsloth--Qwen3.5-35B-A3B/correlations.json` records completed analysis of 4,647 attempts. Source hashes are saved in `report/correlation_tables_sources.json`; the audit separately records the analysis SHA-256. Companion CSVs contain trace, problem, and expert features. The `views-v1` directory contains full reasoning, seven classes, ten position bins, overflow, and coverage summaries. Saved analysis uses 500 bootstrap replicates, seed 42, minimum expert support of four traces, and at most 128 ordered expert pairs.

The position-reference corpus has 4,647 traces and mean reasoning length 12,498.062 tokens. Its SHA-256 is 99b71a50bf12f39ecec05f9034b3b07ff57236d2e53a7343f01caa791e108c1e. View population version 2 pools all reasoning tokens for full and position views, and available tagged sentence tokens for class views. All attempts, including limit hits, are retained with pairwise omission of missing features.

Independent audit. `report/audit_tagged_correlation_run.py` matches all generation records to replay fingerprints, verifies available annotation fingerprints and sentence offsets, checks selected layers and compact values, and validates inclusion, correctness propagation, token-limit flags, and view coverage. It independently recalculates saved binary and trace-level metric-pair coefficients across 20 scopes from CSV observations (52,199 coefficients). It additionally checks 984 problem-mean Spearman and within-problem difference estimates for whole continuation and full reasoning. It also checks reported expert maxima and their continuous-metric associations. Counts and numerical discrepancies are recorded in `report/correlation_run_audit.json`.

These checks do not independently rejudge mathematical answers or recompute model activations, bootstrap intervals, or BH adjustments. The historical `report/audit_correlation_run.py` audits a different untagged population.

Regenerating the report. From the repository root, using Python with NumPy, SciPy, and Matplotlib:

```
python3 report/audit_tagged_correlation_run.py \
  > /tmp/correlation_run_audit.json
cp /tmp/correlation_run_audit.json report/correlation_run_audit.json
MPLCONFIGDIR=/tmp/moe-report-matplotlib \
  python3 report/generate_correlation_tables.py
cd report
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The renderer writes coverage, correctness, class, lexical-event, expert, and repeated-attempt tables, a compact plot atlas, and standalone PDF/PNG figures. The atlas replaces the large numeric appendix in the compiled report. The renderer reuses saved feature correlations and additionally computes descriptive sampled-label counts, adjacent-sentence transitions, class-share associations, and binned outcomes from fingerprinted inputs. These new summaries and bin counts/ranges are saved in `report/tutor_plot_statistics.json`; router-value intervals use 500 source-problem bootstrap resamples with seed 42. No model is fitted. Narrative text and the tutor-coverage checklist are maintained alongside the generated artifacts. Original experiment outputs are unchanged.

Scope of reproducibility. Mutable local Docker tags and checkpoint names do not fix all underlying binaries. Exact reproduction also requires resolved model revisions, image digests, package versions, and the matching code revision. Saved contracts and source fingerprints identify this report's numerical population.

Three-model report update. Run `python3 report/generate_multimodel_correlations.py` from the repository root, then rebuild with the `latexmk` command above. This reads the Qwen run above and the corresponding `reasoning-vllm-v1` runs under `results/correlation_pipeline/gemma-nvfp4-nf4` and `results/`

correlation_pipeline/gpt-oss-20b. The generator verifies coverage and recomputes reported aggregate coefficients from saved CSVs, then writes comparison tables, figures, and both additional model atlases. Source hashes and check results are recorded separately in report/multimodel_correlation_sources.json and report/multimodel_correlation_audit.json.

B Coverage of the current tutor feedback

This section refers to the current Call with Lorenzo Molfetta (2).vtt, automatically transcribed in Italian. Requests below are interpreted as report changes; proposals for future model interventions are not results.

Table 31. Current-call requests and available evidence.

Call time	Request	Implementation and limitation
01:21–01:59; 04:37–05:49	Explicit targets, common metric names, uncertainty	Class plot titles name correctness; router definitions distinguish leading, selection-boundary, and selected/rest gaps. Position panels now show saved problem-bootstrap intervals (Figure 8).
06:52–07:56	Replace the trace-coverage plot with reasoning-length statistics	Table 17 reports observed mean, median, and 90th percentile tokens for each benchmark.
16:18–16:47; 19:08–19:39	Compact correlation plots; remove obsolete lexical targets	The appendix plots all 20 scopes as triangles plus correctness columns. Regex-event panels are removed; original numeric artifacts remain available.
24:41–29:57; 36:55–37:32	Class proportions against accuracy and token count	Figures 5 and 6 use sampled sentence shares, with benchmark sample sizes and actual outcome levels. Full-sequence proportions are unavailable.
26:38–27:12; 29:57–30:20	Class-state transition matrices	Figure 4 counts only originally adjacent labelled sentences, with row percentages and counts. Sampling gaps are not bridged.
30:40–34:25	Cumulative class counts as reasoning progresses	Figure 3 starts at zero and ends at observed sampled class totals. Token progress is normalized within each attempt.
35:35–36:26; 39:34–40:08	Selected-pool expert identity versus outcomes	Figure 11 shows all 256 IDs against correctness and token count. IDs still pool six layer-local modules.
42:06–45:17	Actual router values associated with higher accuracy	Figure 12 gives value ranges, accuracy, counts, and source-problem bootstrap intervals; no causal threshold is inferred.
20:11–21:07; 37:29–39:20	Rank-prefix selected/rest curves and subset entropy	Only the all-selected gap is available. Top-2/4/6 probability summaries and selected/unselected entropies require new replay because raw probabilities were not saved.

The final agreement keeps class dynamics descriptive and defers routing conditioned on class trajectories (41:45–42:05; 45:23–48:10). Any intervention comparison must evaluate the baseline on the same problem subset (40:55–41:29); no intervention is claimed here.

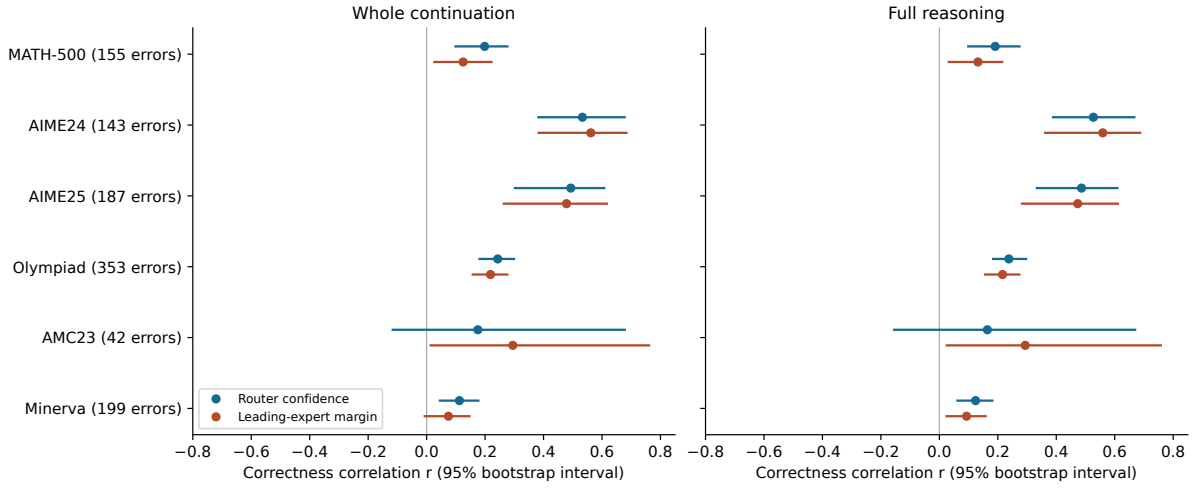


Fig. 1. Qwen3.5 router confidence and leading-expert margin versus correctness, with saved pointwise 95% problem-bootstrap intervals across all six benchmarks. Both views retain all attempts, including limit hits; full reasoning excludes non-reasoning continuation tokens. Error counts refer to attempts, not independent source problems.

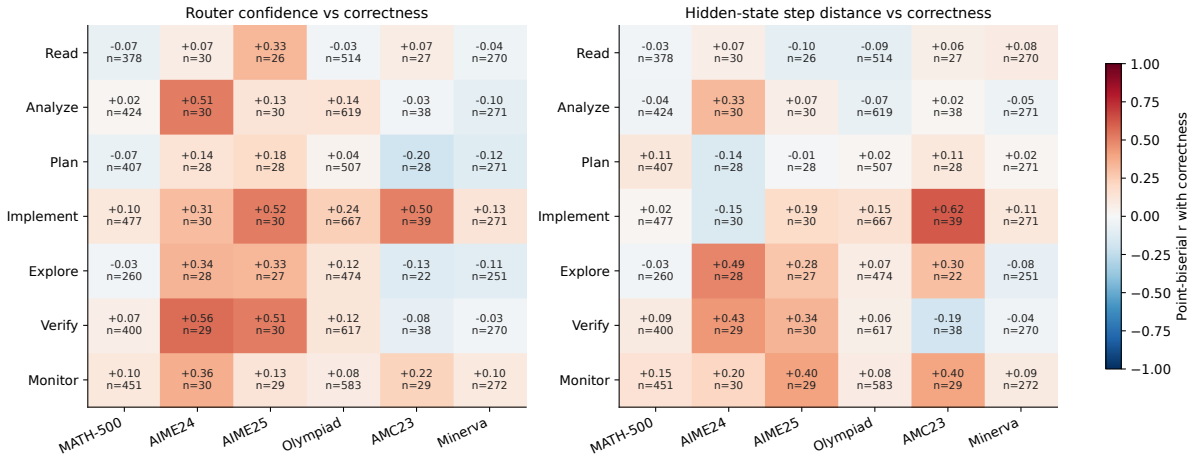


Fig. 2. Class-conditioned confidence and hidden-state step distance versus whole-attempt correctness. Cells show point-biserial r and finite trace count. Classes use sampled Qwen3.8 labels on Qwen3.5 generations, with different token-bearing cohorts. No significance markings are implied.

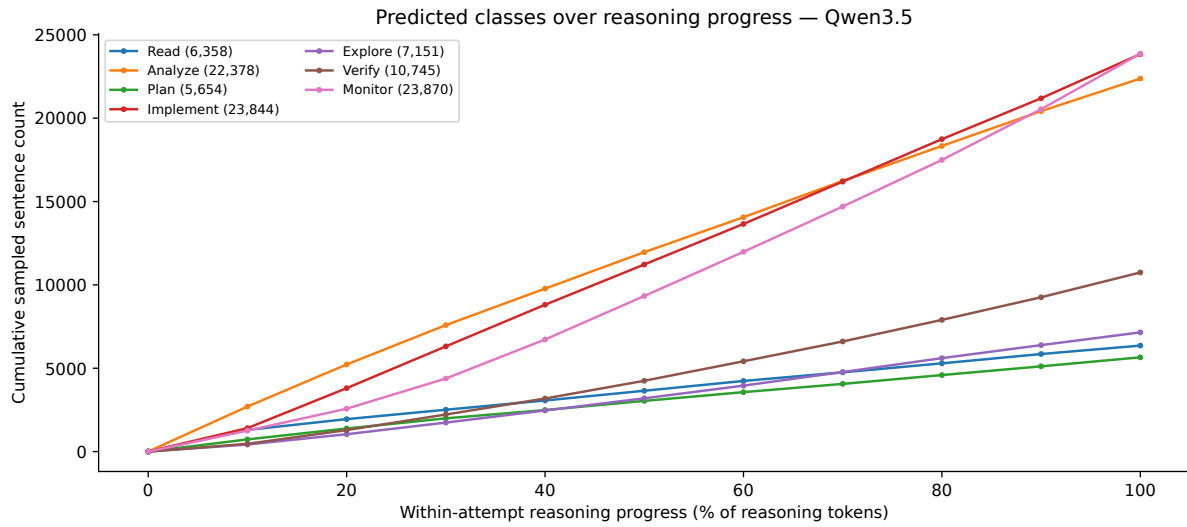


Fig. 3. Cumulative counts of sampled sentences by predicted class, pooled across benchmarks. Counts start at zero; a sentence is counted when its last reasoning token is reached. Progress is normalized by each attempt's full reasoning-token length, unlike the absolute windows in Figure 8. Legend totals equal the rightmost endpoints. These are raw sampled counts, not cumulative percentages or estimates of full-corpus counts; longer traces and benchmark sampling quotas affect them.

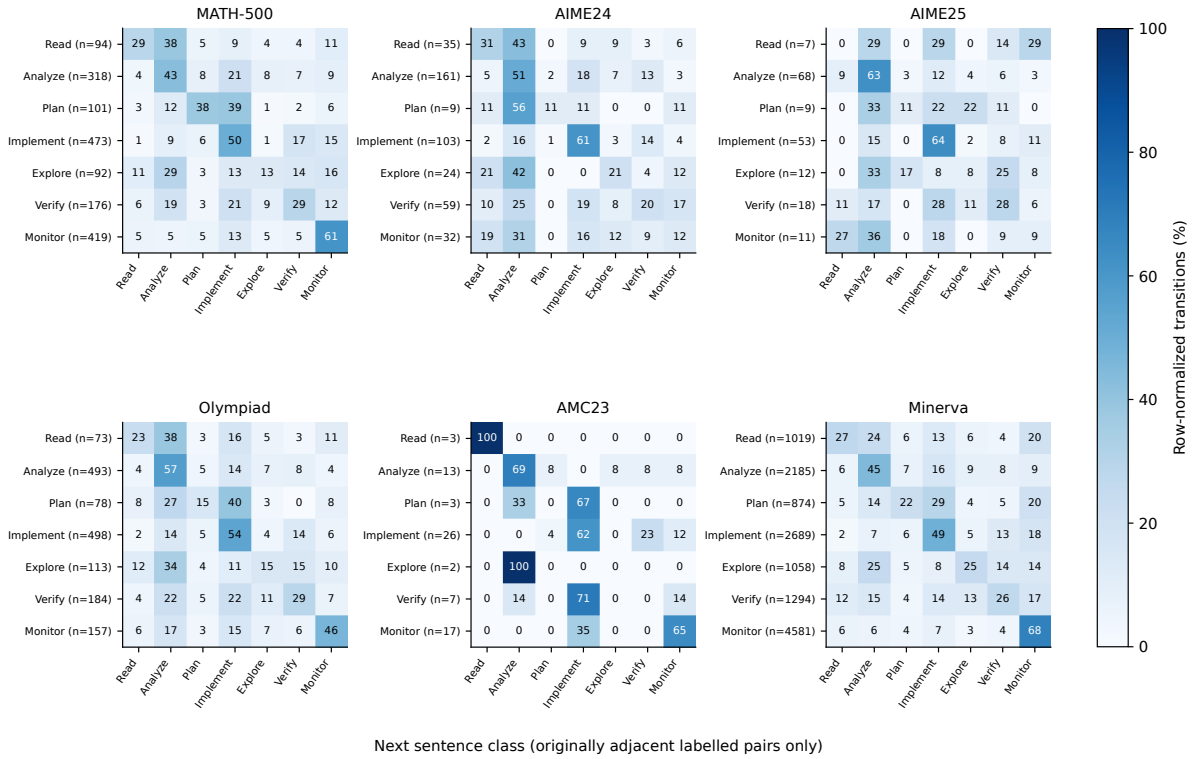


Fig. 4. Observed class-state transitions by benchmark. Rows are the current class, columns the next class; cells are row percentages and row labels give pair counts. Only originally consecutive sentence indices with both labels are counted. Gaps in sampling are never bridged. Self-transitions are retained; empty rows are undefined. This describes the observed adjacent-pair sample, not the complete transition process or an effect of changing state.

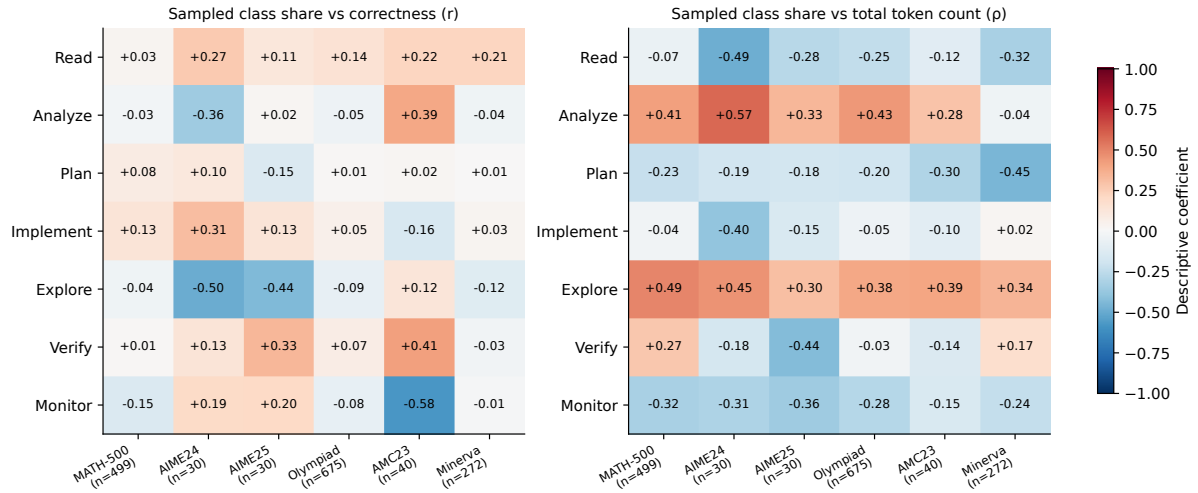


Fig. 5. Sampled class proportions versus whole-attempt correctness (point-biserial r) and total continuation-token count (Spearman ρ). All labelled attempts enter each class, including observed zero proportions. These differ from correlations of router features within class-selected tokens. No significance claim is made; class shares sum to one and cannot be interpreted as independent effects.

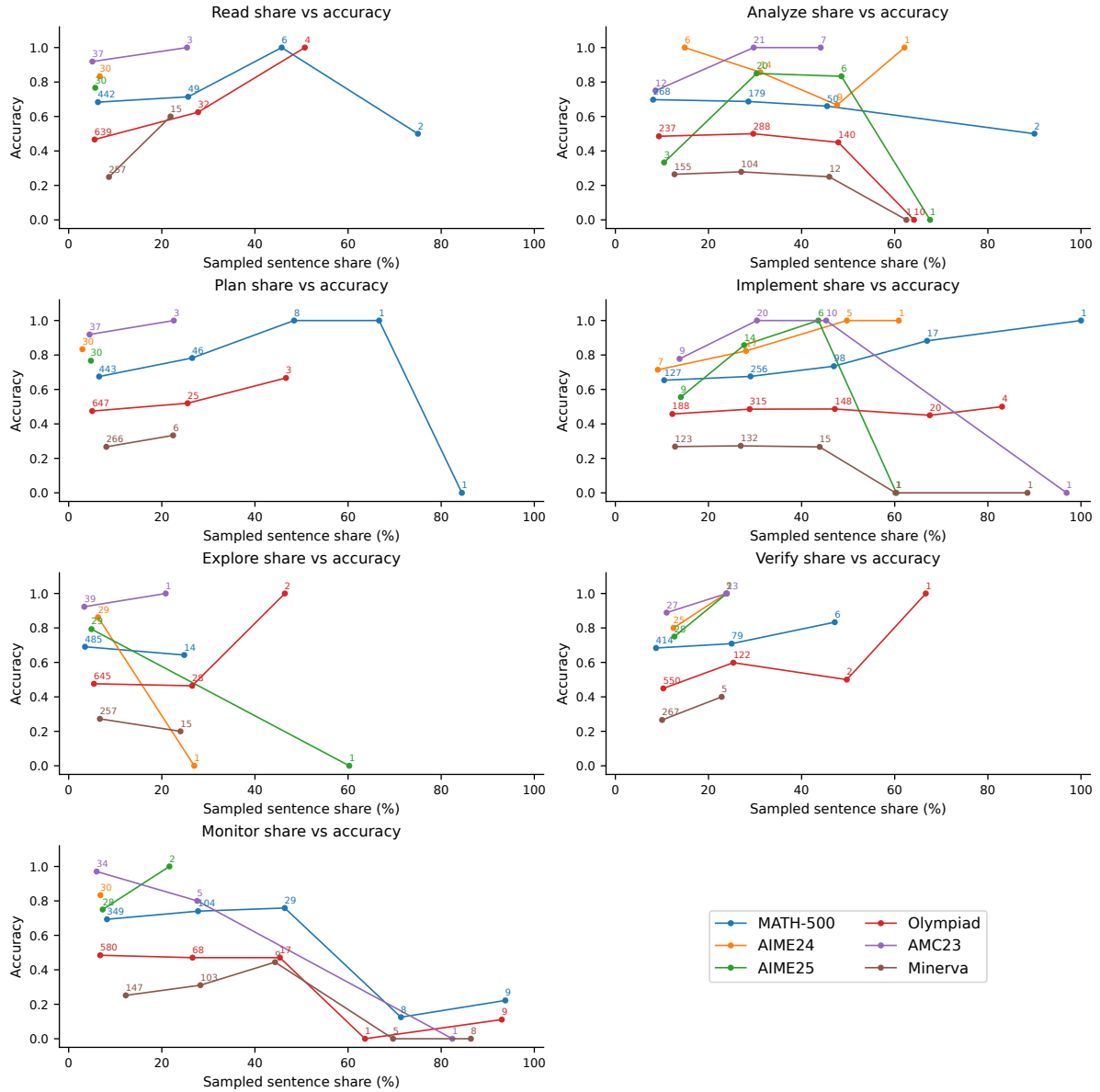


Fig. 6. Actual outcome levels versus sampled class proportions. Fixed share bins are $[0, 20)$, $[20, 40)$, $[40, 60)$, $[60, 80)$, and $[80, 100]$ percent. Each point is at the mean observed share and shows accuracy; small labels give attempt counts. Empty bins are omitted. Colours distinguish benchmarks; lines only connect occupied bins. Sparse bins, sentence sampling, length, and difficulty can explain apparent trends; these descriptive plots do not identify optimal class proportions or a validated accuracy-length Pareto frontier.

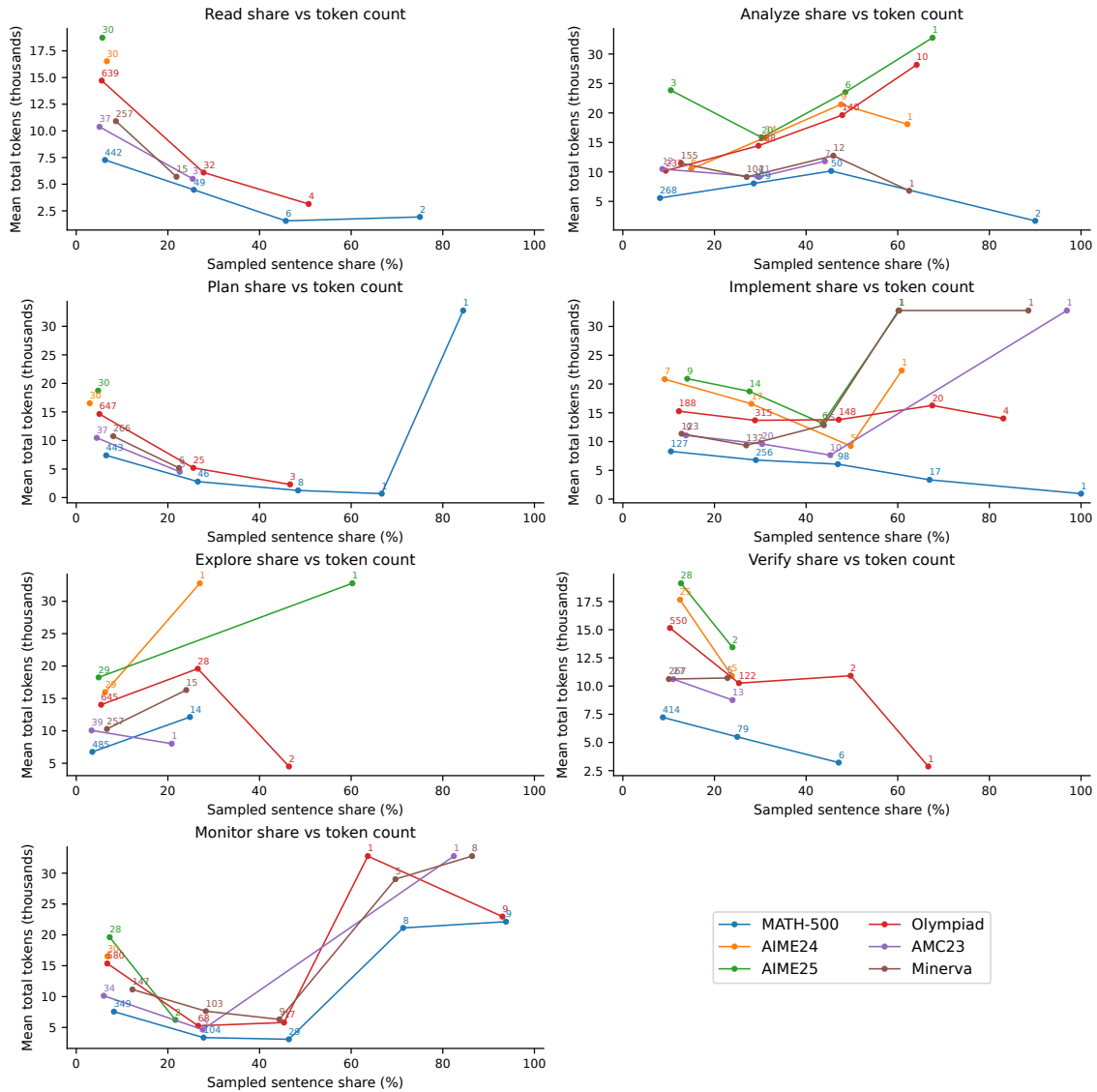


Fig. 7. Mean total continuation tokens versus sampled class share. Bin boundaries, colours, sample-count labels, and sampling limitations are as in Figure 6. Token count includes the final answer and token-limit completions; lines are descriptive, not intervention effects.

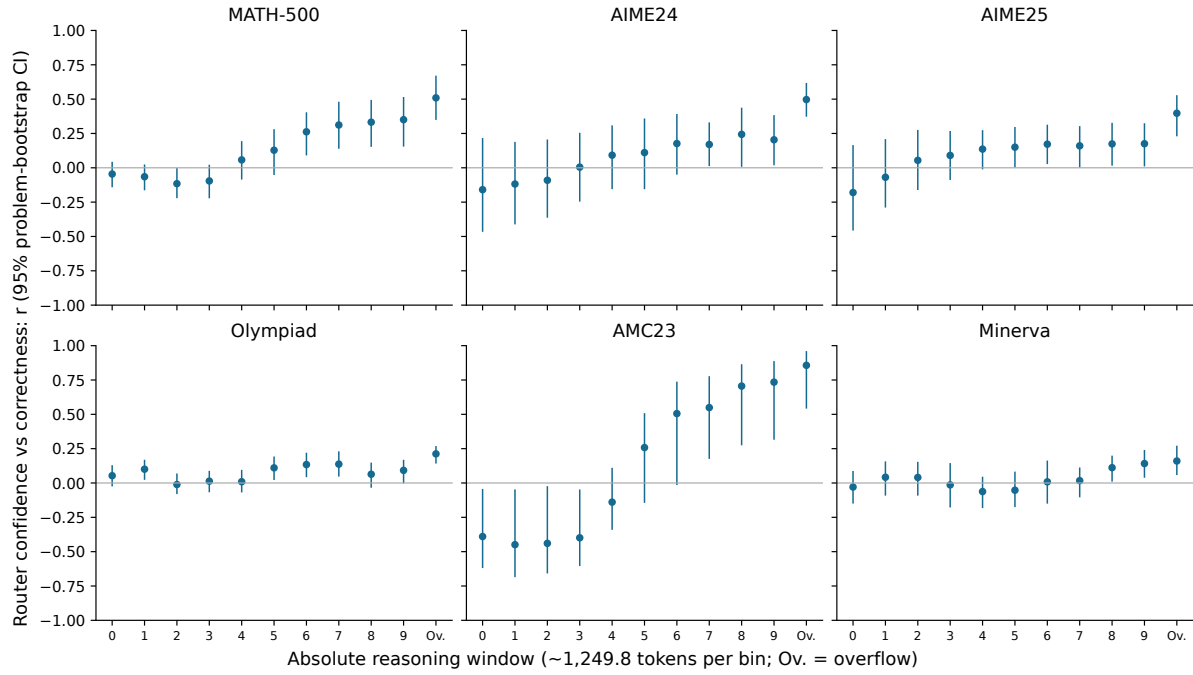


Fig. 8. Router confidence versus whole-attempt correctness in absolute reasoning windows, with saved pointwise 95% source-problem bootstrap intervals. Each benchmark has its own panel. Missing estimates remain blank. Later windows use changing token-bearing cohorts; these are not within-trace causal effects.

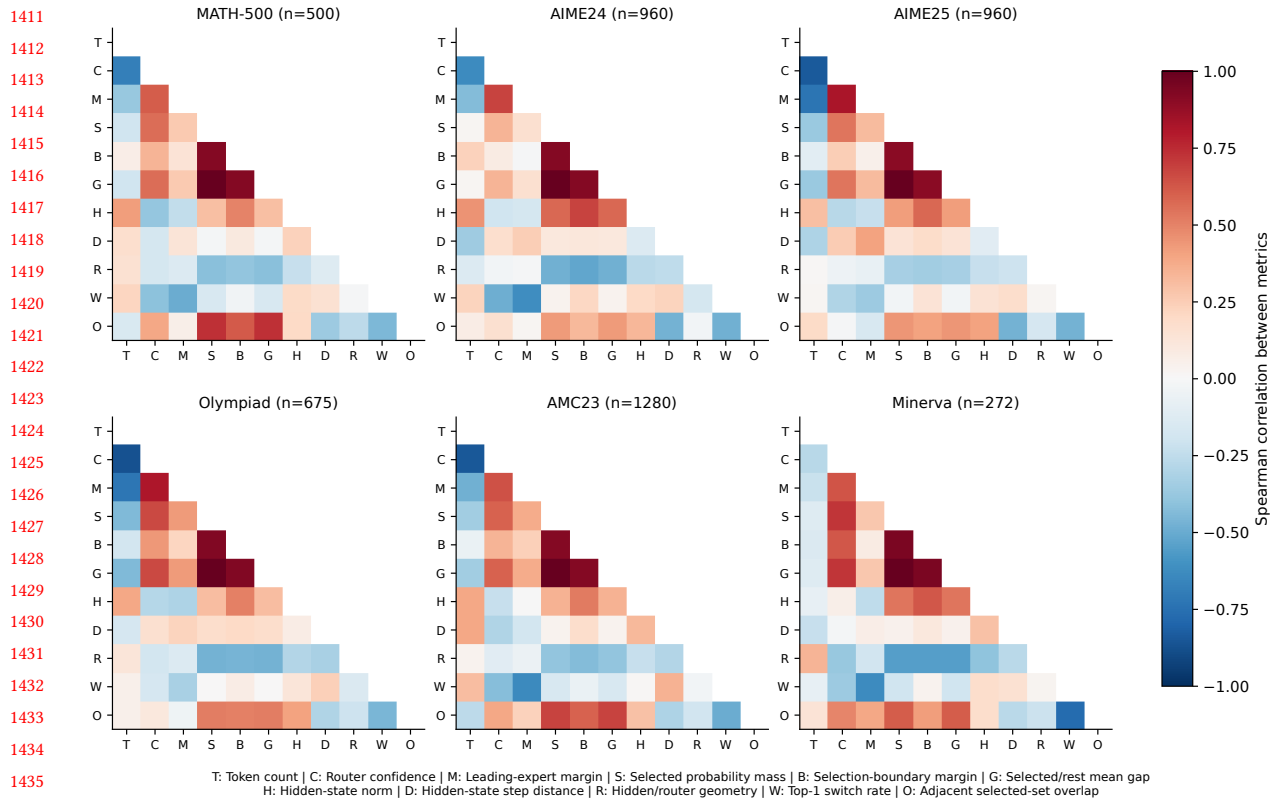


Fig. 9. Whole-continuation metric–metric Spearman correlations across all six benchmarks. Lower triangles display each association once; diagonals and upper triangles are omitted. Repeated attempts and limit hits remain included. These are descriptive metric relationships, not correctness effects.

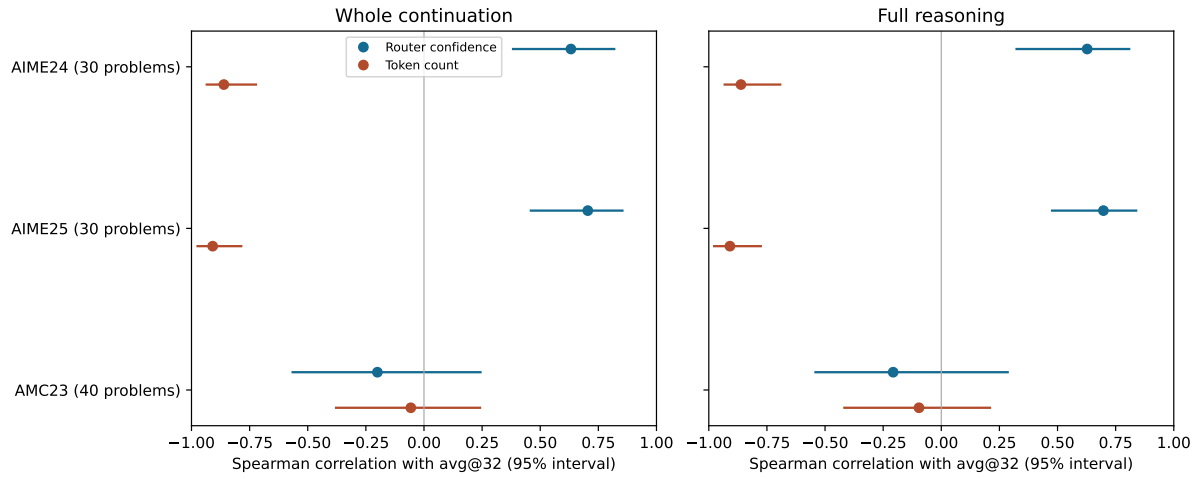


Fig. 10. Mean confidence and token count versus problem-level avg@32, with saved 95% whole-problem bootstrap intervals. Whole-continuation and full-reasoning views use the same complete 30/30/40 problem groups. These estimates remain unadjusted for difficulty, length, and termination.

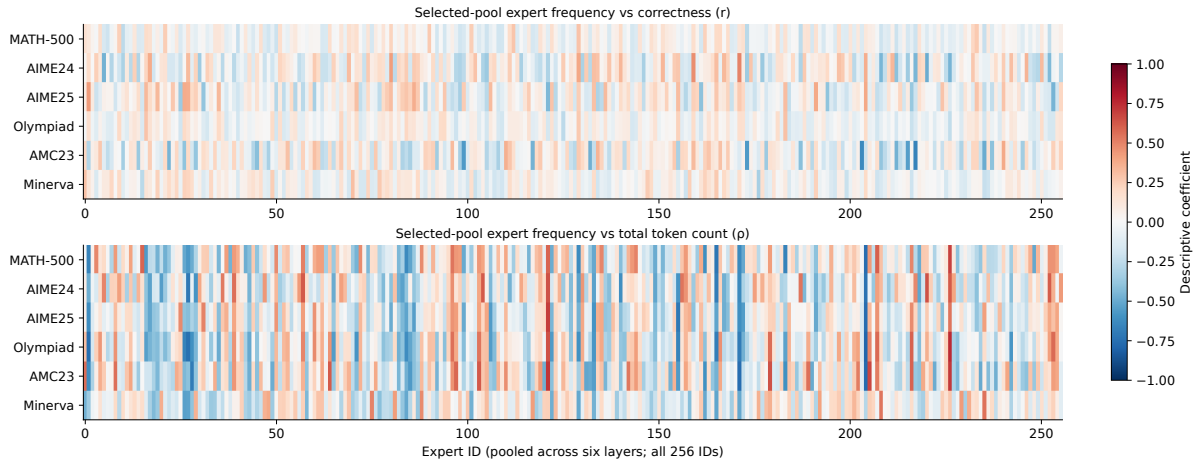


Fig. 11. Selected-pool membership frequency for every expert ID versus correctness (r , top) and total continuation tokens (ρ , bottom). All 256 IDs are shown in their original order for each benchmark. Rates pool tokens and six layers; the same ID refers to distinct layer-local modules. These associations do not establish that activating an expert will improve accuracy or reduce length.

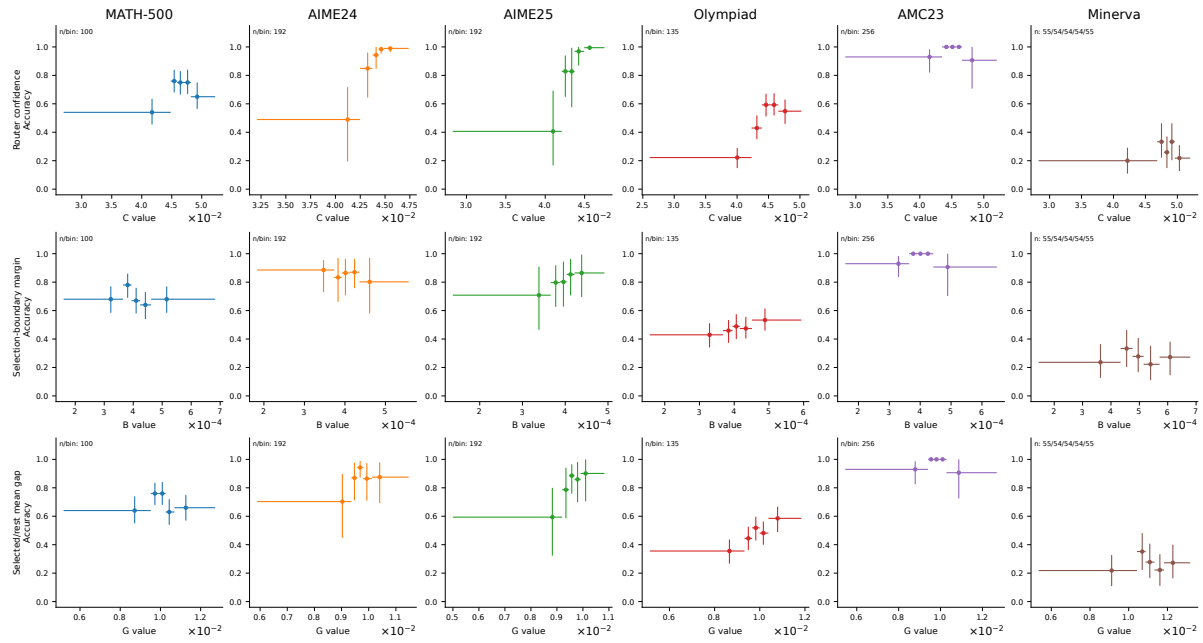


Fig. 12. Accuracy at observed router-metric values. Each benchmark uses up to five quantile bins; horizontal bars show actual value ranges, points show mean values and accuracy, and panel labels give attempt counts per bin (left to right, or a shared count when equal). Vertical bars are pointwise 95% percentile intervals from 500 whole-source-problem bootstrap resamples within each fixed bin (seed 42). Degenerate intervals can occur in all-correct/all-incorrect bins and do not imply certainty. All attempts and limit hits are retained. Bins can share source problems, so their estimates are not independent; neither bin selection nor length and termination confounding is addressed by these intervals.

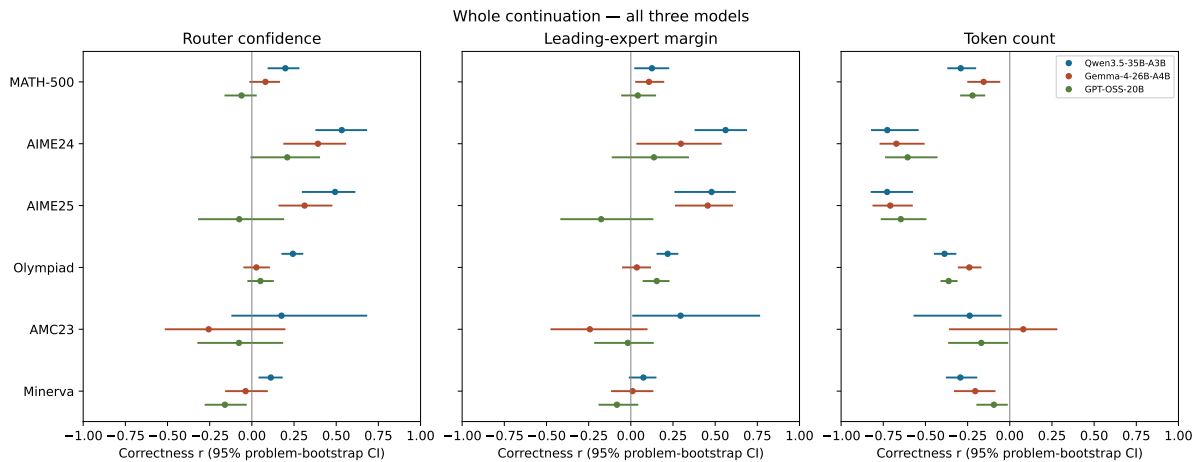


Fig. 13. Whole continuation: saved correctness correlations and pointwise source-problem bootstrap intervals. Overlap or separation of intervals is not a paired test of model differences.

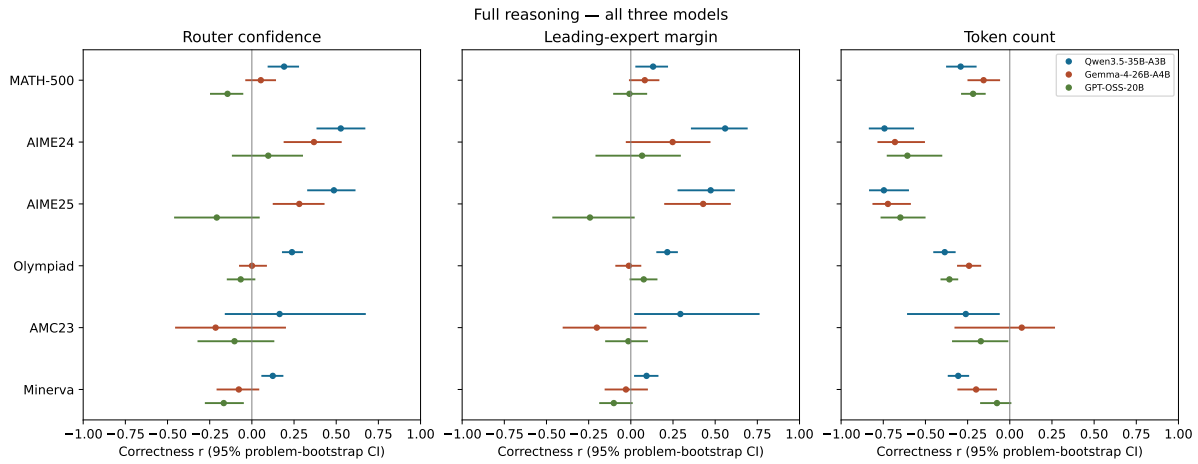


Fig. 14. Full reasoning: saved correctness correlations and pointwise source-problem bootstrap intervals. Overlap or separation of intervals is not a paired test of model differences.

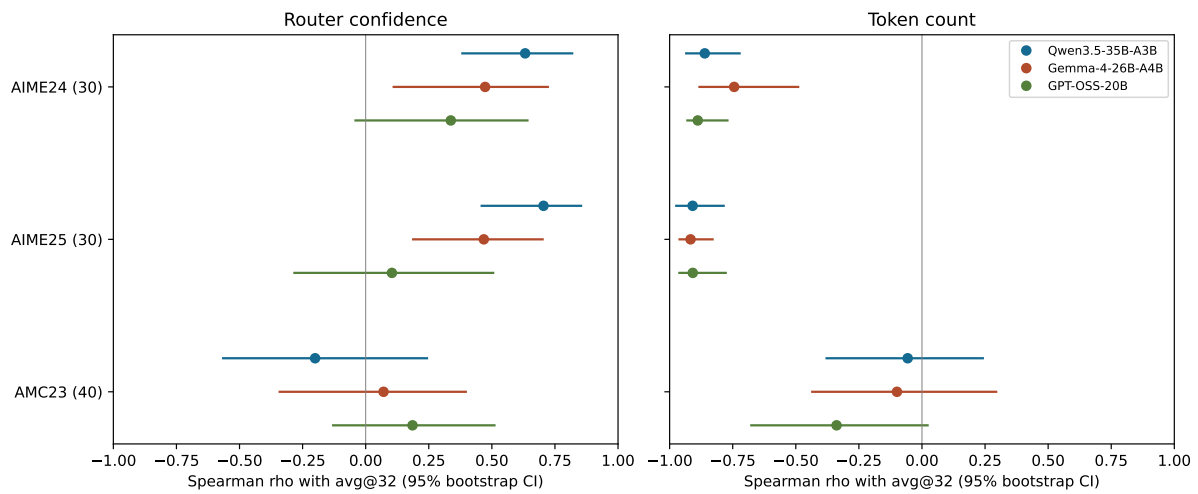


Fig. 15. All models: whole-continuation problem means versus avg@32, with saved pointwise problem-bootstrap intervals. All 32 attempts enter each problem mean.

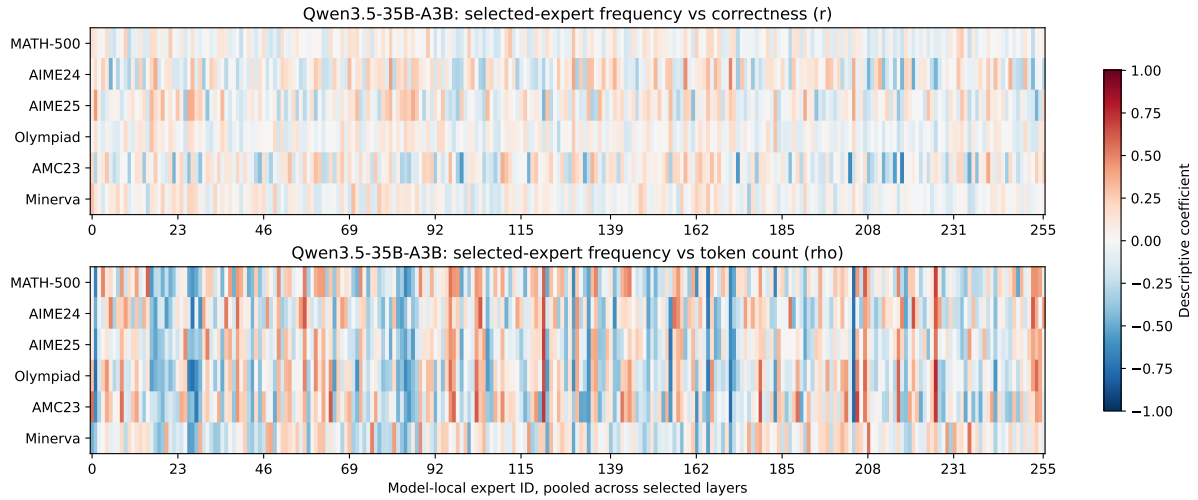


Fig. 16. Qwen3.5-35B-A3B: selected-pool expert frequency correlations with correctness and token count. IDs are model-local and pooled across selected layers; they do not identify matching experts across models. Missing coefficients are blank.

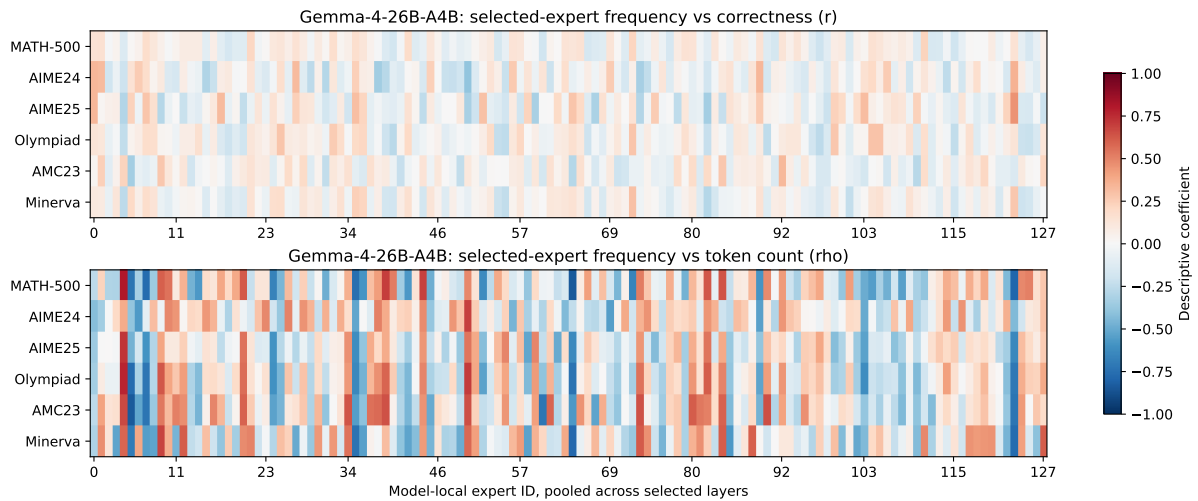


Fig. 17. Gemma-4-26B-A4B: selected-pool expert frequency correlations with correctness and token count. IDs are model-local and pooled across selected layers; they do not identify matching experts across models. Missing coefficients are blank.

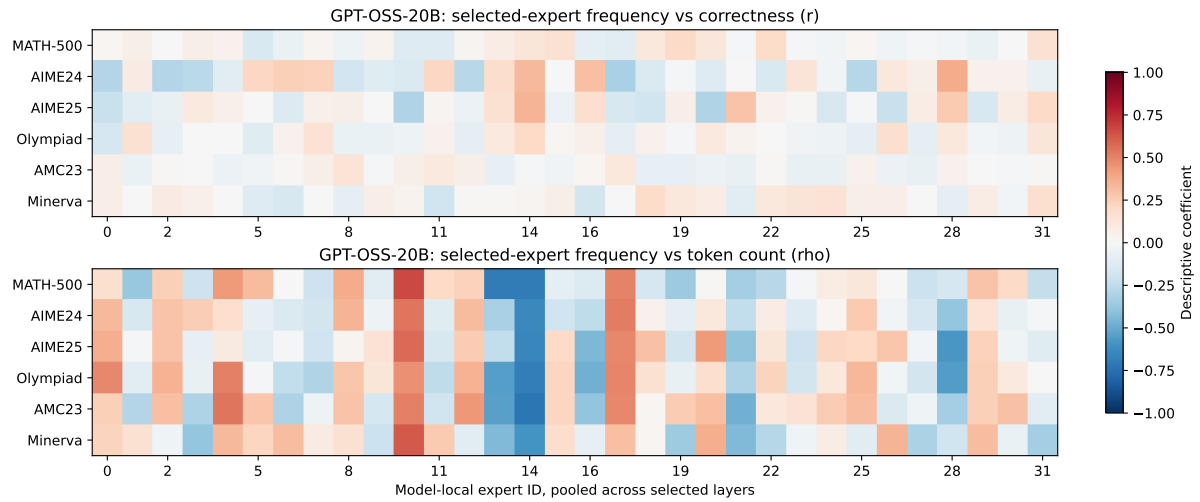


Fig. 18. GPT-OSS-20B: selected-pool expert frequency correlations with correctness and token count. IDs are model-local and pooled across selected layers; they do not identify matching experts across models. Missing coefficients are blank.

C Aggregate correlation plot atlas

Each page fixes a token scope. Lower triangles show metric–metric Spearman ρ ; the last column (Acc) shows point-biserial r with whole-attempt correctness. A dash means undefined or unavailable, not zero. Panels are descriptive; no significance threshold is implied. Sample sizes and saved intervals remain in the fingerprinted source JSON. Legacy lexical targets are omitted. Detailed numeric tables remain in `correlation_tables.tex` as a separate generated artifact.

Table 32. Common metric notation; $k = 8$ for this model.

Code	Metric
T	Token count
C	Router confidence
M	Leading-expert margin
S	Selected probability mass
B	Selection-boundary margin
G	Selected/rest mean gap
H	Hidden-state norm
D	Hidden-state step distance
R	Hidden/router geometry
W	Top-1 switch rate
O	Adjacent selected-set overlap
Ch	Character count
St	Step count

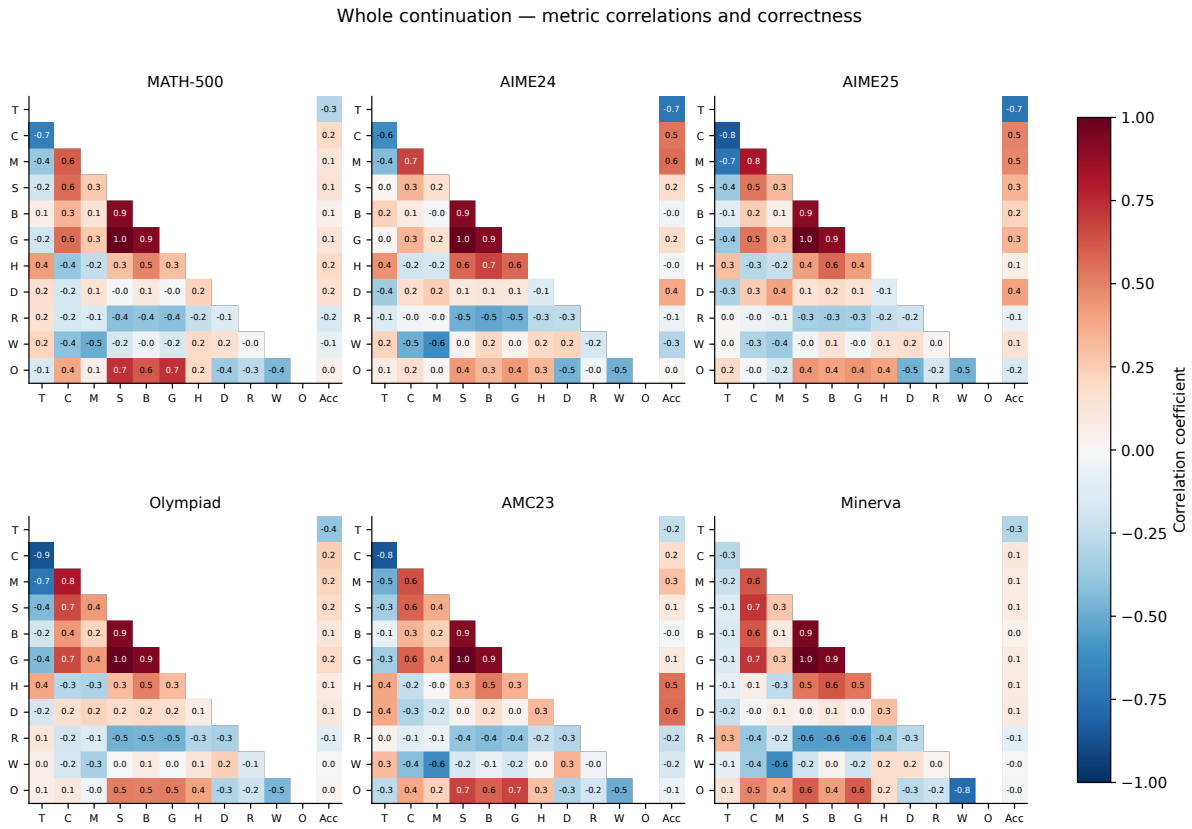


Fig. 19. Whole continuation: metric relationships and correctness. Codes are defined in Table 32.

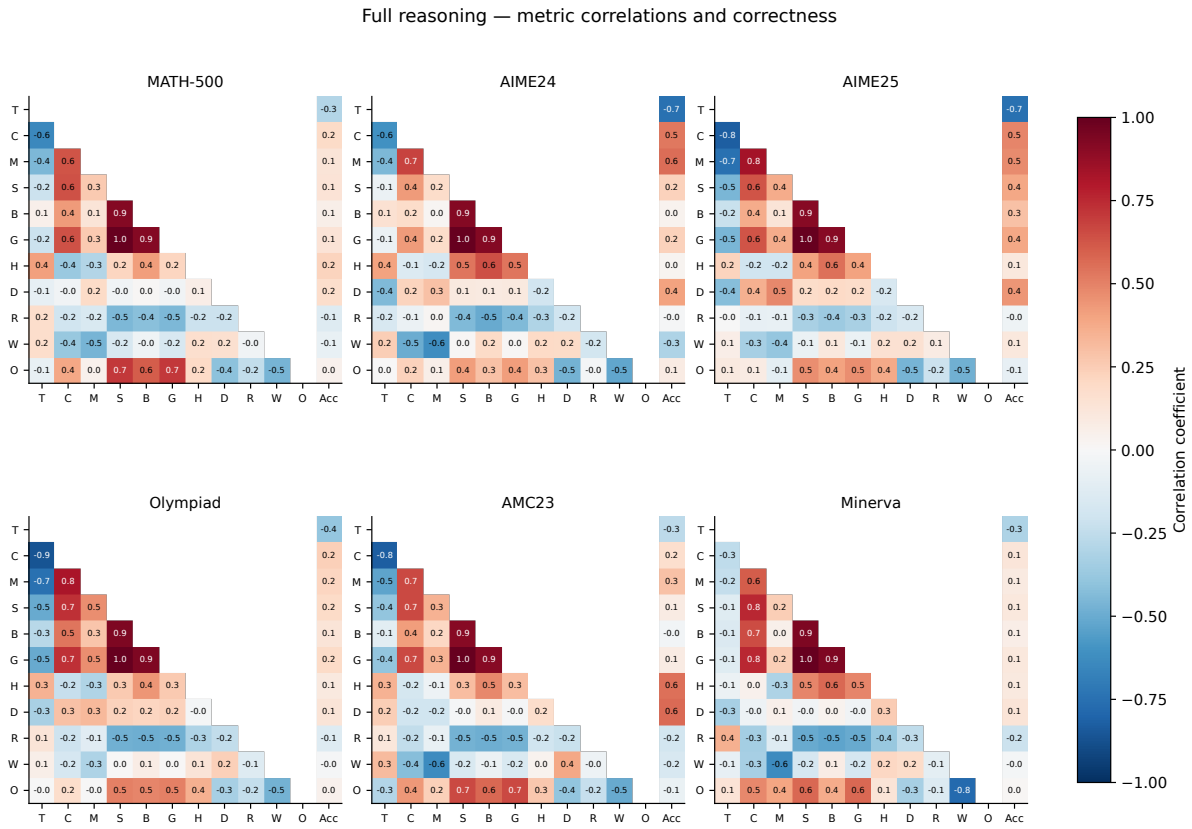


Fig. 20. Full reasoning: metric relationships and correctness. Codes are defined in Table 32.

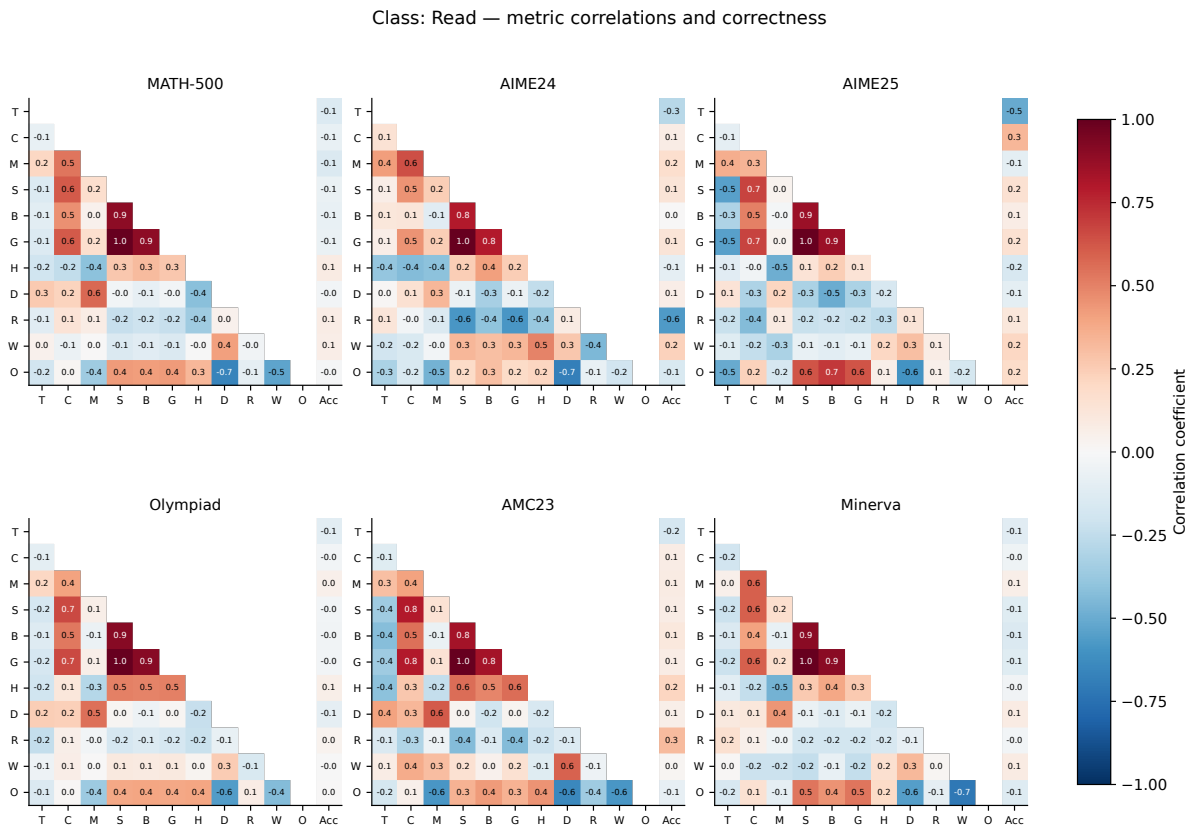


Fig. 21. Class: Read: metric relationships and correctness. Codes are defined in Table 32.

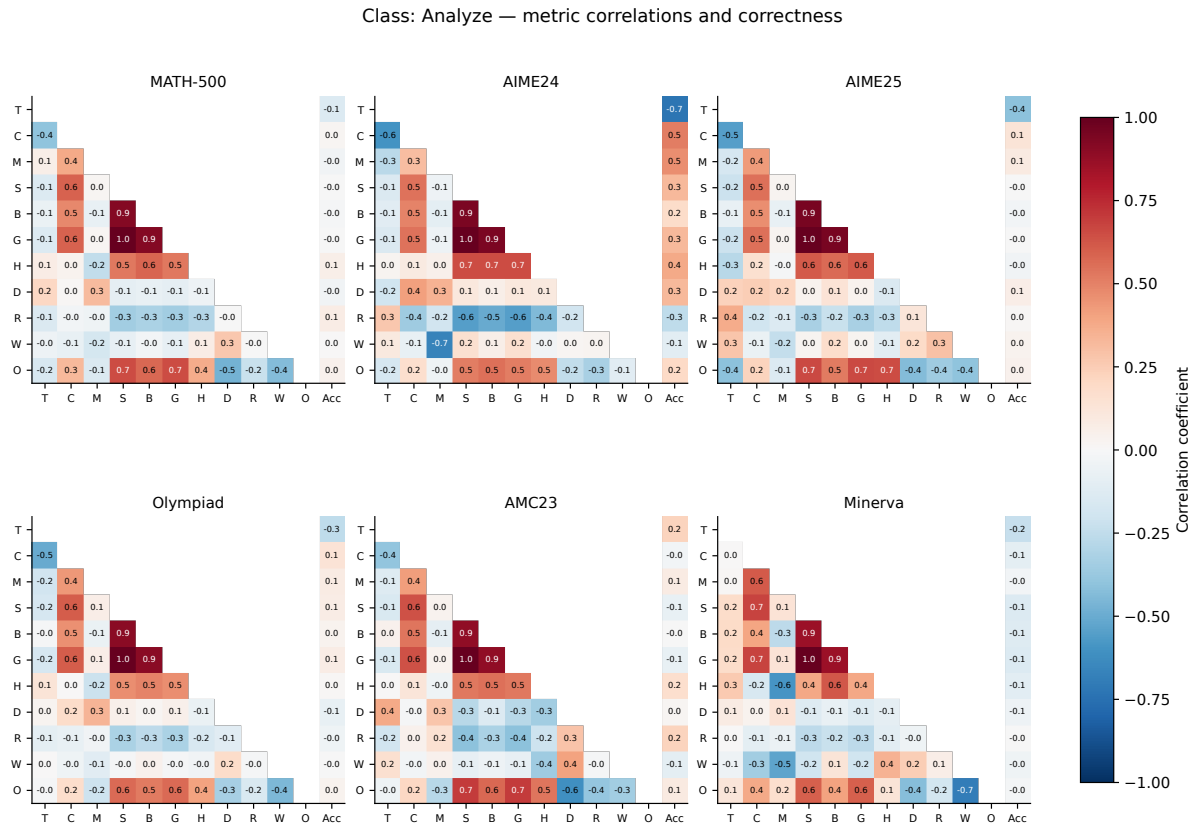


Fig. 22. Class: Analyze: metric relationships and correctness. Codes are defined in Table 32.

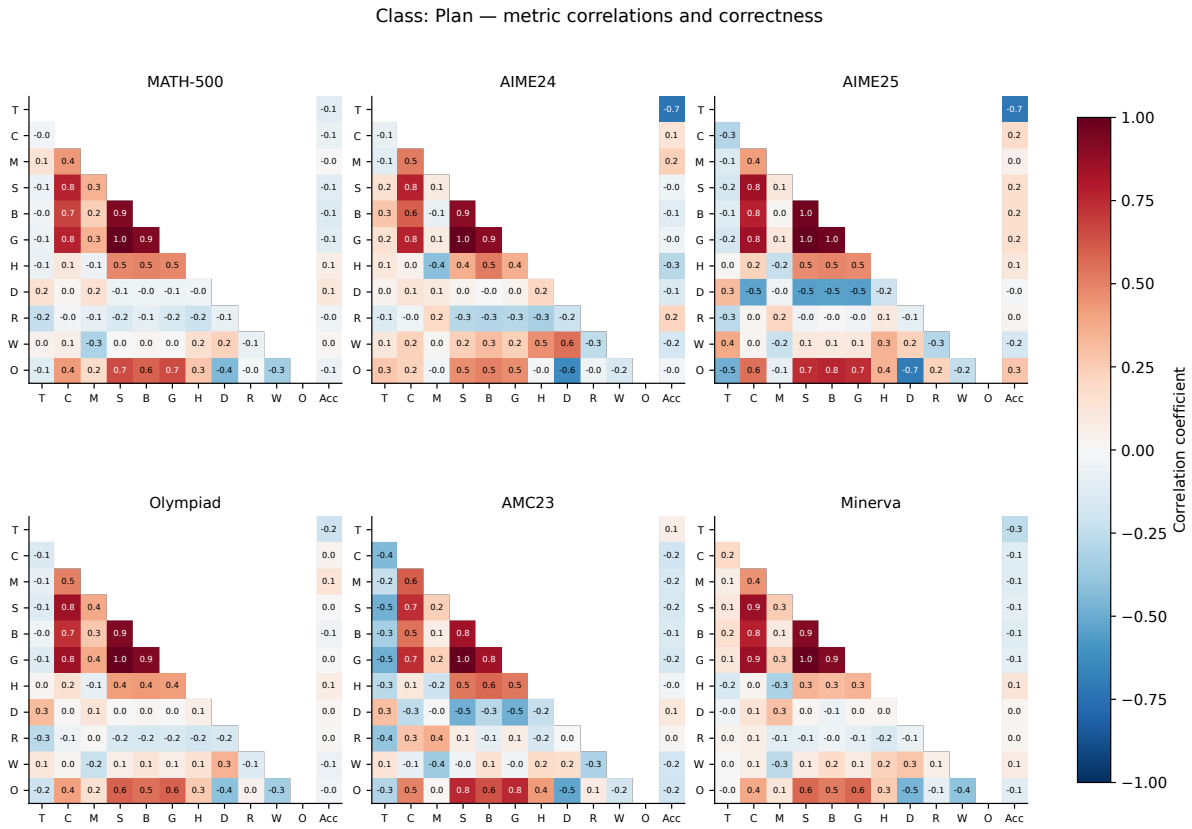


Fig. 23. Class: Plan: metric relationships and correctness. Codes are defined in Table 32.

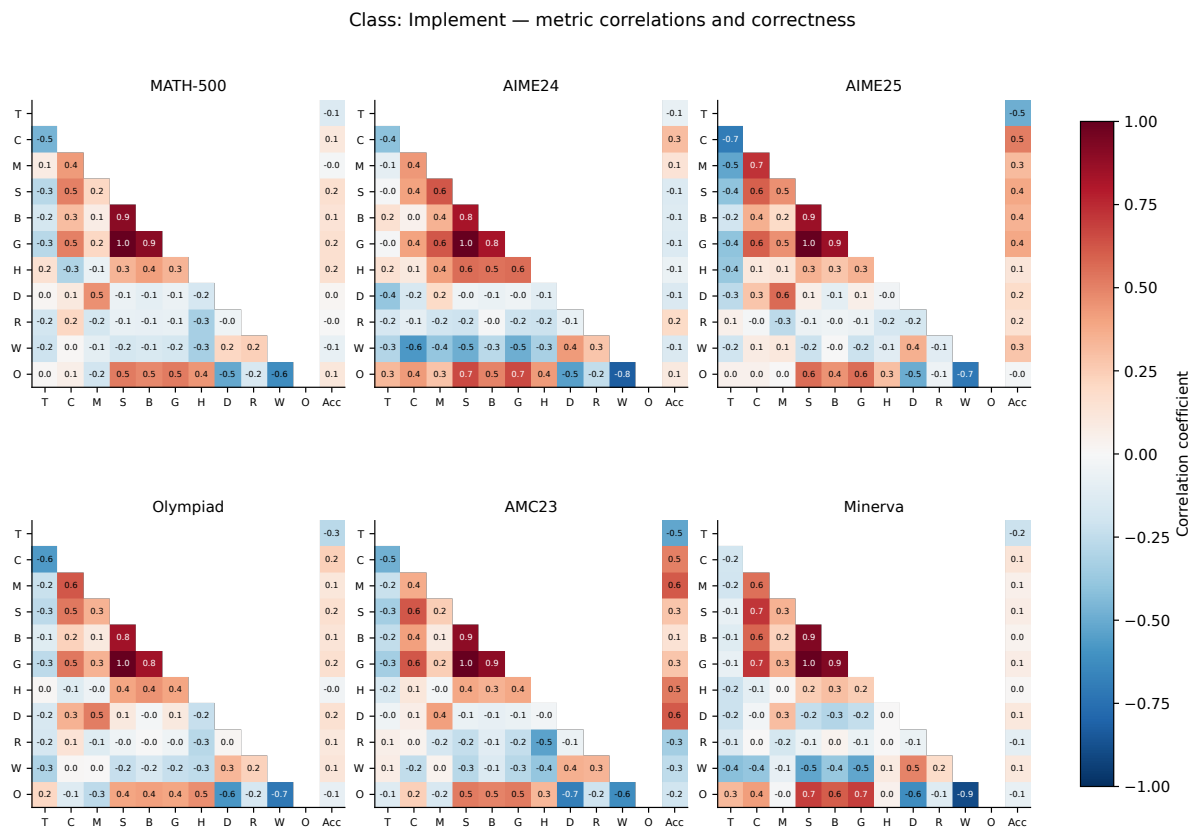


Fig. 24. Class: Implement: metric relationships and correctness. Codes are defined in Table 32.

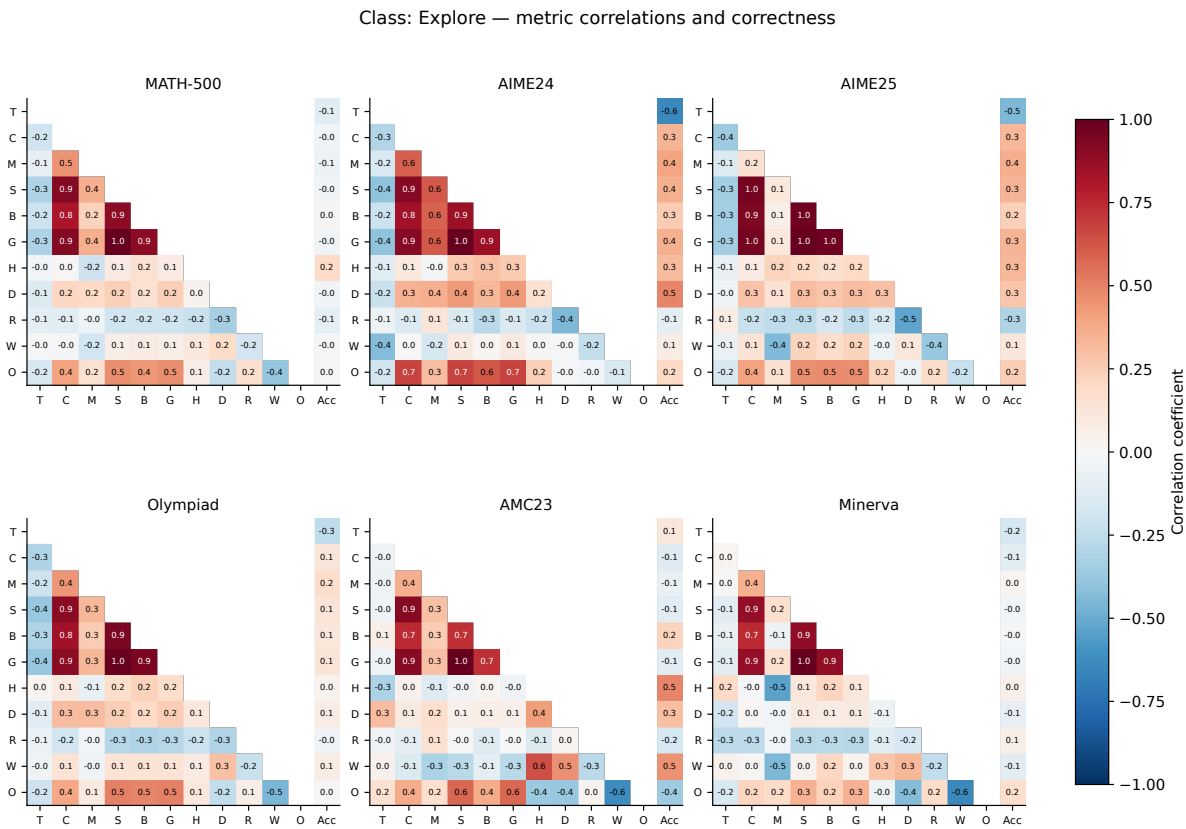


Fig. 25. Class: Explore: metric relationships and correctness. Codes are defined in Table 32.

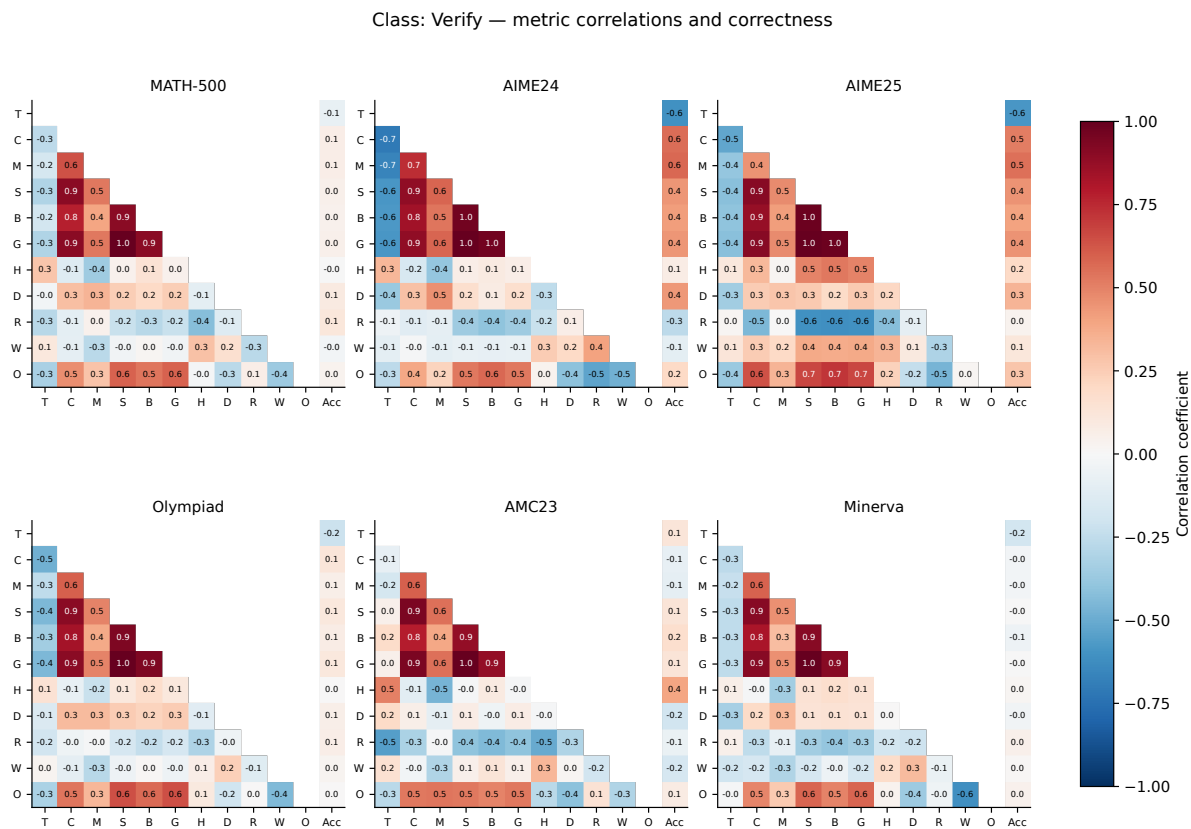


Fig. 26. Class: Verify: metric relationships and correctness. Codes are defined in Table 32.

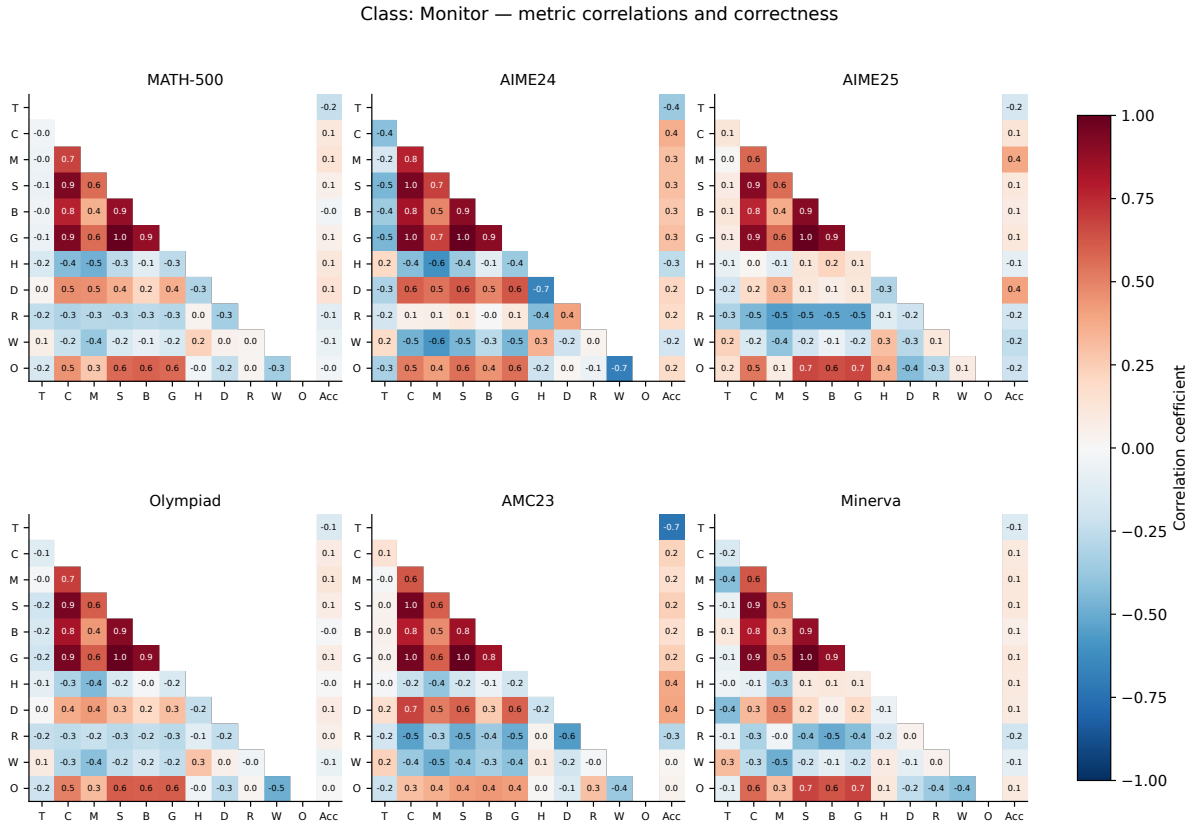


Fig. 27. Class: Monitor: metric relationships and correctness. Codes are defined in Table 32.

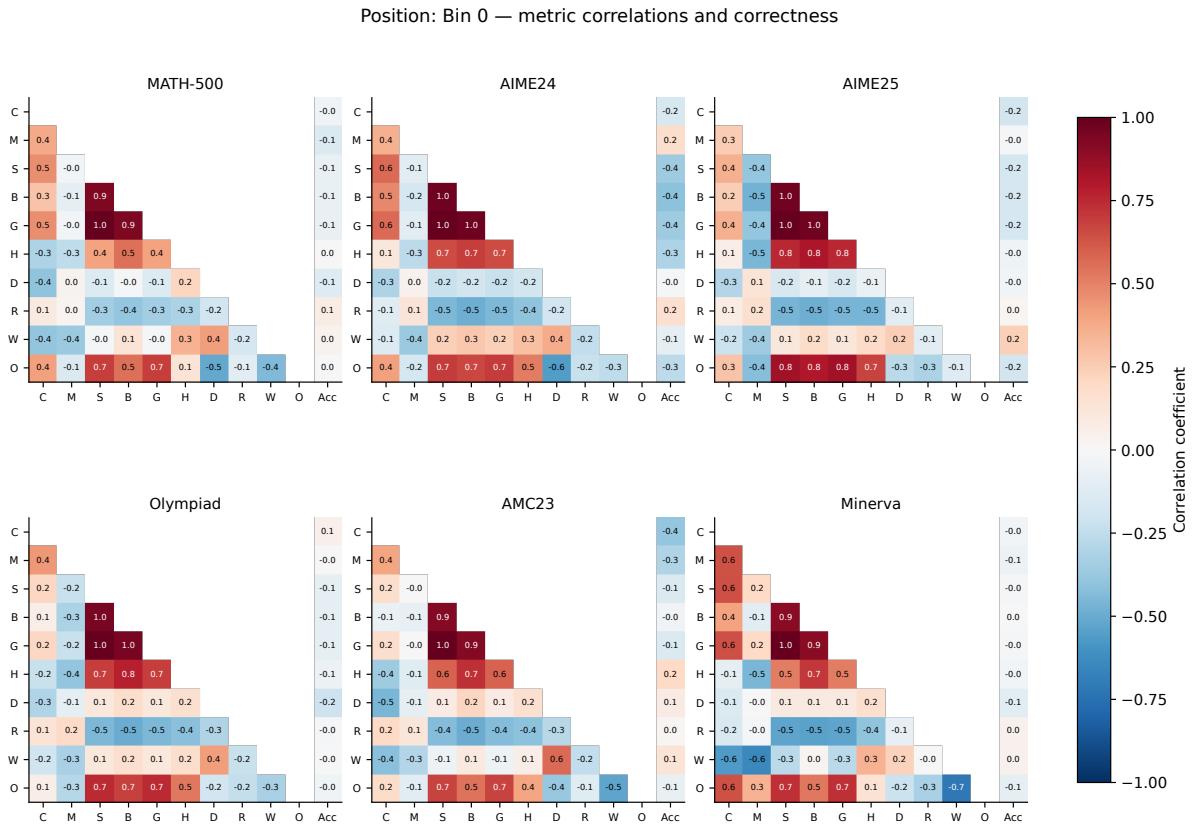


Fig. 28. Position: Bin 0: metric relationships and correctness. Codes are defined in Table 32.

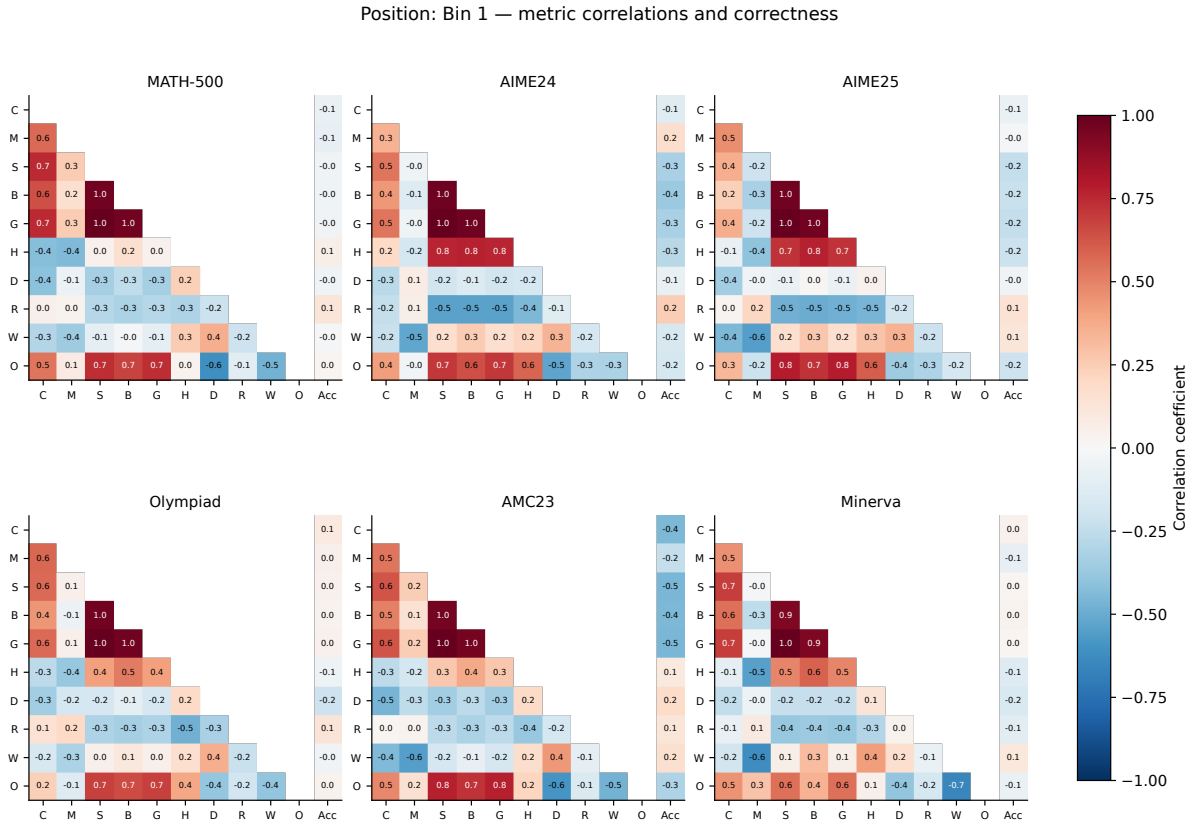


Fig. 29. Position: Bin 1: metric relationships and correctness. Codes are defined in Table 32.

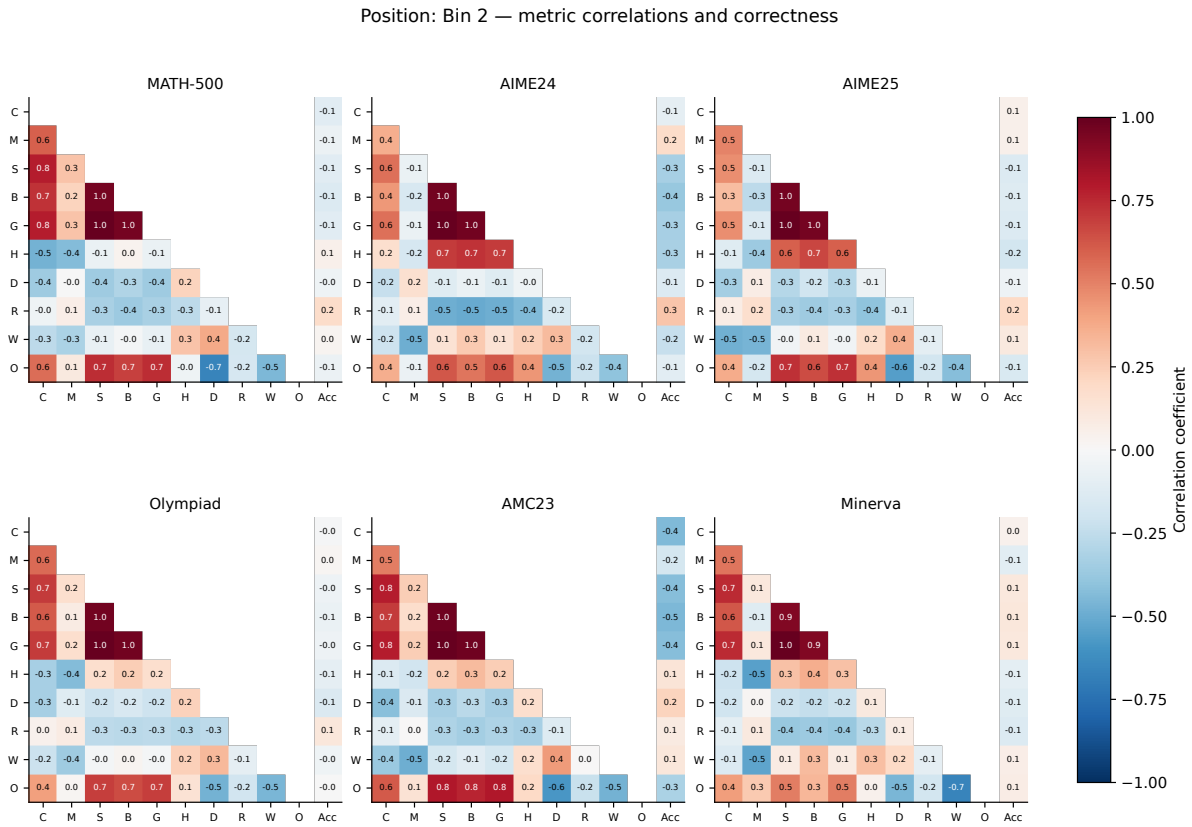


Fig. 30. Position: Bin 2: metric relationships and correctness. Codes are defined in Table 32.

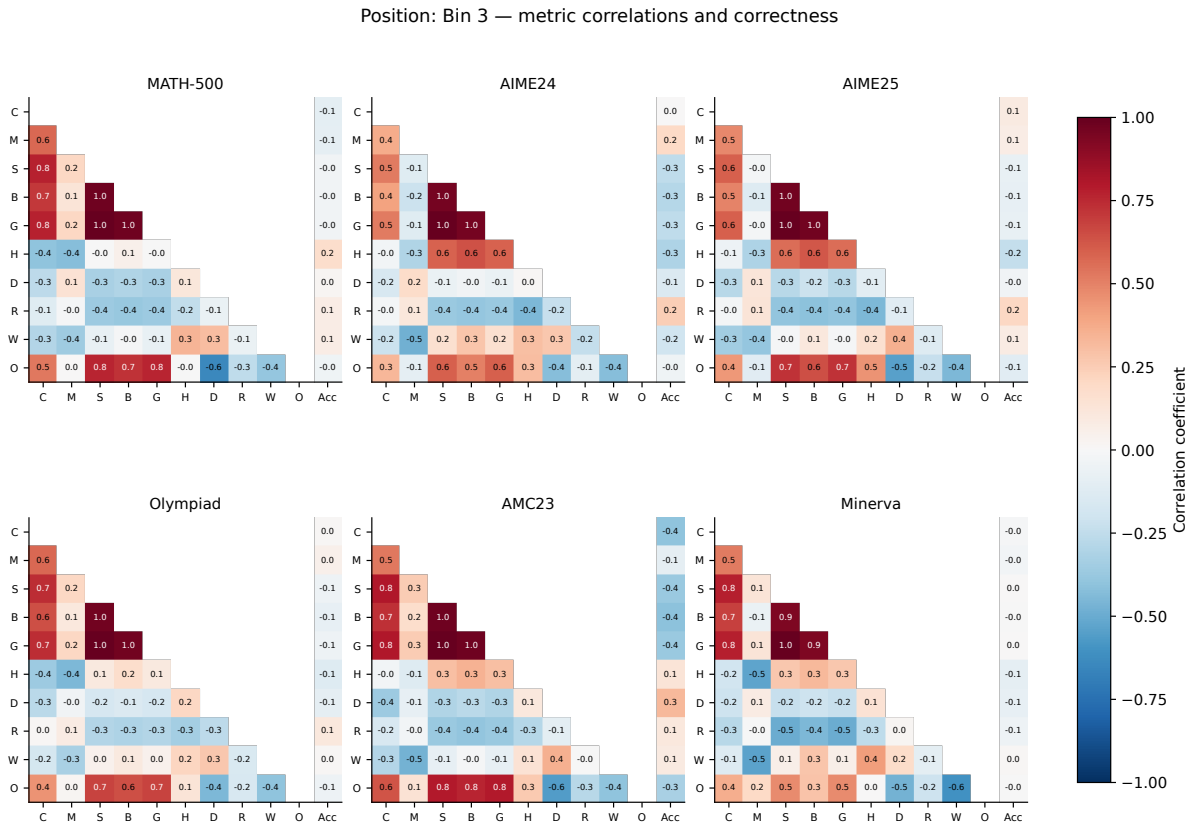


Fig. 31. Position: Bin 3: metric relationships and correctness. Codes are defined in Table 32.

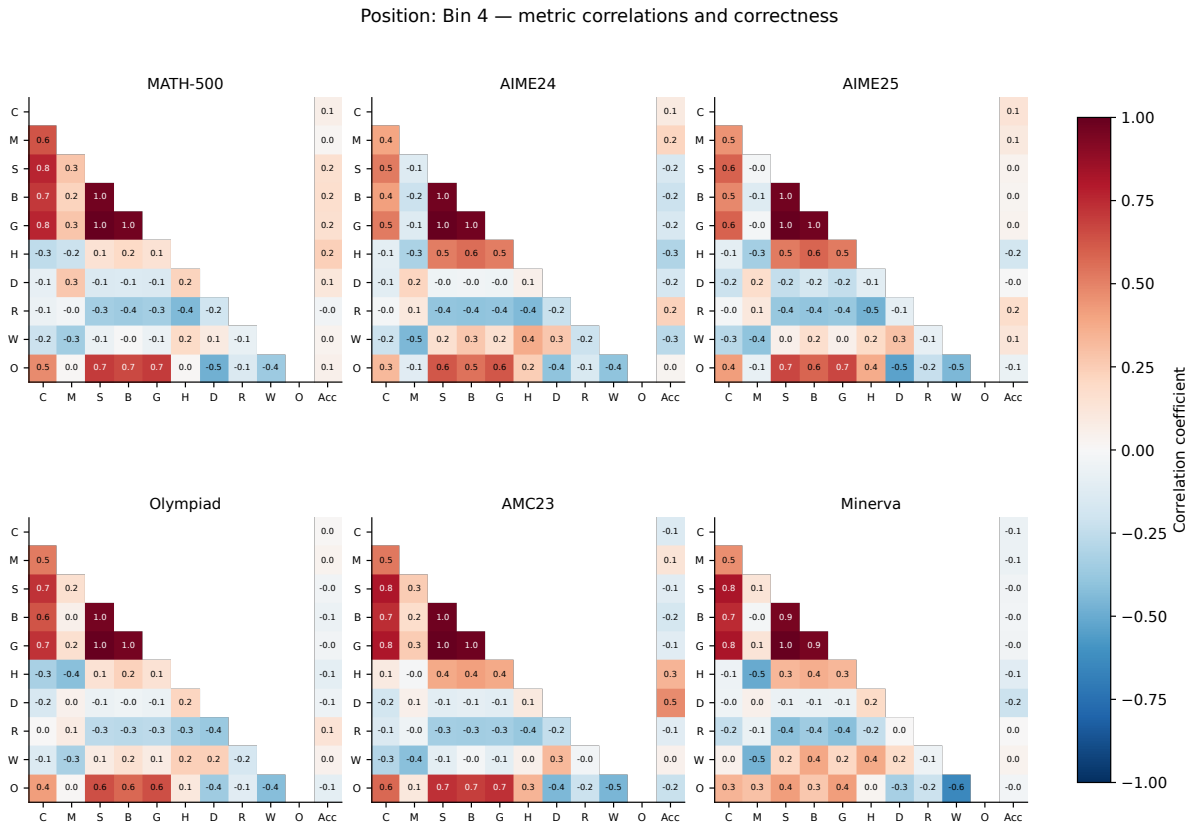


Fig. 32. Position: Bin 4: metric relationships and correctness. Codes are defined in Table 32.

Position: Bin 5 — metric correlations and correctness

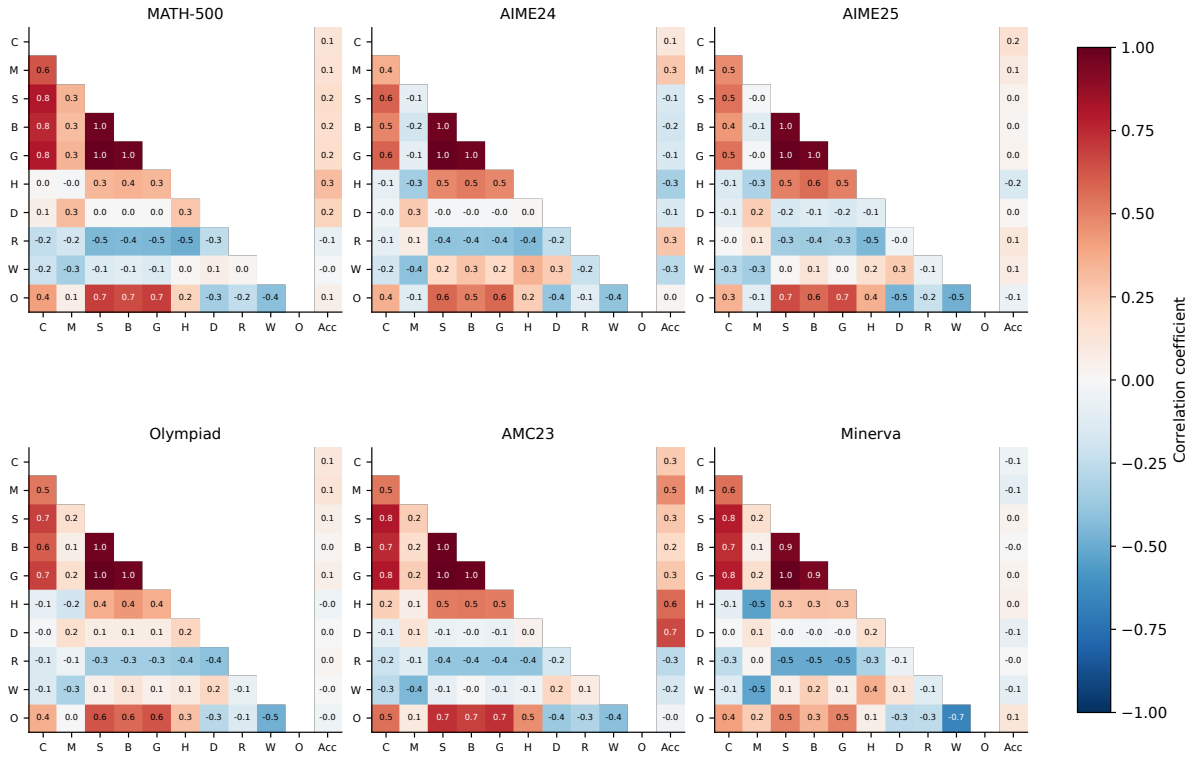


Fig. 33. Position: Bin 5: metric relationships and correctness. Codes are defined in Table 32.

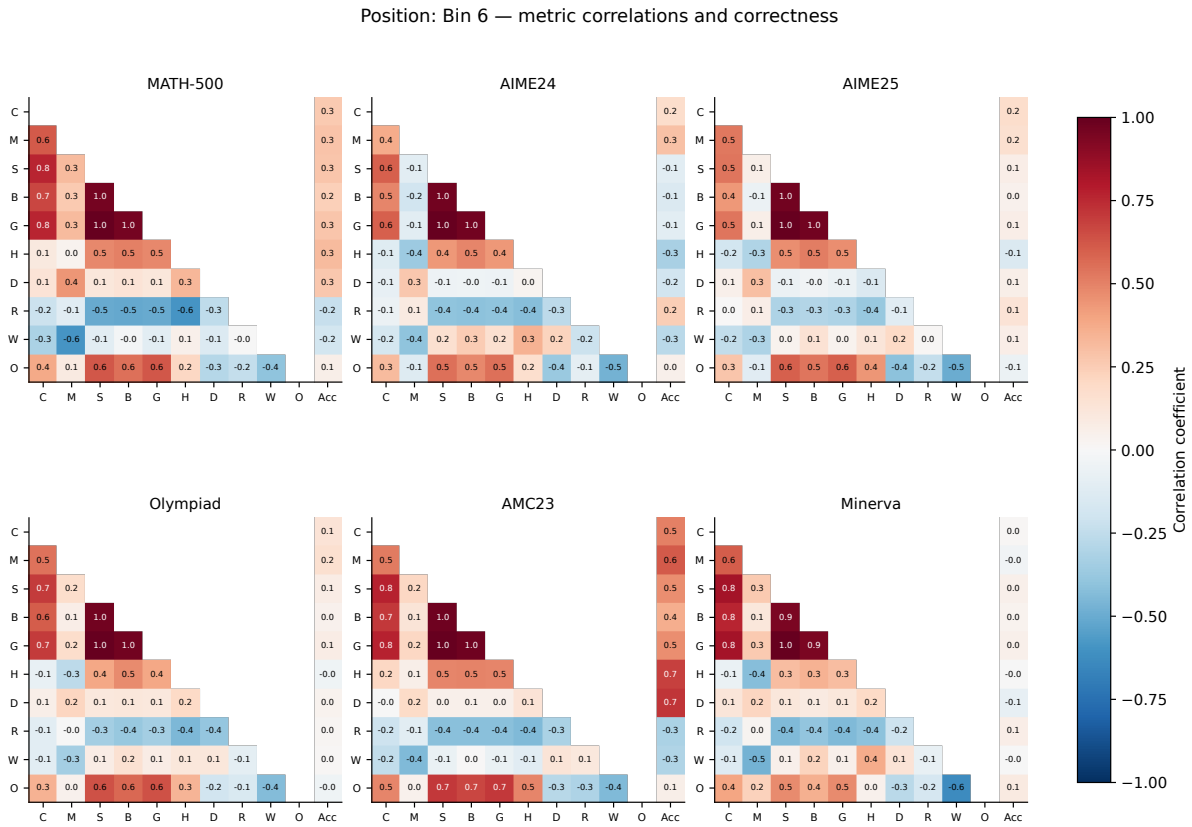


Fig. 34. Position: Bin 6: metric relationships and correctness. Codes are defined in Table 32.

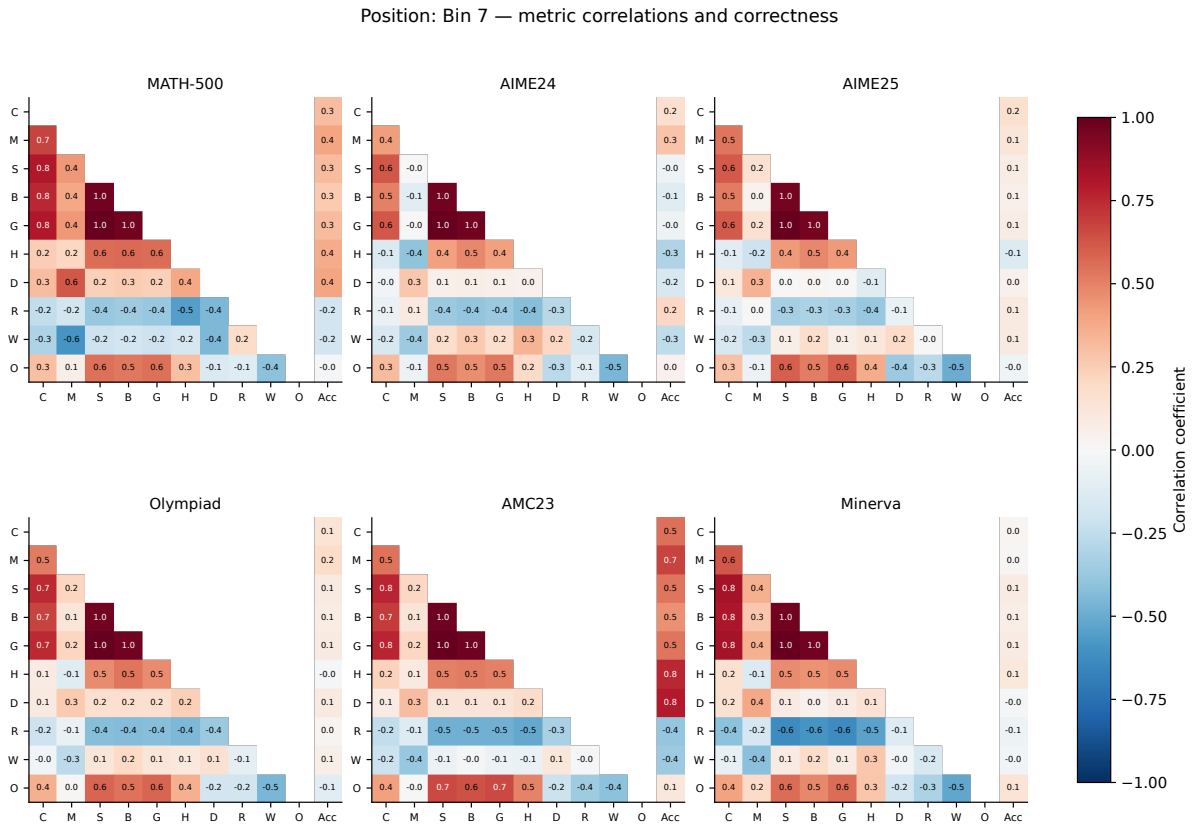


Fig. 35. Position: Bin 7: metric relationships and correctness. Codes are defined in Table 32.

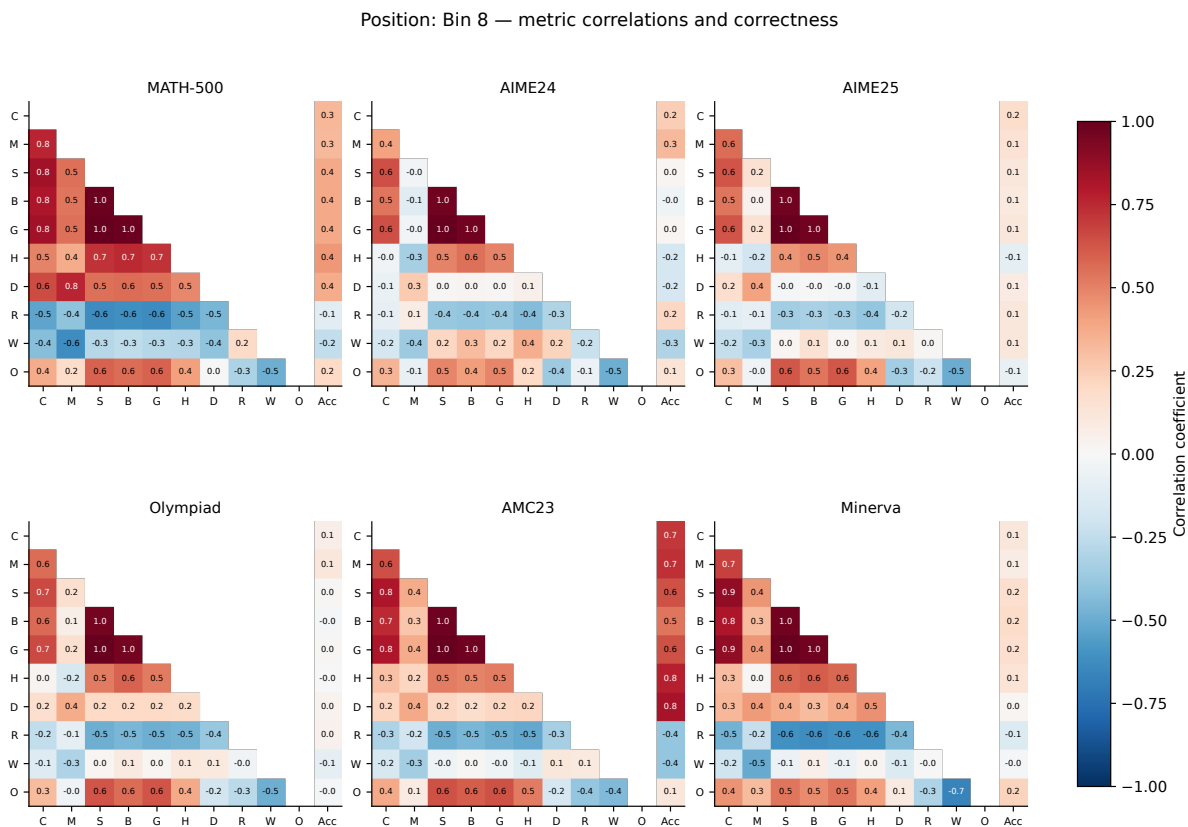


Fig. 36. Position: Bin 8: metric relationships and correctness. Codes are defined in Table 32.

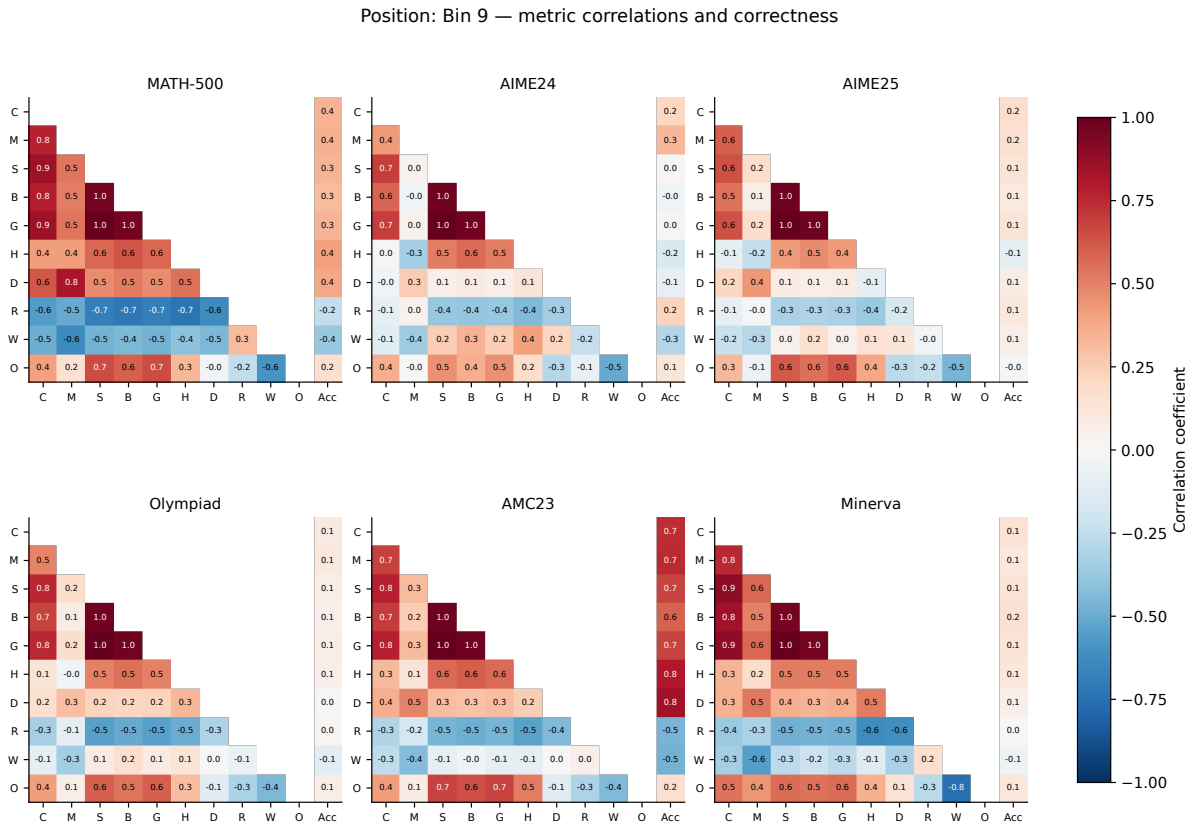


Fig. 37. Position: Bin 9: metric relationships and correctness. Codes are defined in Table 32.

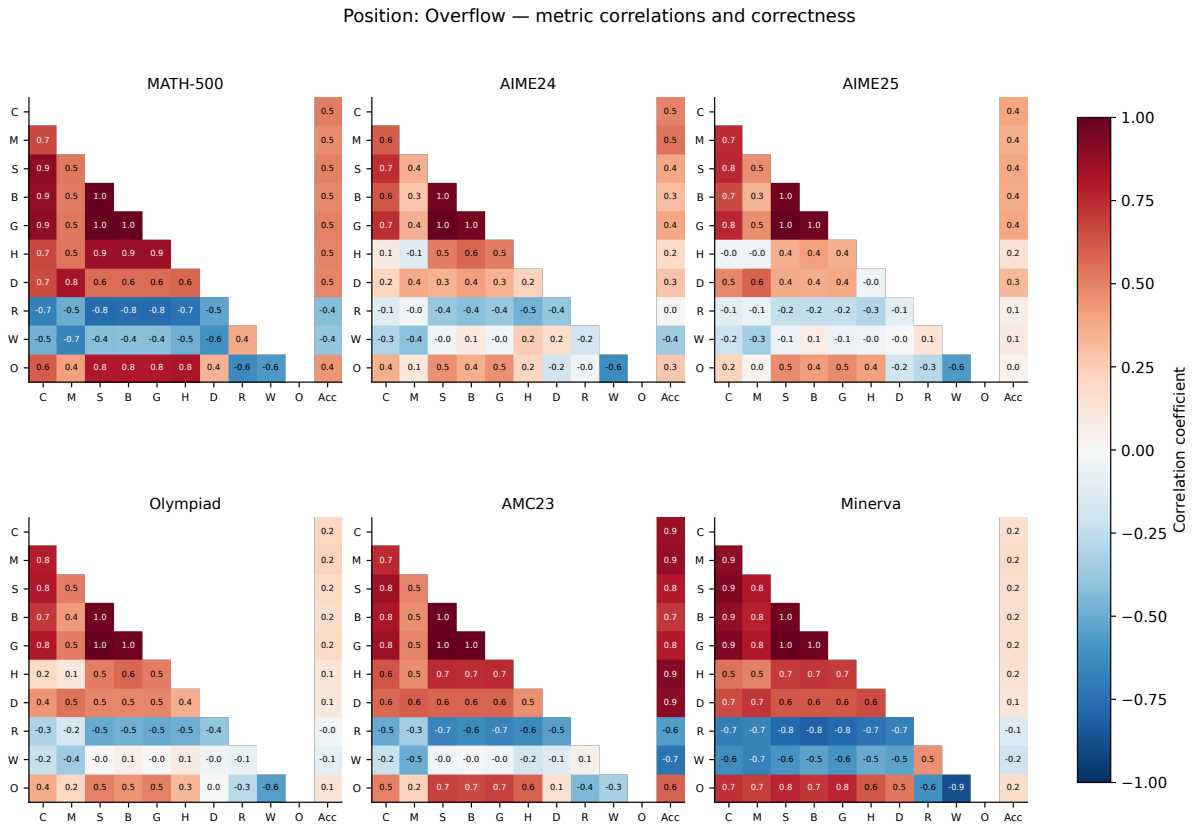


Fig. 38. Position: Overflow: metric relationships and correctness. Codes are defined in Table 32.

D Gemma and GPT-OSS correlation plot atlas

Lower triangles show metric–metric Spearman ρ ; Acc shows point-biserial r with correctness. Missing or undefined values are dashes, not zero. Token-bearing counts are upper bounds on pairwise finite counts. Each model has whole-continuation, full-reasoning, seven class, and eleven position scopes. Qwen plots appear in the preceding atlas. These are descriptive screens without significance selection.

Table 33. Metric key for all models.

Code	Metric
T	Token count
C	Router confidence
M	Leading-expert margin
S	Selected probability mass
B	Selection-boundary margin
G	Selected/rest mean gap
H	Hidden-state norm
D	Hidden-state step distance
R	Hidden/router geometry
W	Top-1 switch rate
O	Adjacent selected-set overlap

Gemma-4-26B-A4B: Whole continuation

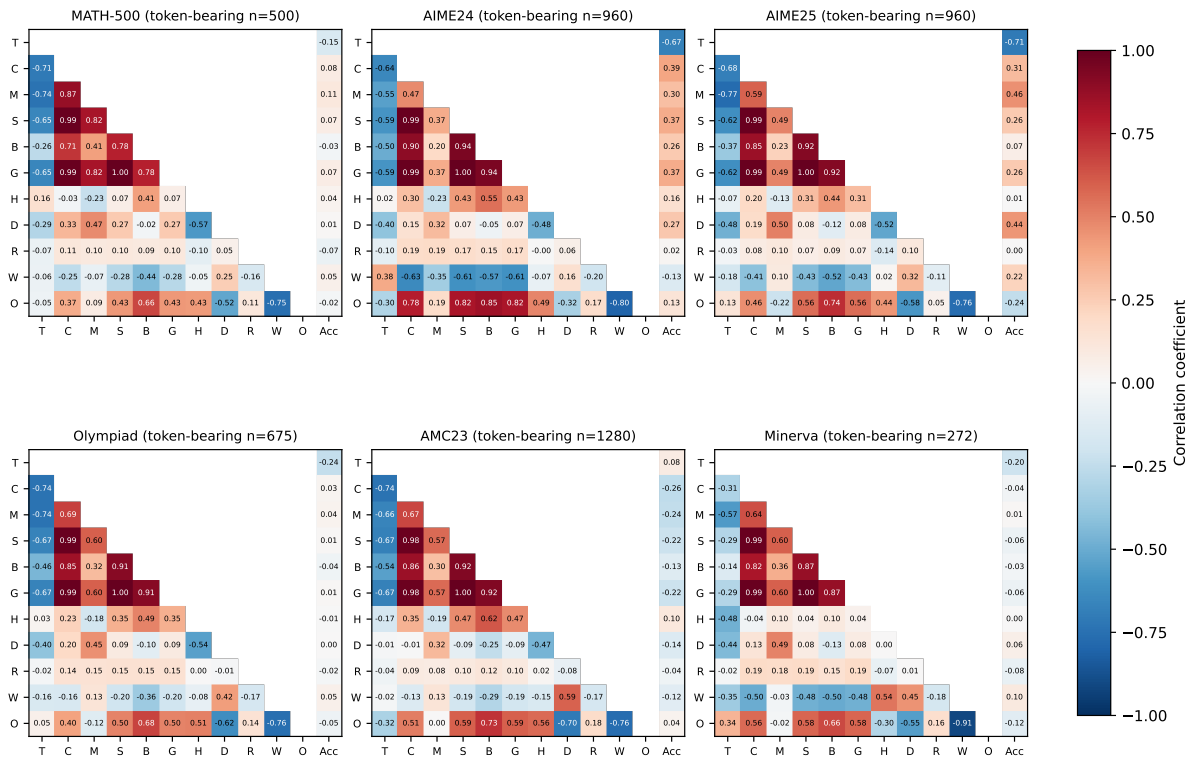


Fig. 39. Gemma-4-26B-A4B / Whole continuation. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Full reasoning

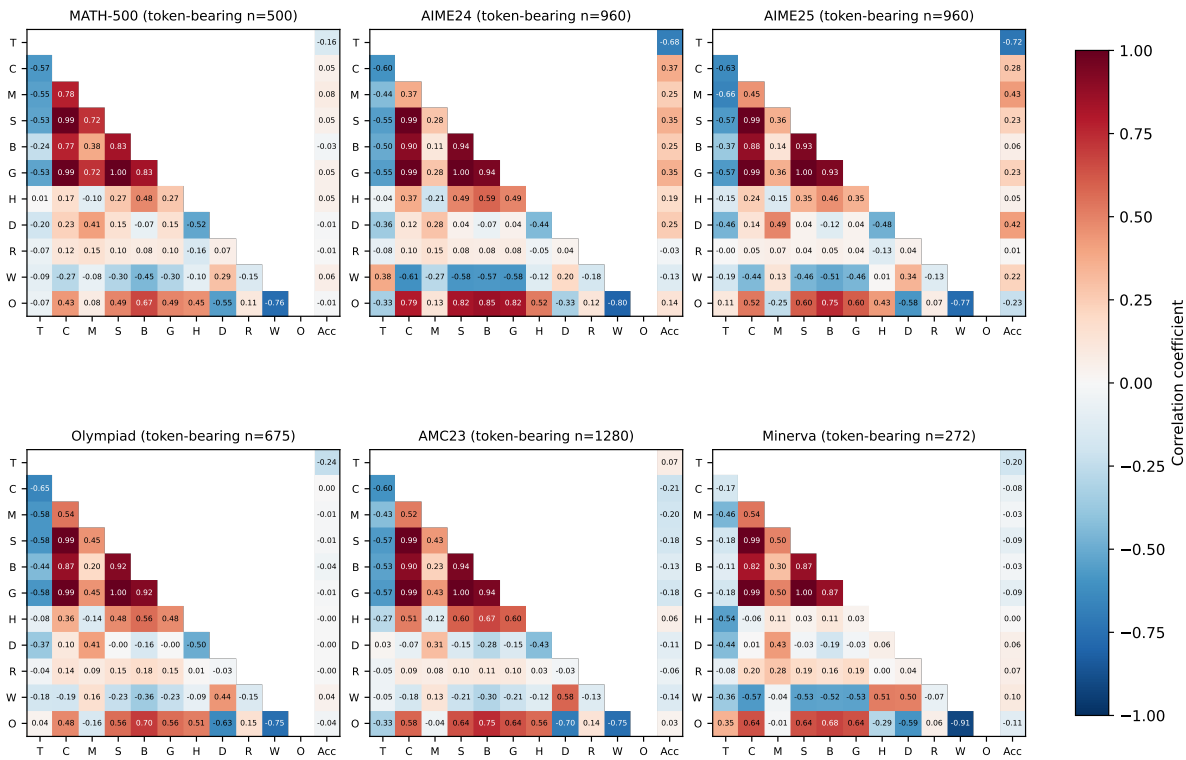


Fig. 40. Gemma-4-26B-A4B / Full reasoning. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Read

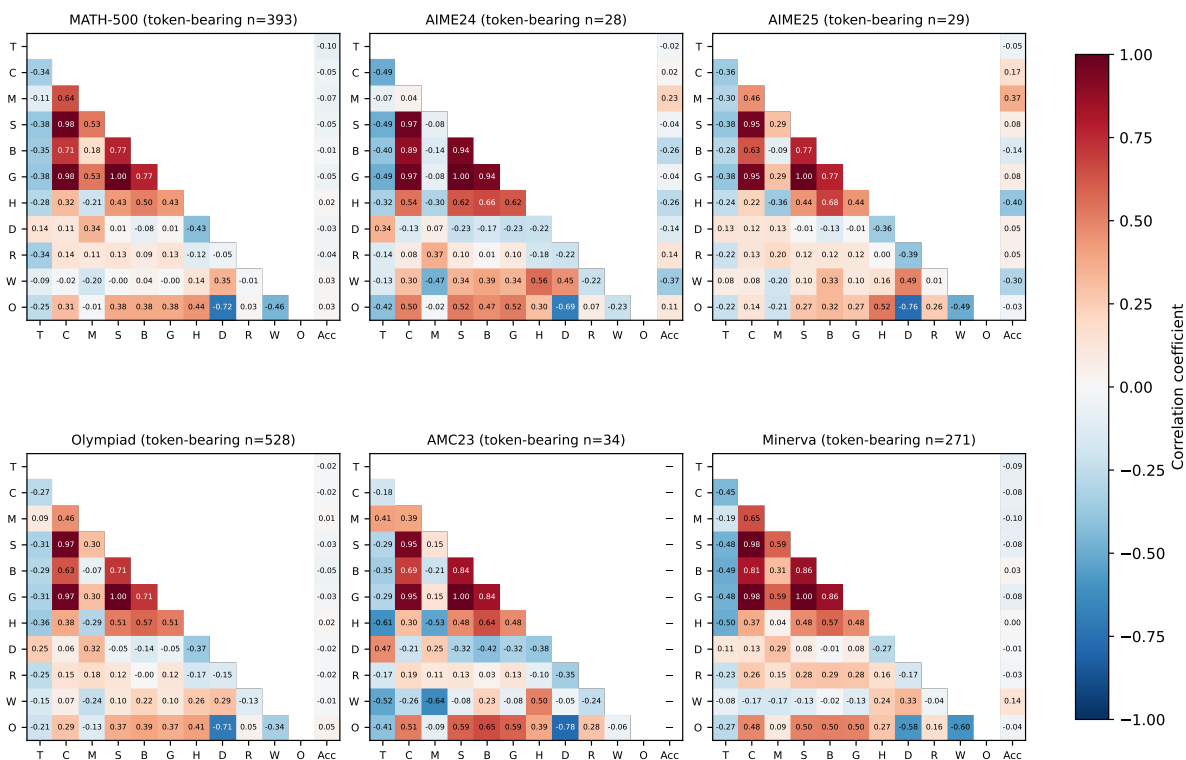


Fig. 41. Gemma-4-26B-A4B / Class: Read. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Analyze

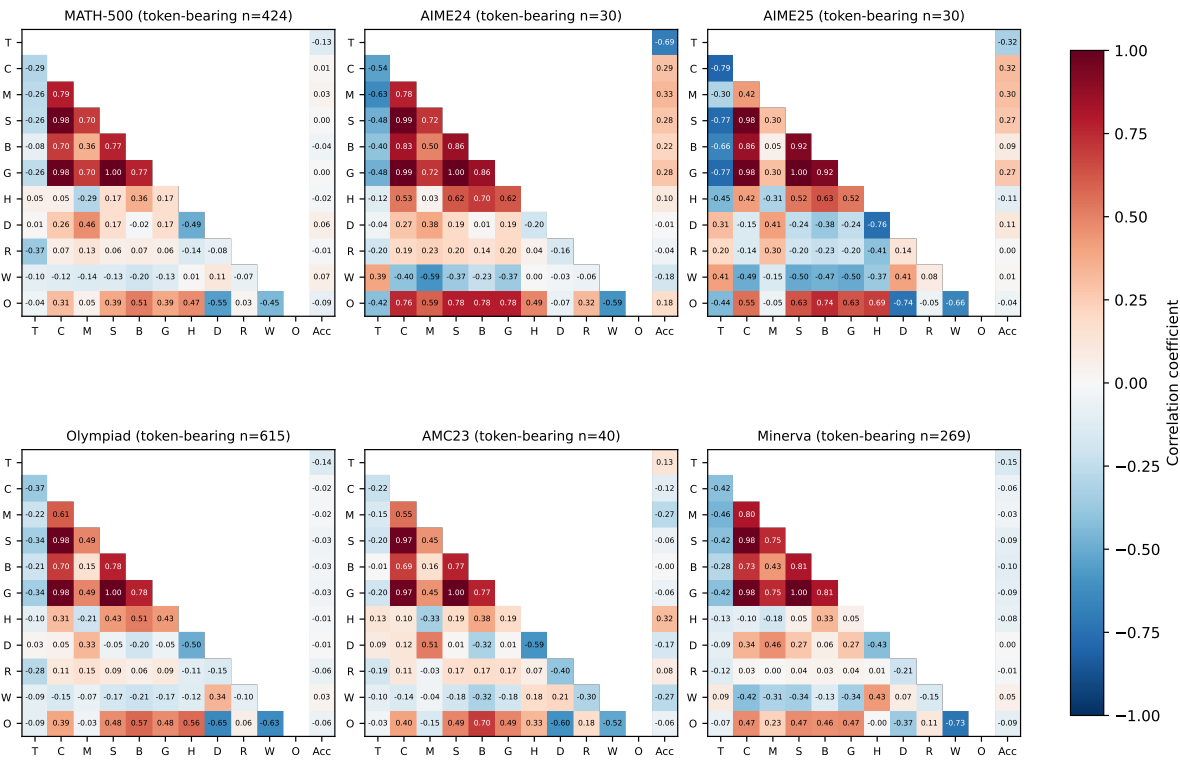


Fig. 42. Gemma-4-26B-A4B / Class: Analyze. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Plan

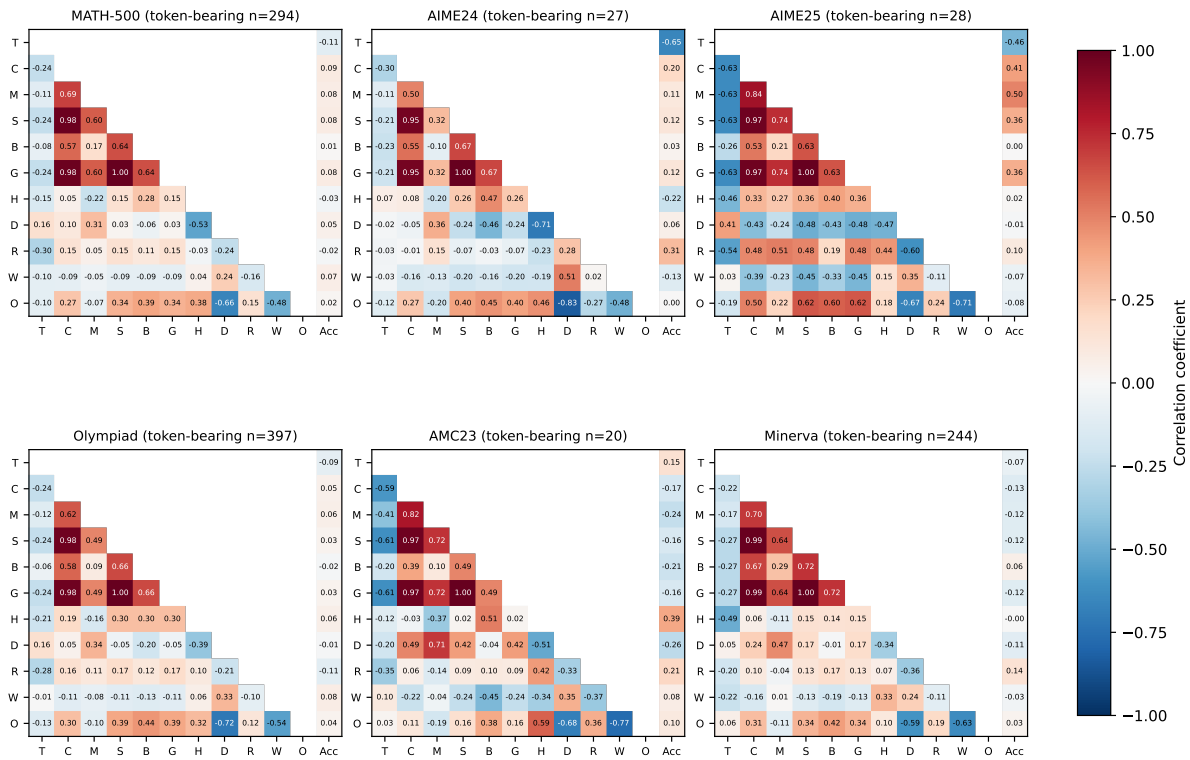


Fig. 43. Gemma-4-26B-A4B / Class: Plan. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Implement

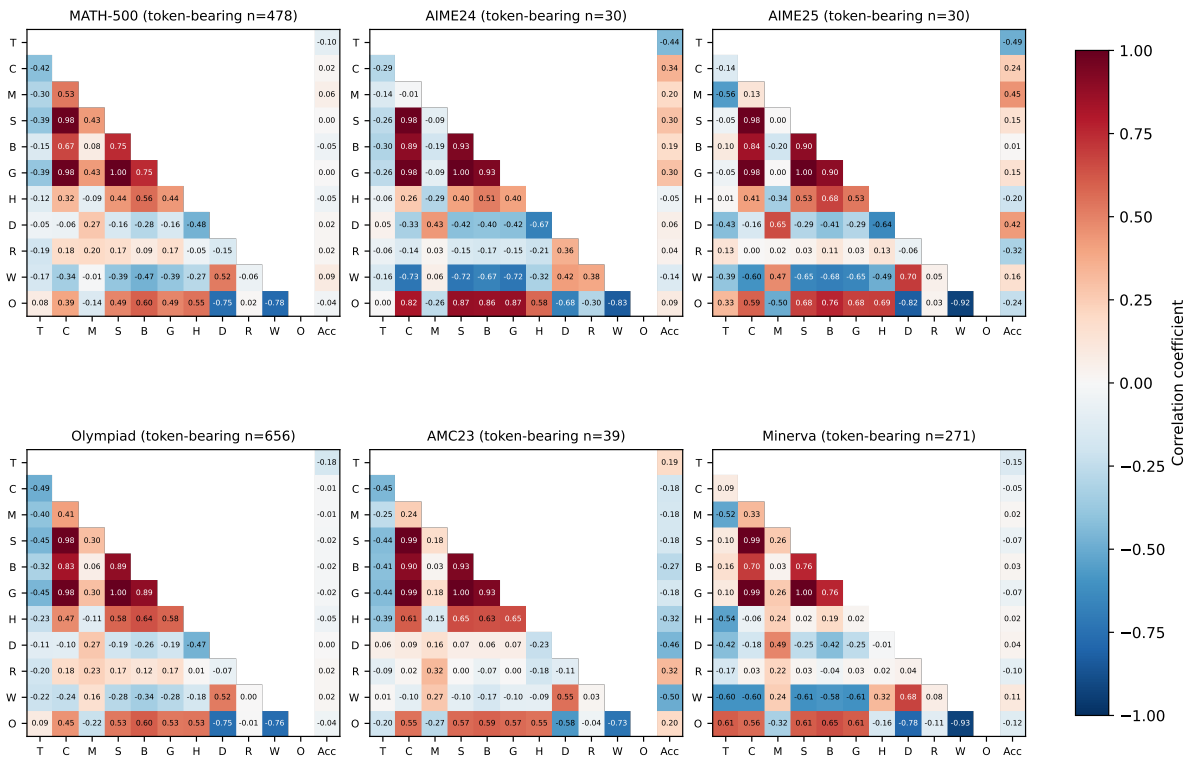


Fig. 44. Gemma-4-26B-A4B / Class: Implement. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Explore

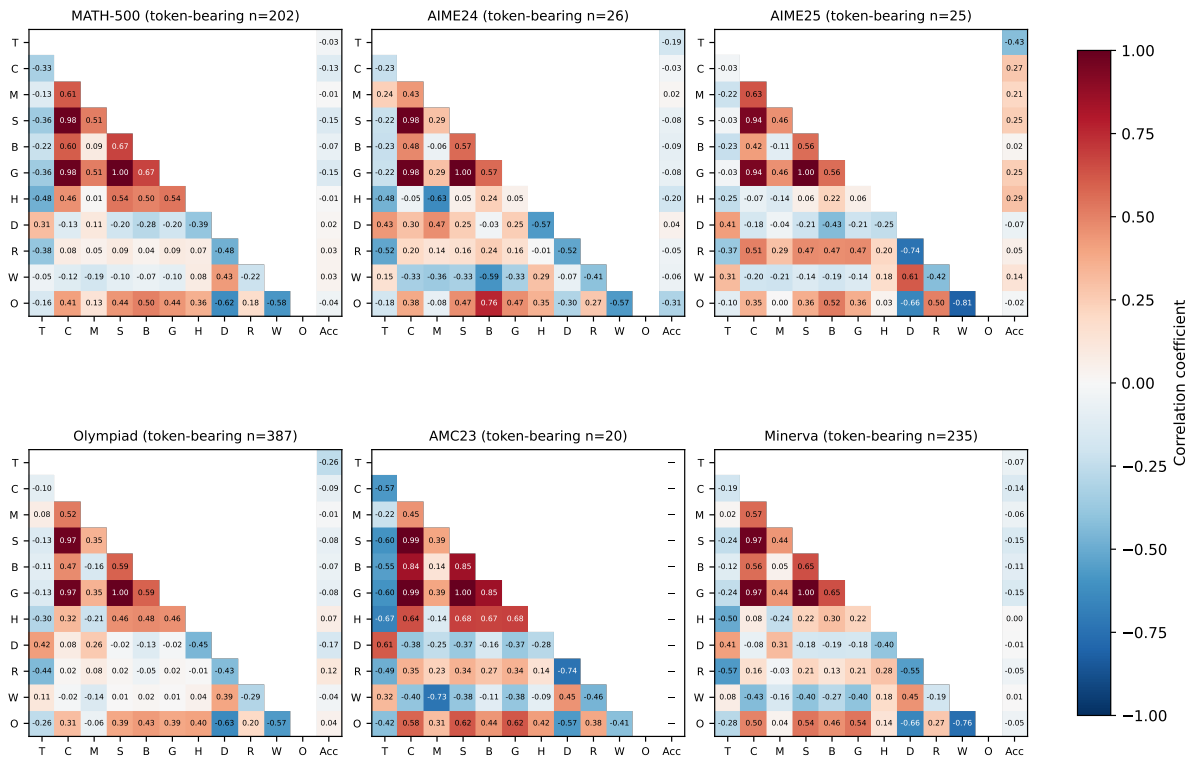


Fig. 45. Gemma-4-26B-A4B / Class: Explore. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Verify

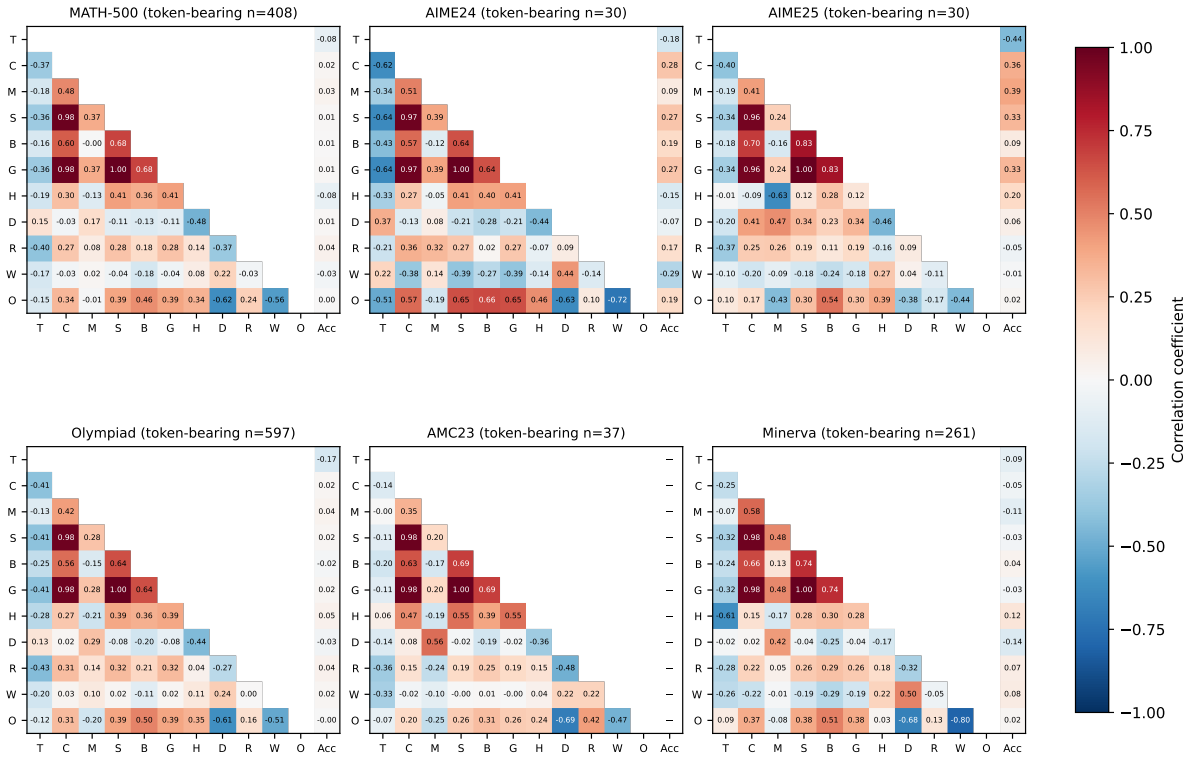


Fig. 46. Gemma-4-26B-A4B / Class: Verify. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Monitor

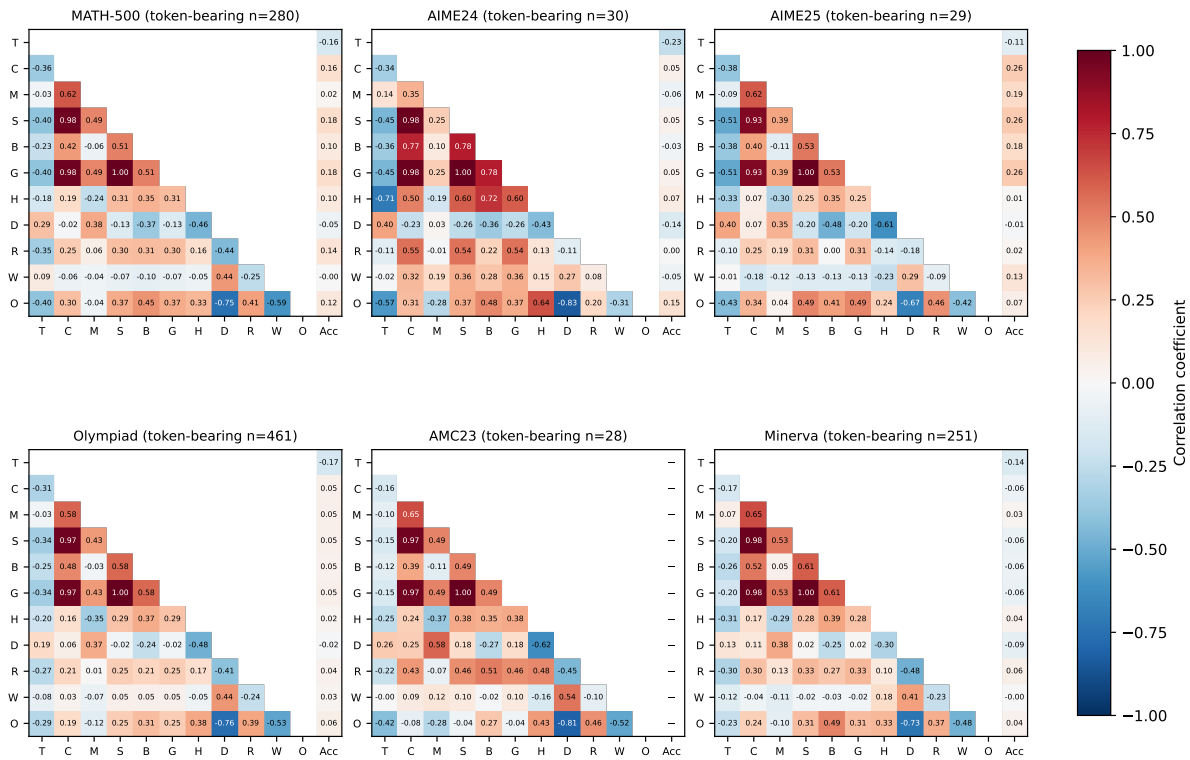


Fig. 47. Gemma-4-26B-A4B / Class: Monitor. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 00

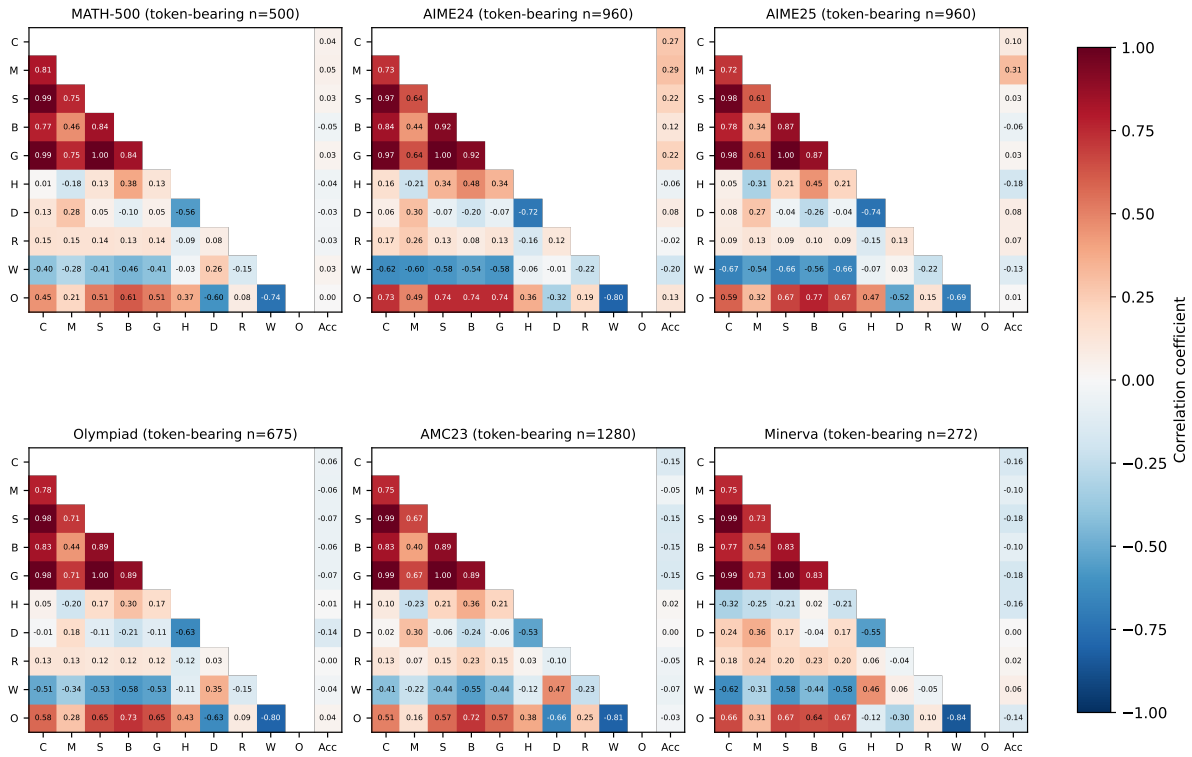


Fig. 48. Gemma-4-26B-A4B / Position: bin 00. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 01

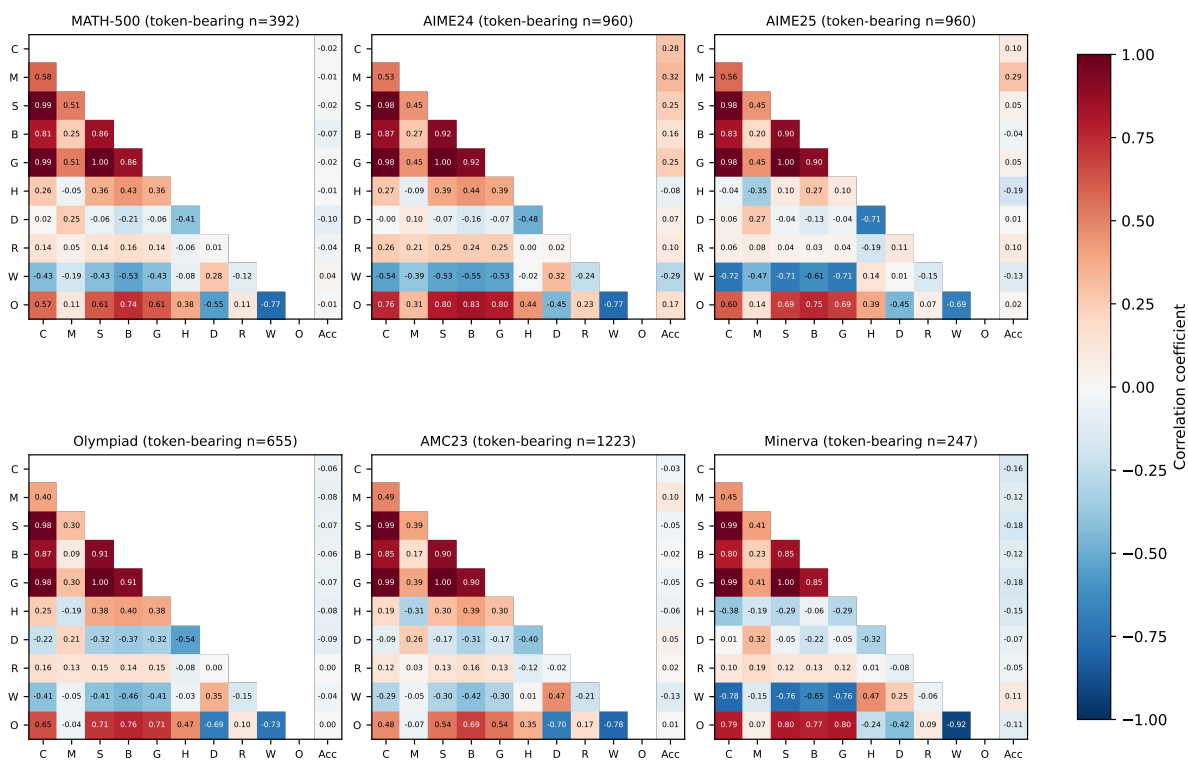


Fig. 49. Gemma-4-26B-A4B / Position: bin 01. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 02

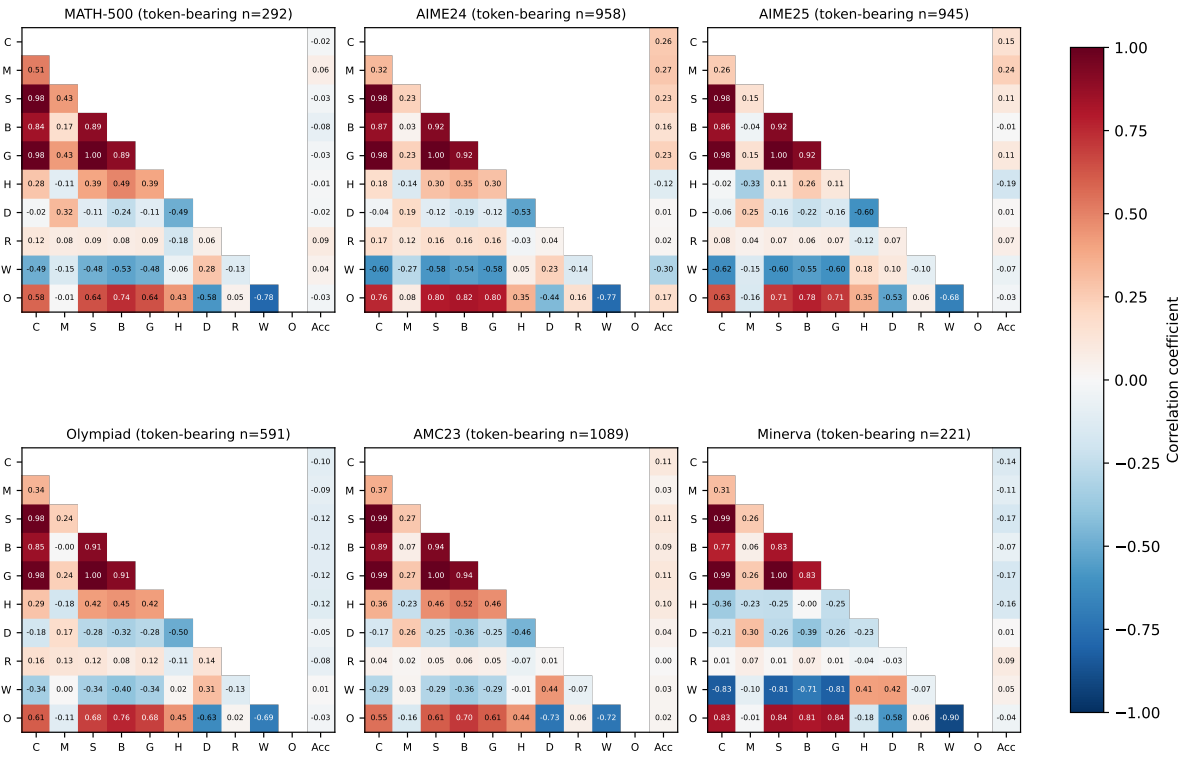


Fig. 50. Gemma-4-26B-A4B / Position: bin 02. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 03

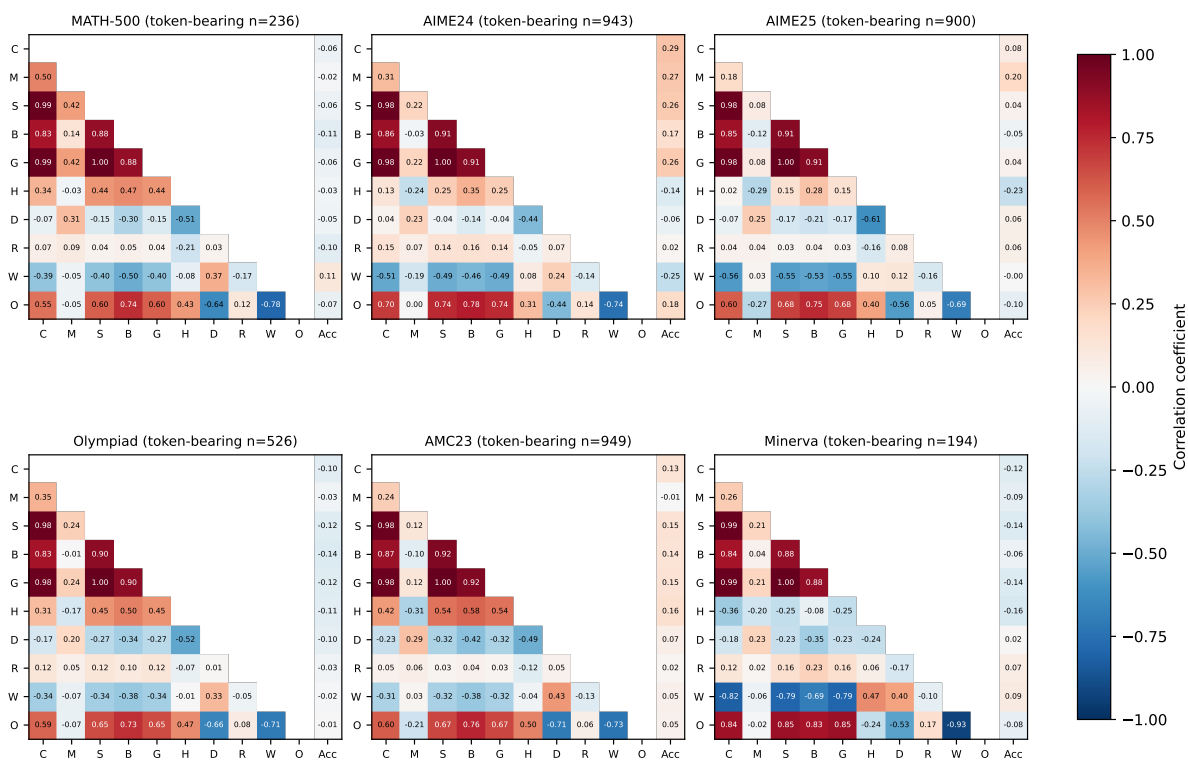


Fig. 51. Gemma-4-26B-A4B / Position: bin 03. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 04

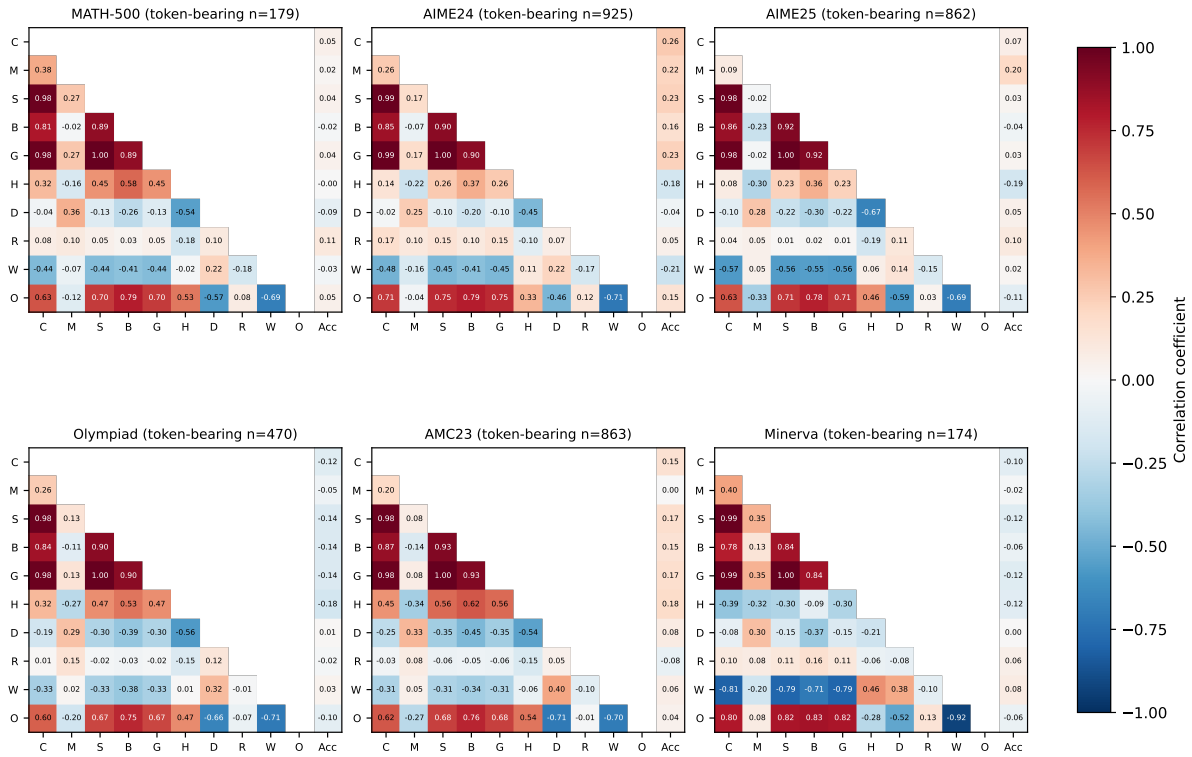


Fig. 52. Gemma-4-26B-A4B / Position: bin 04. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 05

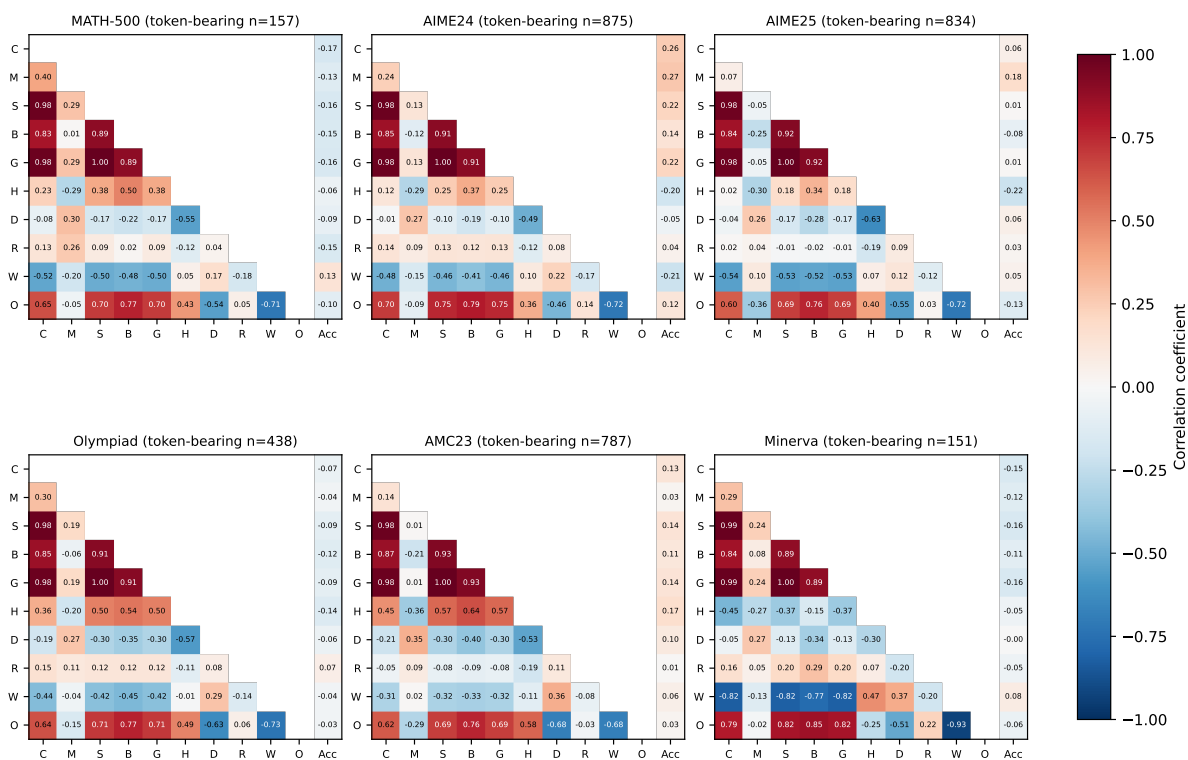


Fig. 53. Gemma-4-26B-A4B / Position: bin 05. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 06

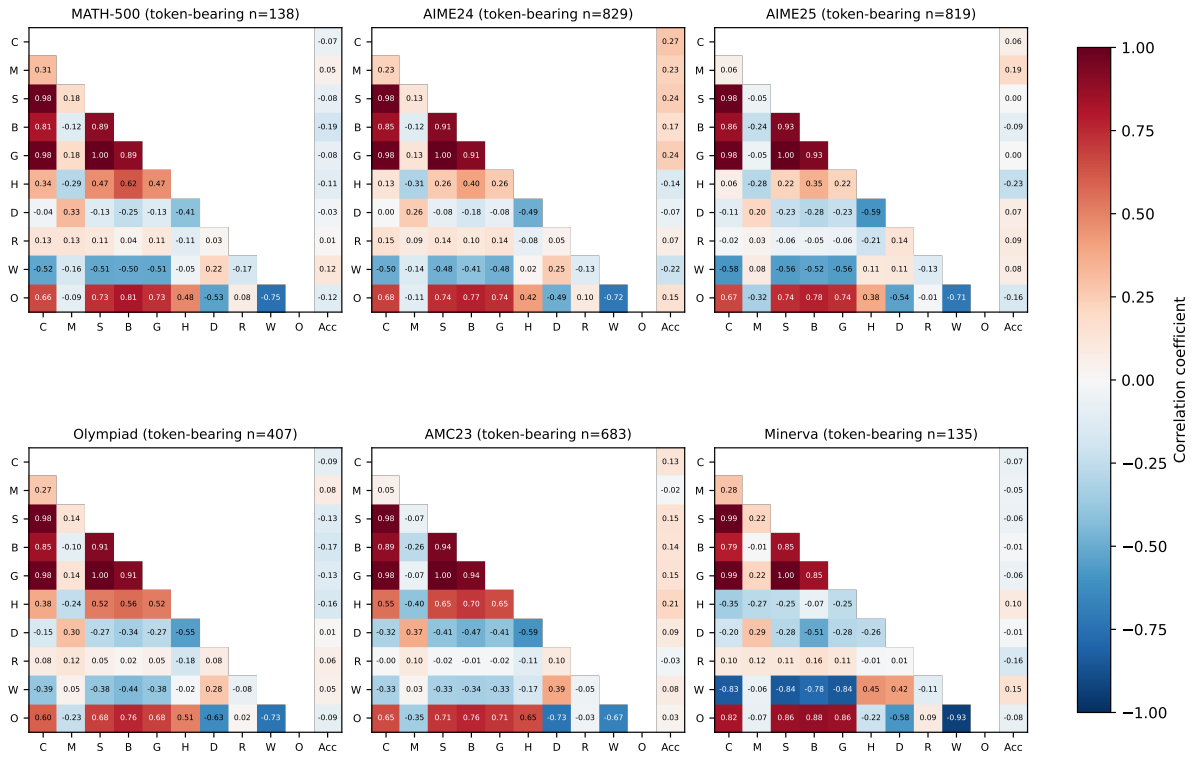


Fig. 54. Gemma-4-26B-A4B / Position: bin 06. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 07

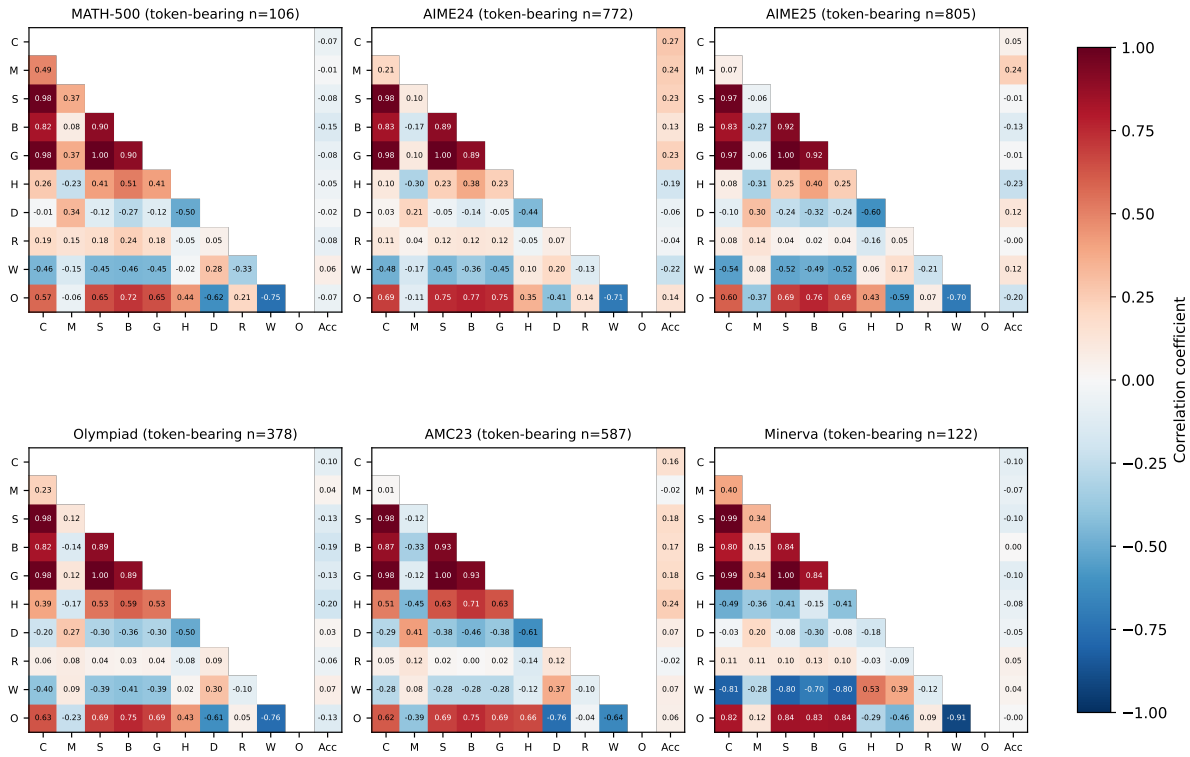


Fig. 55. Gemma-4-26B-A4B / Position: bin 07. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 08

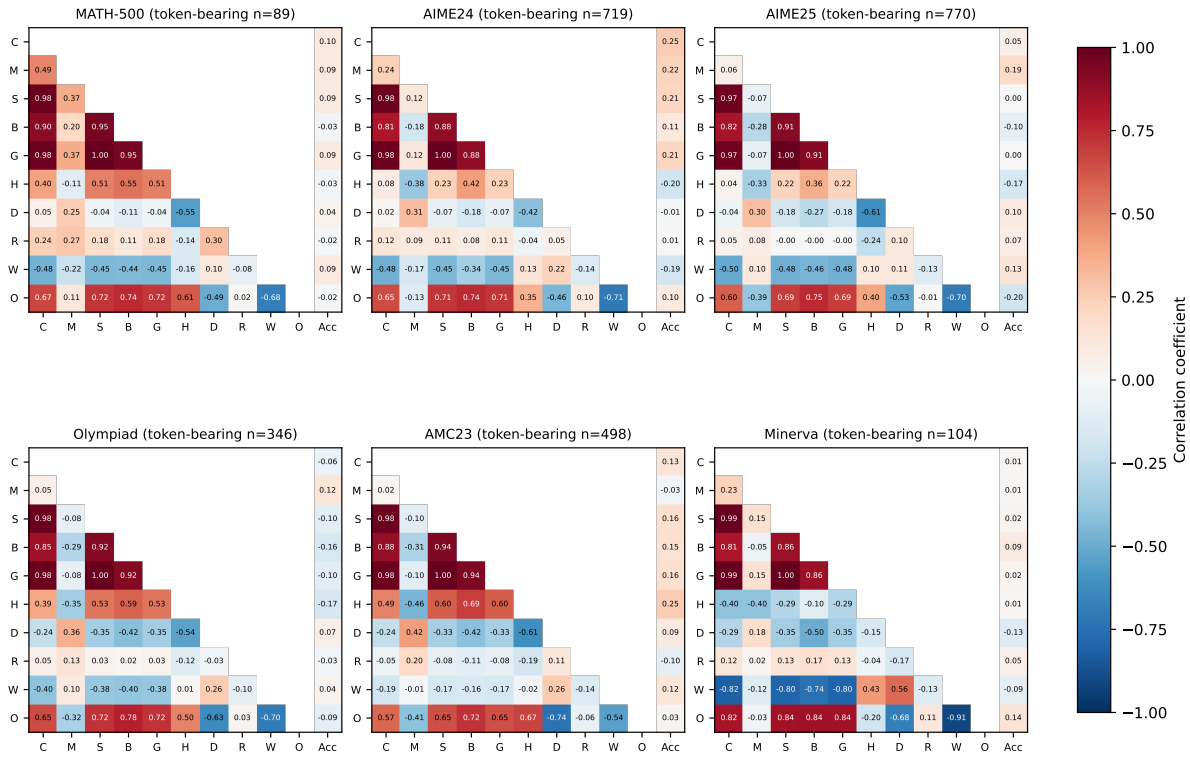


Fig. 56. Gemma-4-26B-A4B / Position: bin 08. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 09

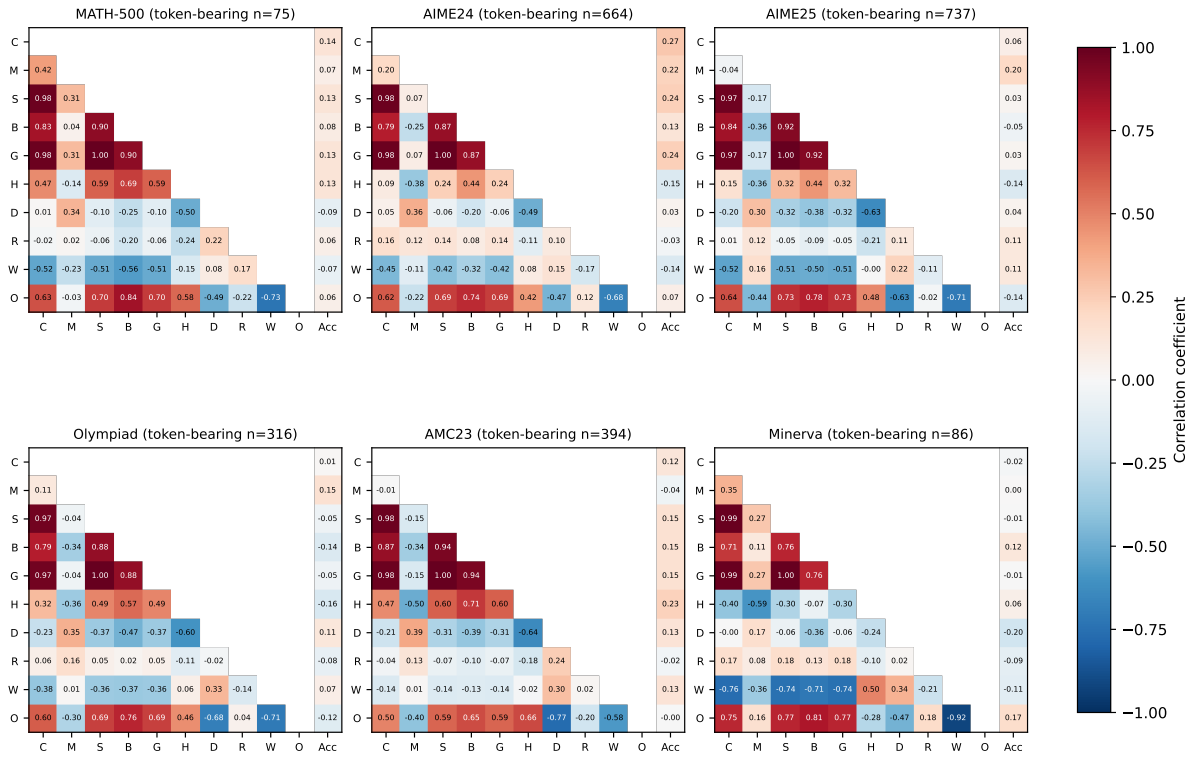


Fig. 57. Gemma-4-26B-A4B / Position: bin 09. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: overflow

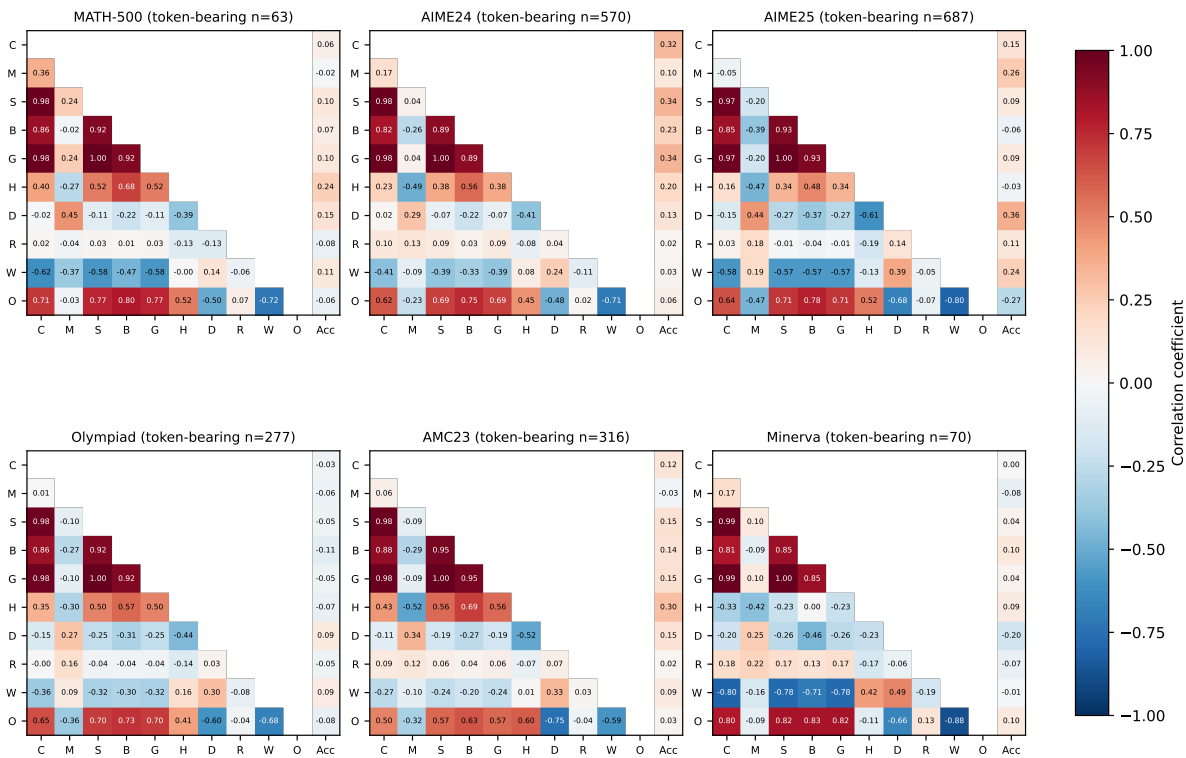


Fig. 58. Gemma-4-26B-A4B / Position: overflow. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Whole continuation

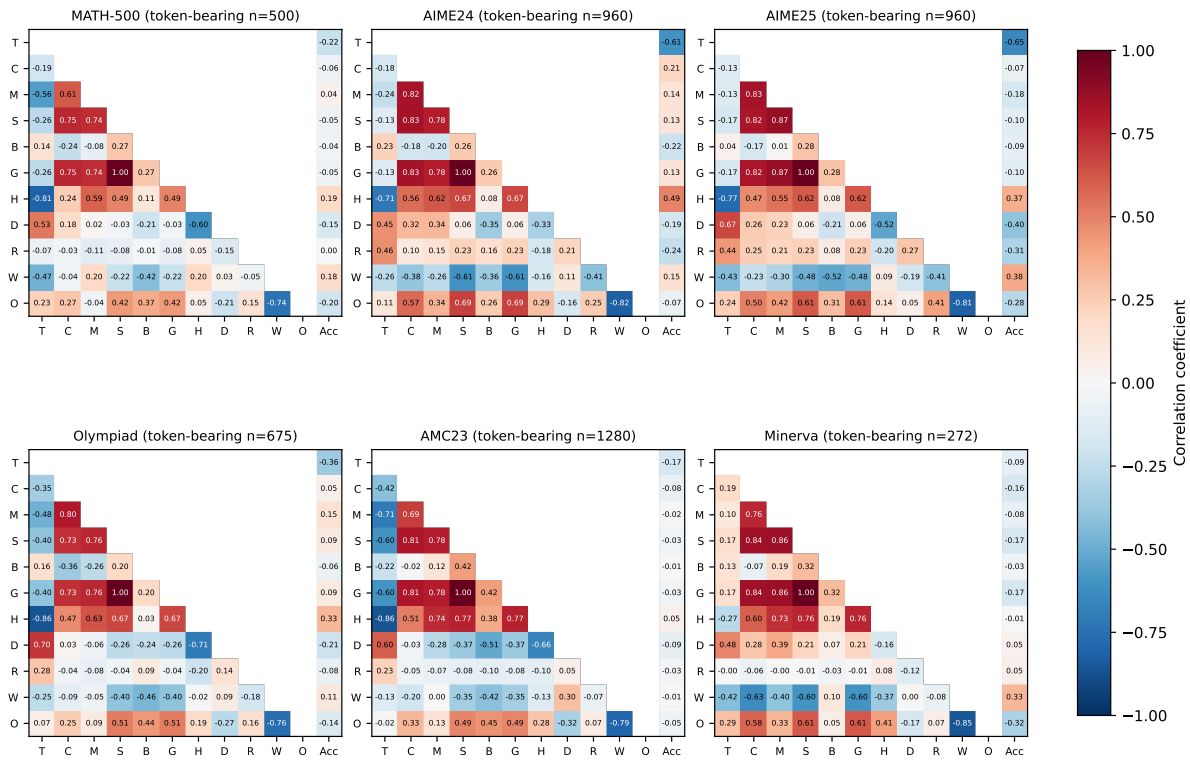


Fig. 59. GPT-OSS-20B / Whole continuation. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Full reasoning

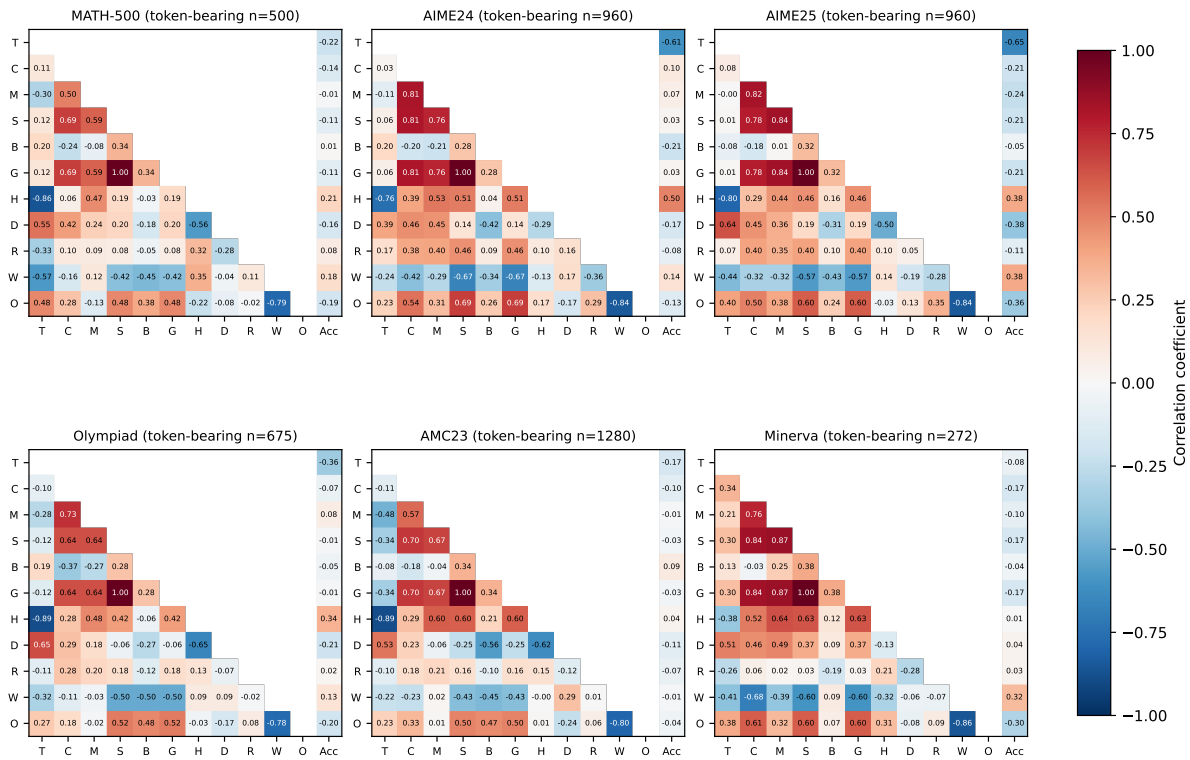


Fig. 60. GPT-OSS-20B / Full reasoning. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Read

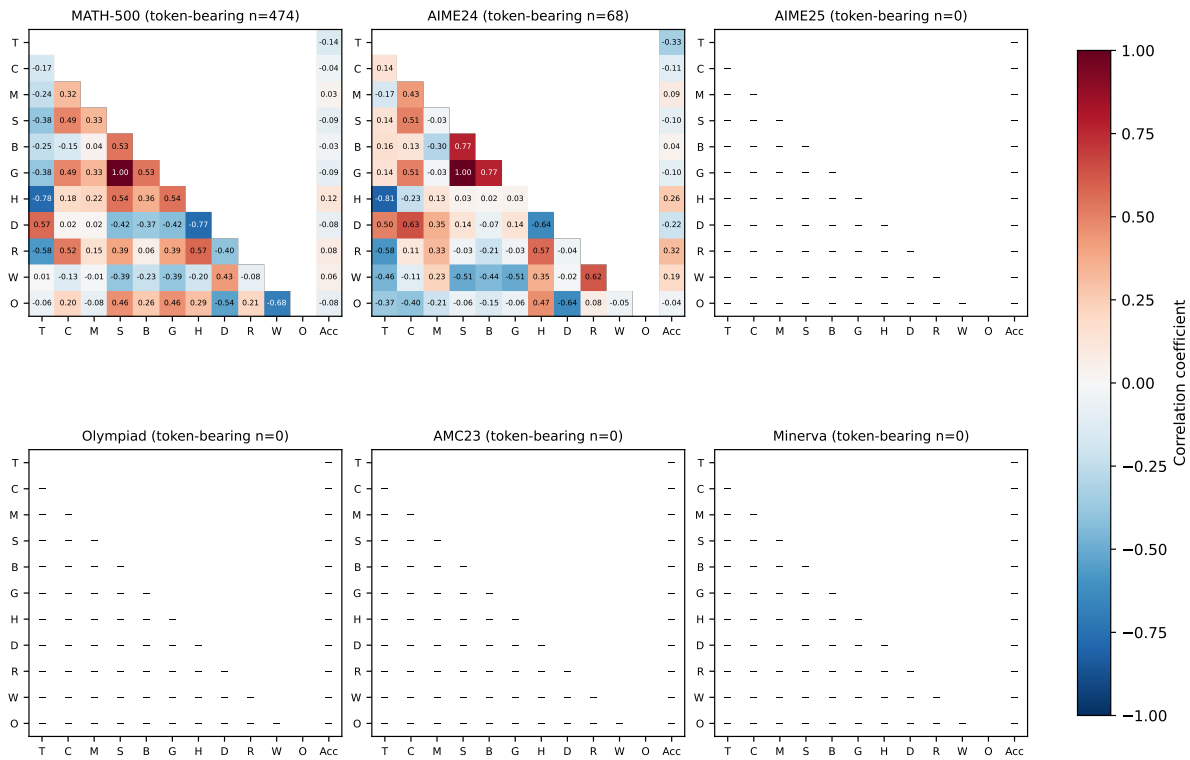


Fig. 61. GPT-OSS-20B / Class: Read. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Analyze

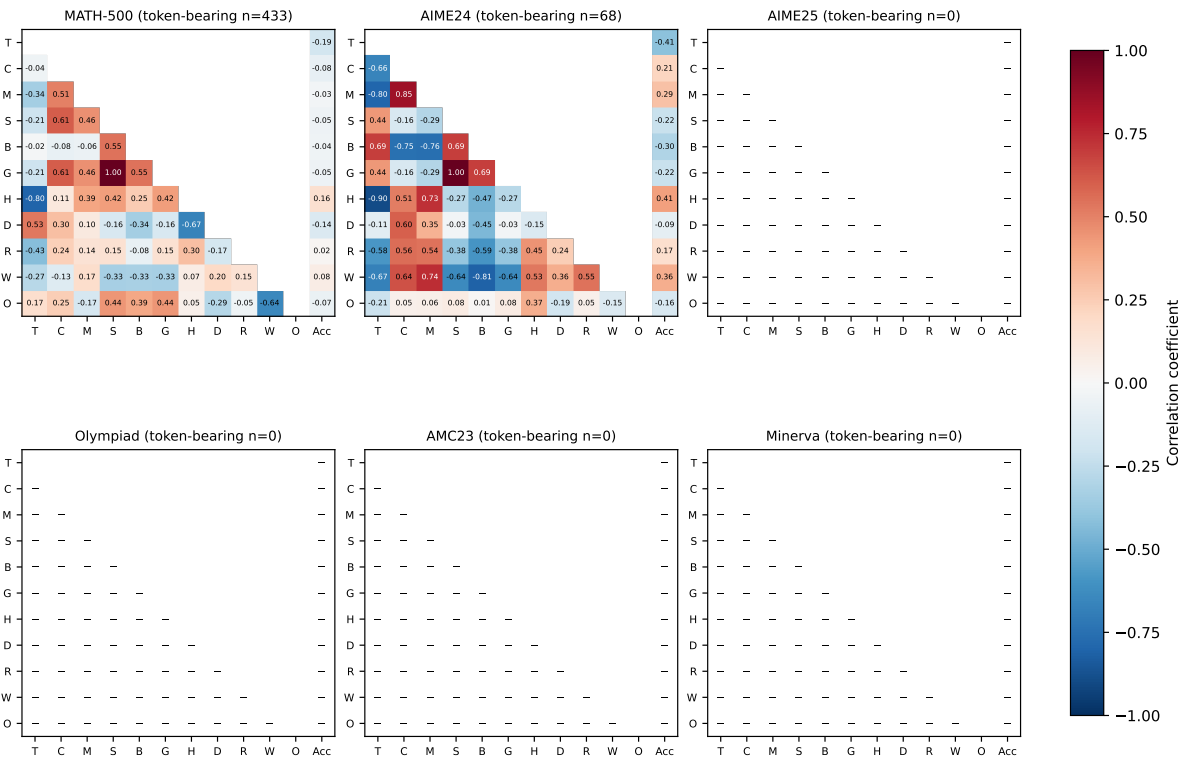


Fig. 62. GPT-OSS-20B / Class: Analyze. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Plan

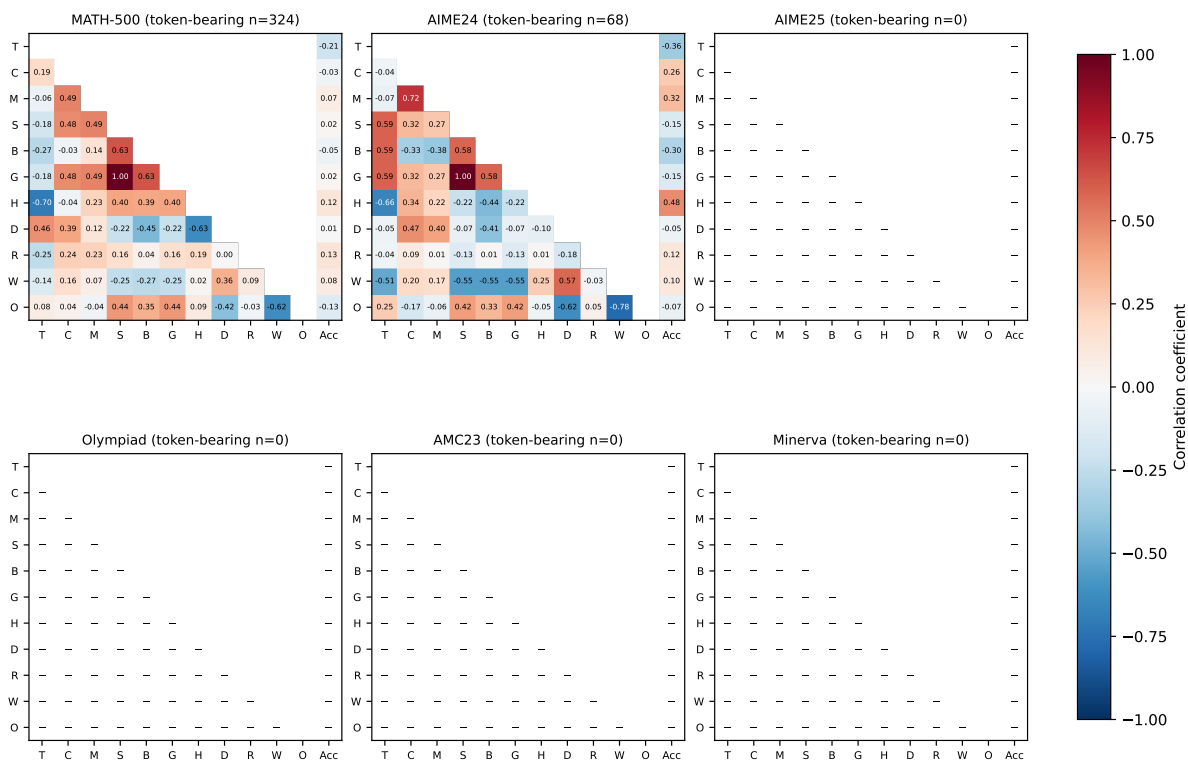


Fig. 63. GPT-OSS-20B / Class: Plan. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Implement

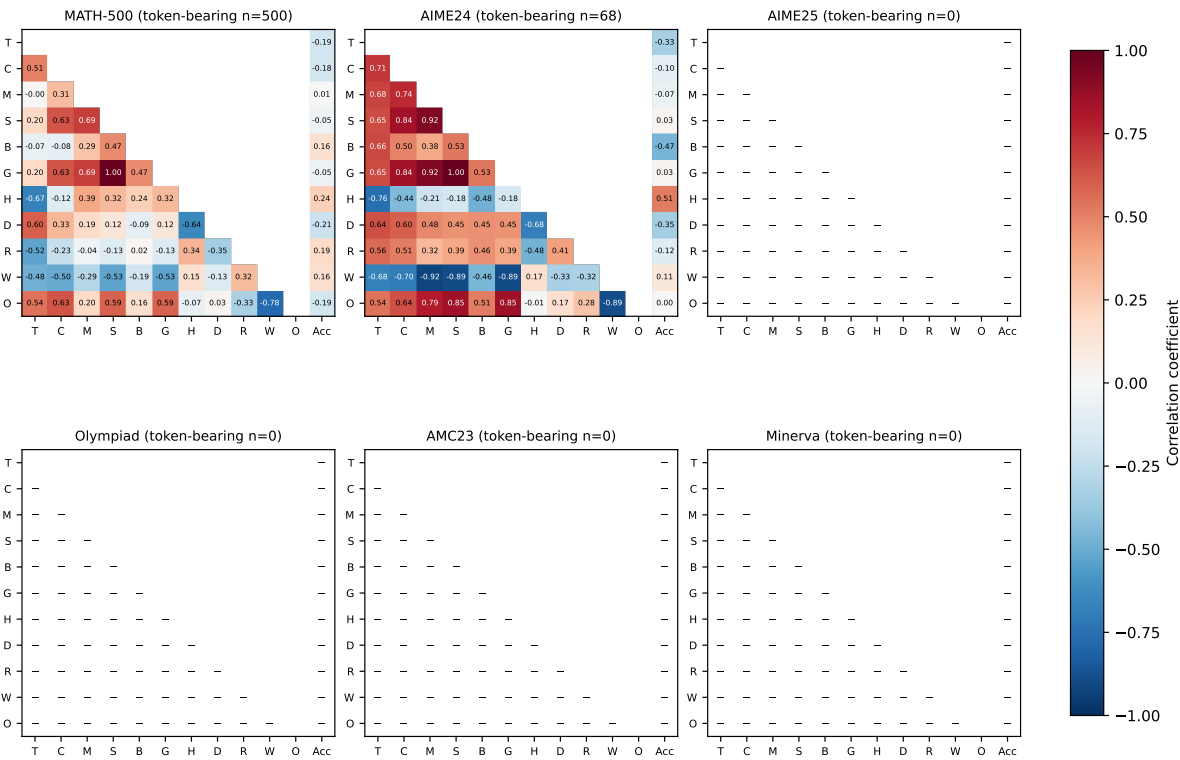


Fig. 64. GPT-OSS-20B / Class: Implement. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Explore

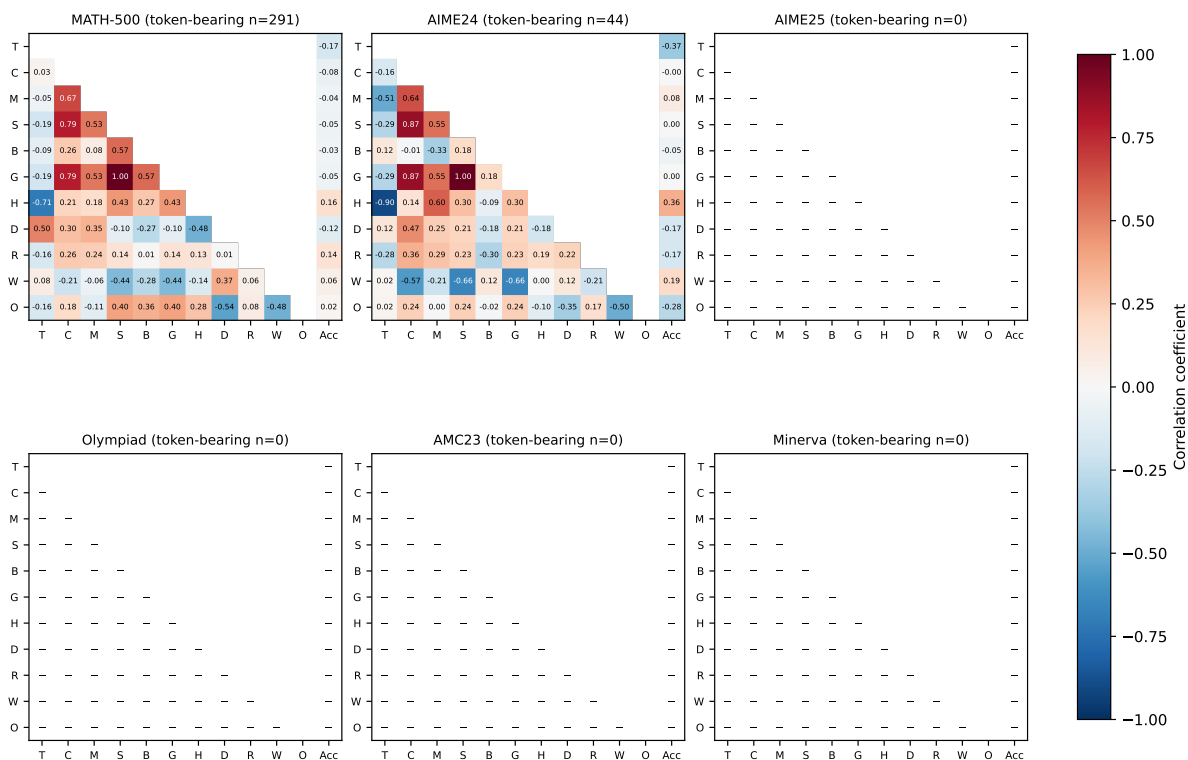


Fig. 65. GPT-OSS-20B / Class: Explore. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Verify

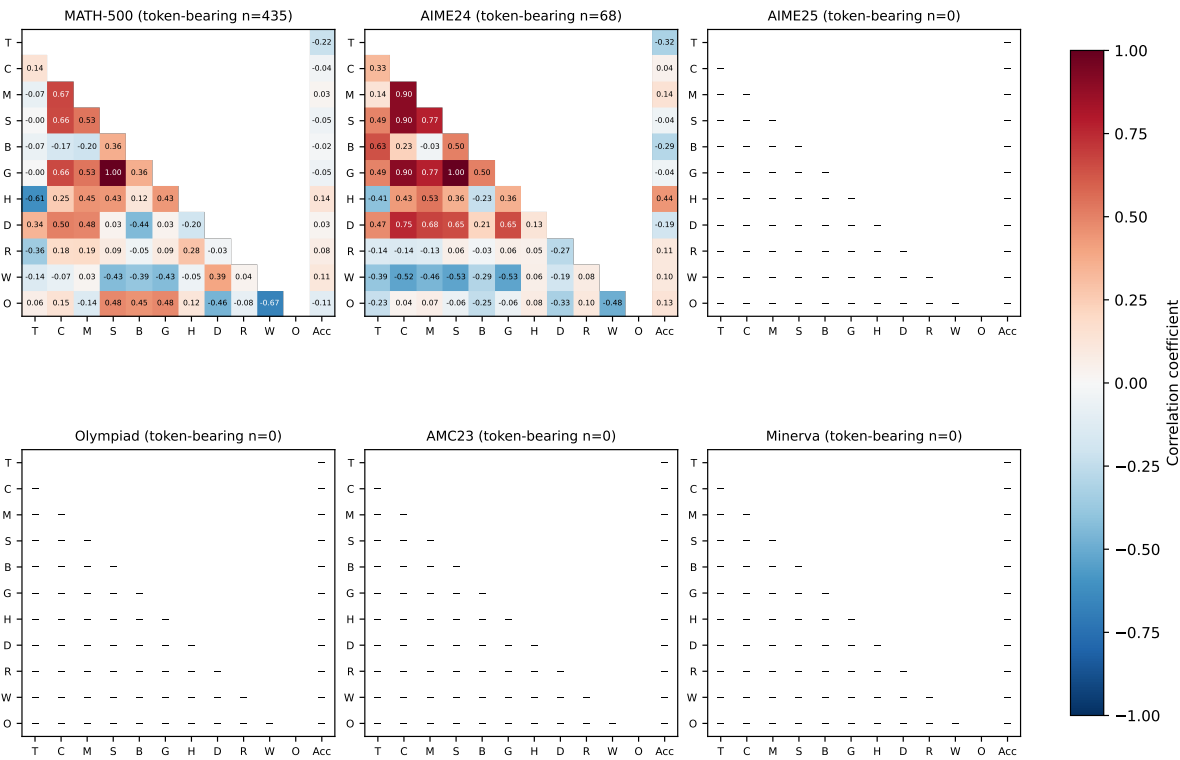


Fig. 66. GPT-OSS-20B / Class: Verify. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Monitor

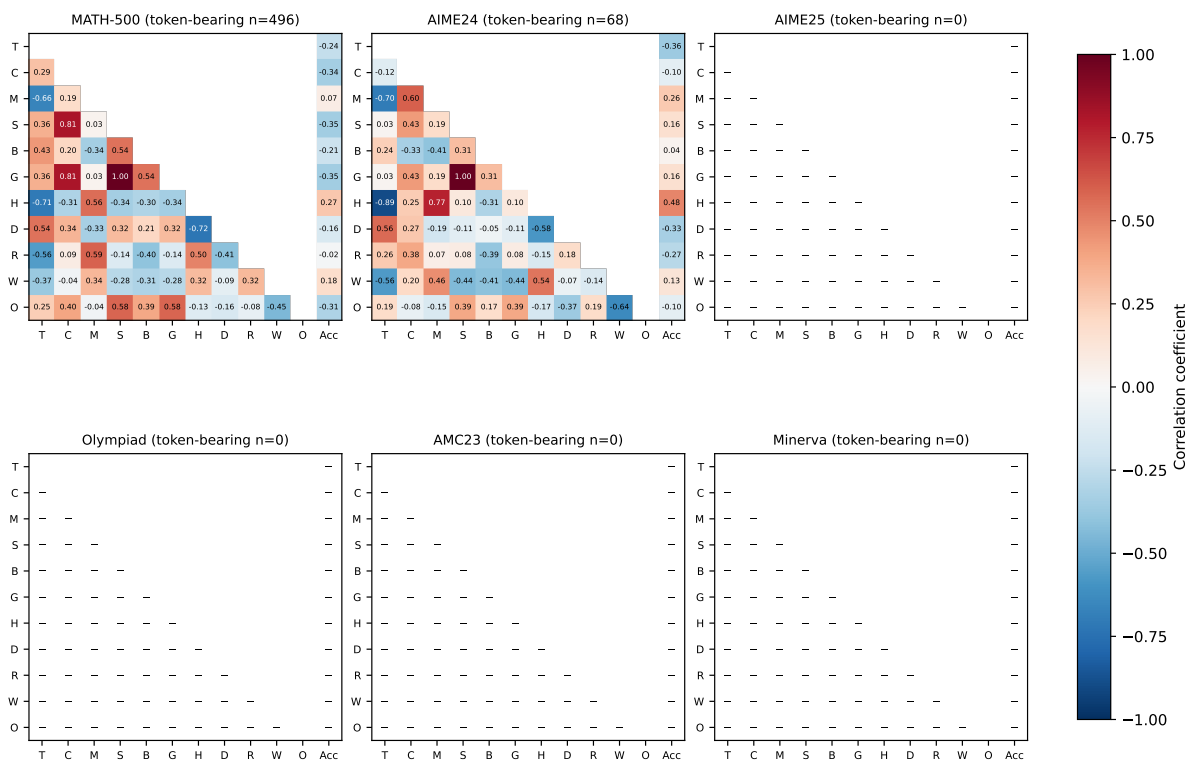


Fig. 67. GPT-OSS-20B / Class: Monitor. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 00

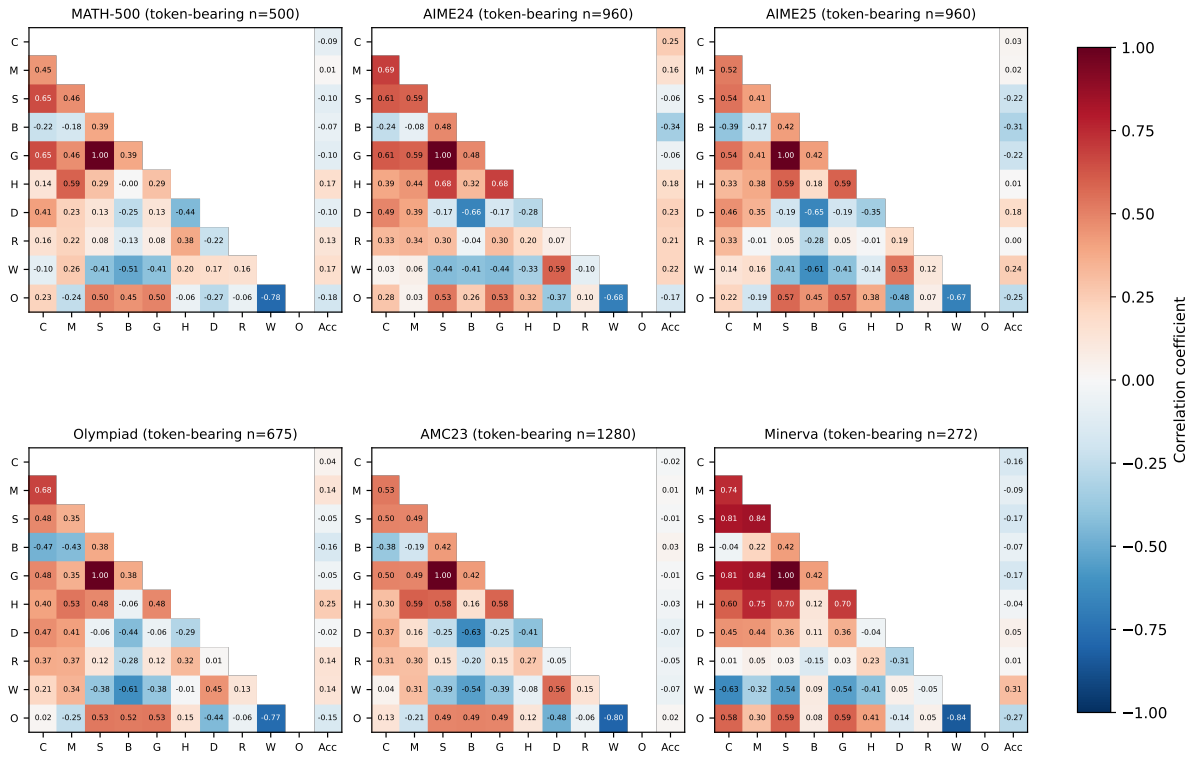


Fig. 68. GPT-OSS-20B / Position: bin 00. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 01

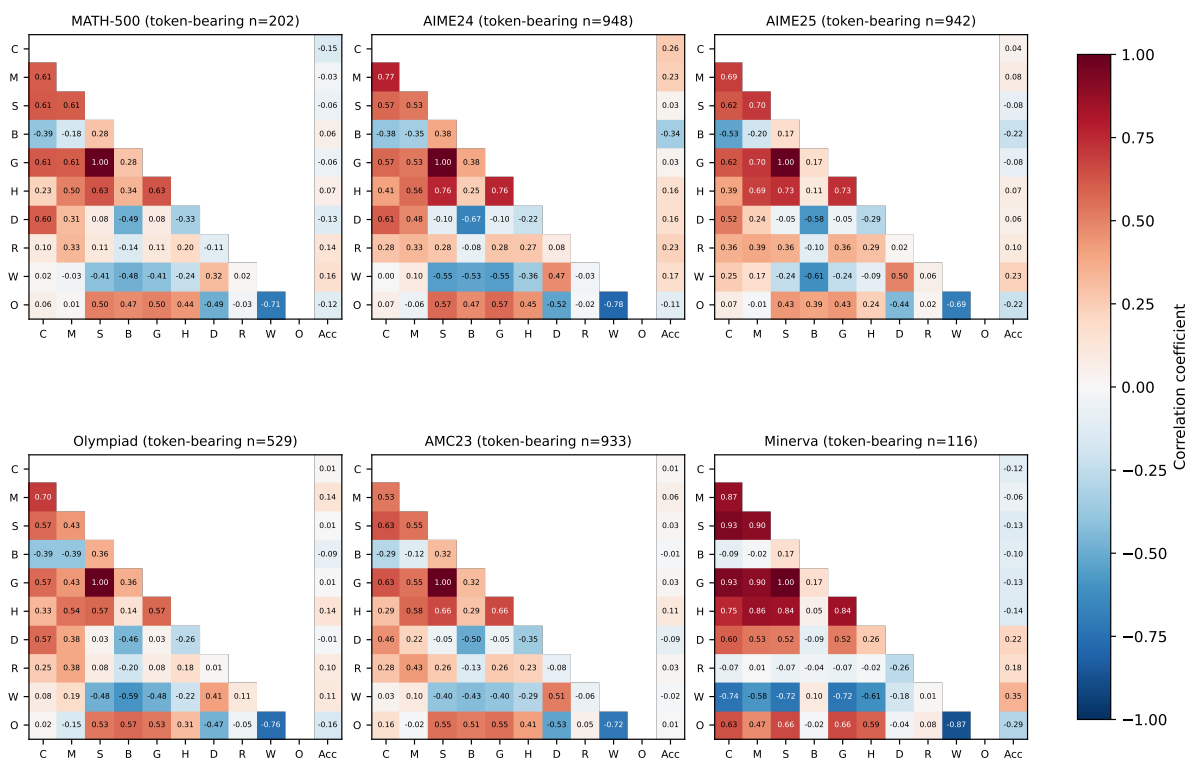


Fig. 69. GPT-OSS-20B / Position: bin 01. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 02

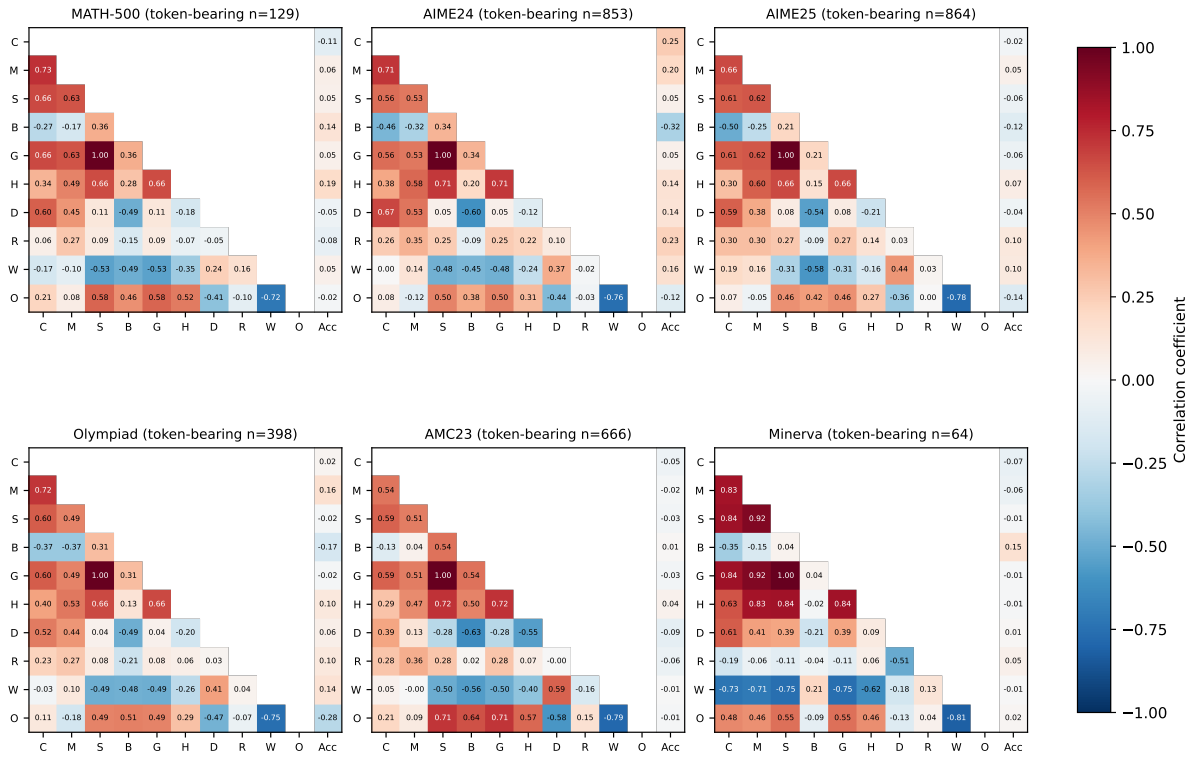


Fig. 70. GPT-OSS-20B / Position: bin 02. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 03

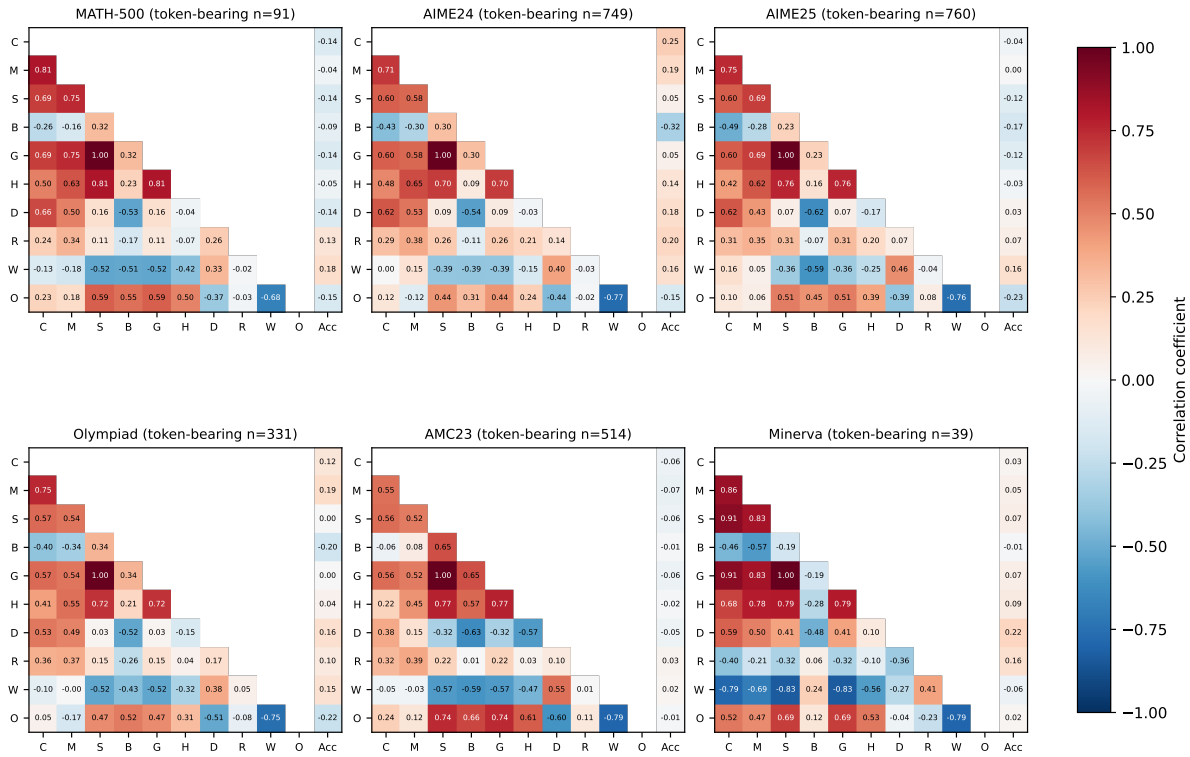


Fig. 71. GPT-OSS-20B / Position: bin 03. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 04

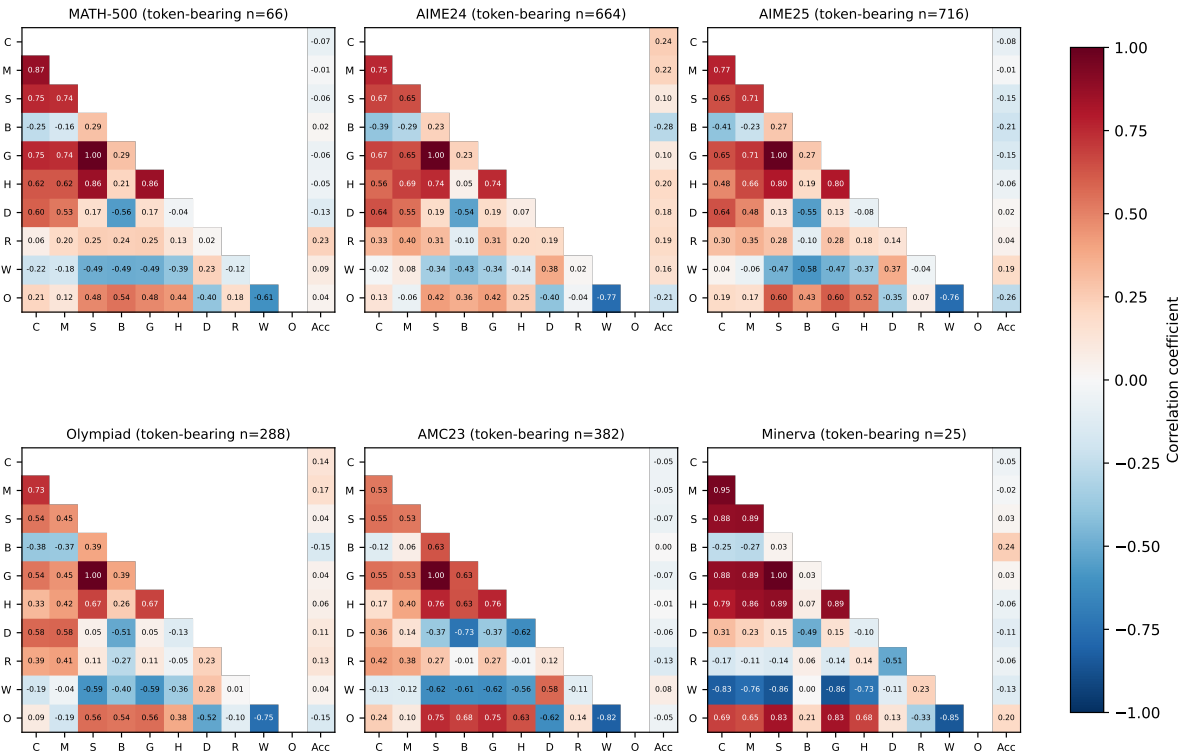


Fig. 72. GPT-OSS-20B / Position: bin 04. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 05

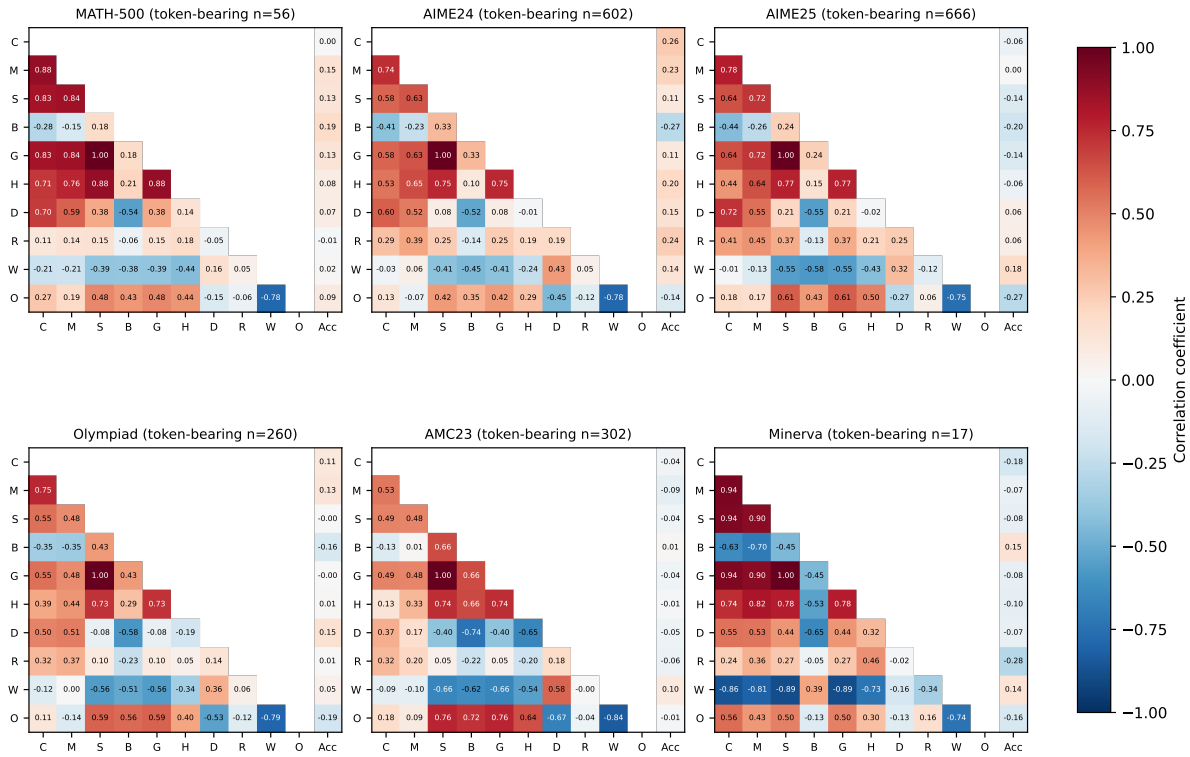


Fig. 73. GPT-OSS-20B / Position: bin 05. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 06

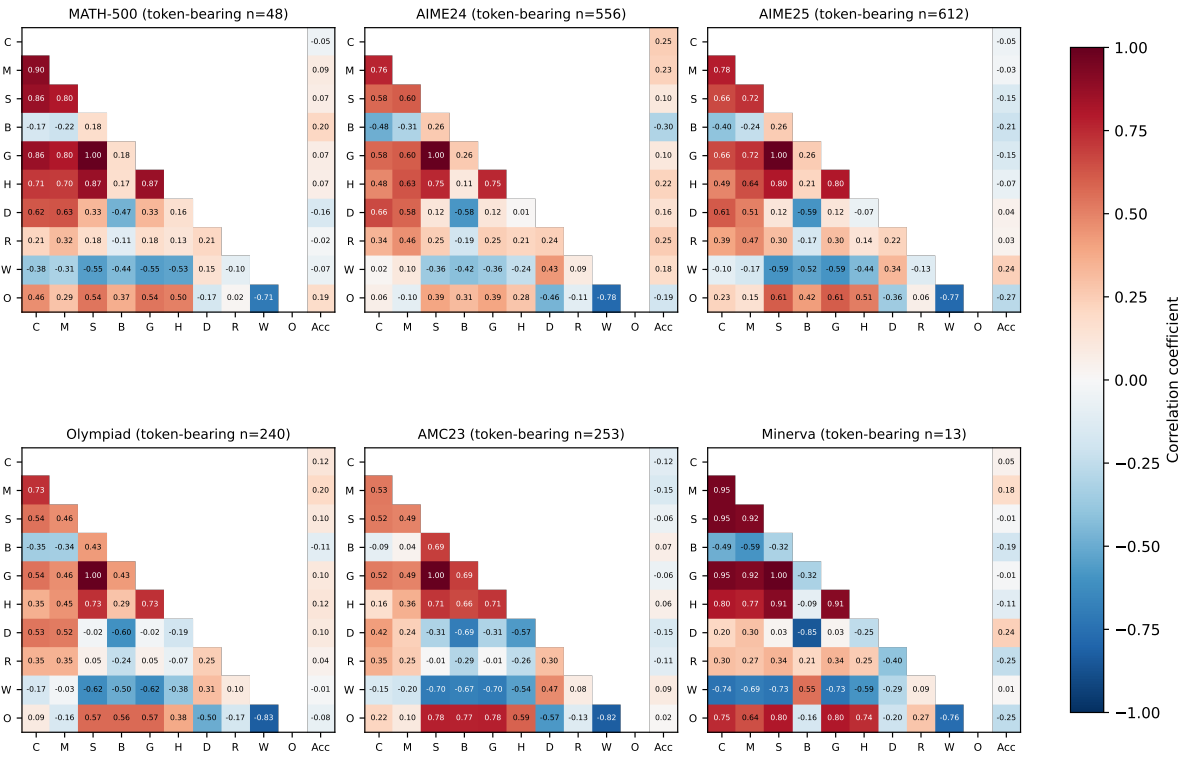


Fig. 74. GPT-OSS-20B / Position: bin 06. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 07

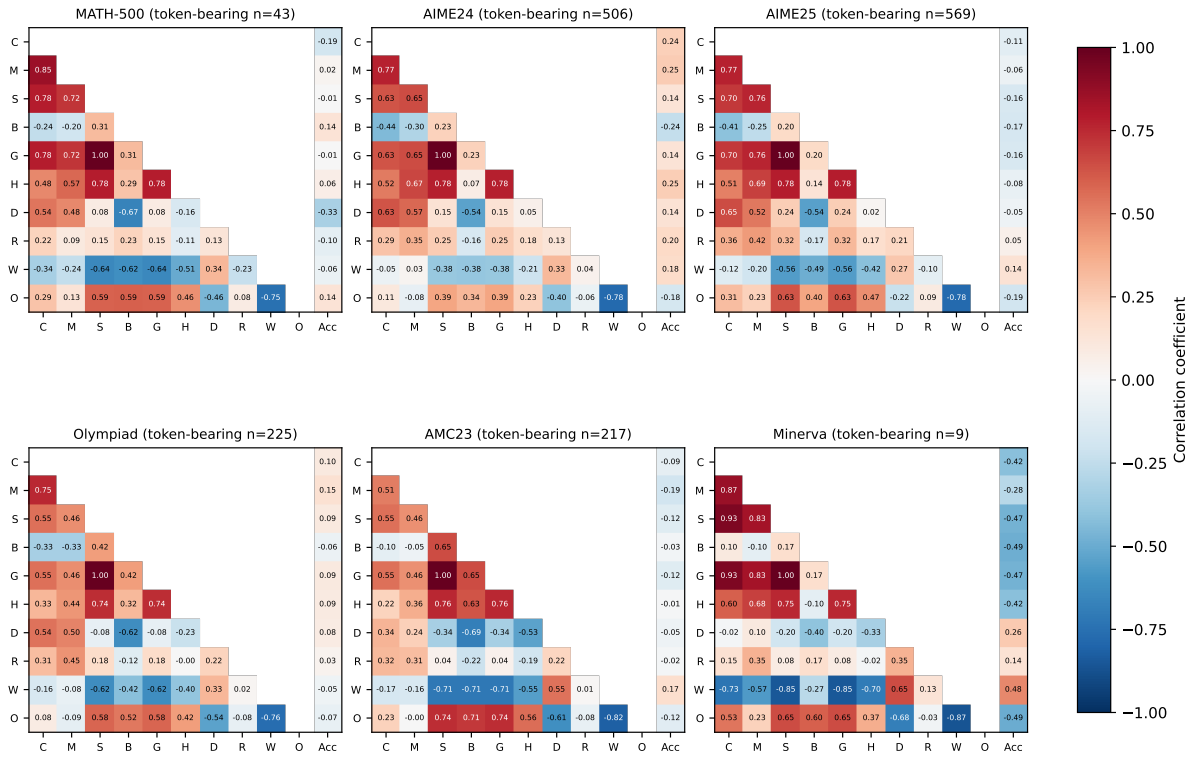


Fig. 75. GPT-OSS-20B / Position: bin 07. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 08

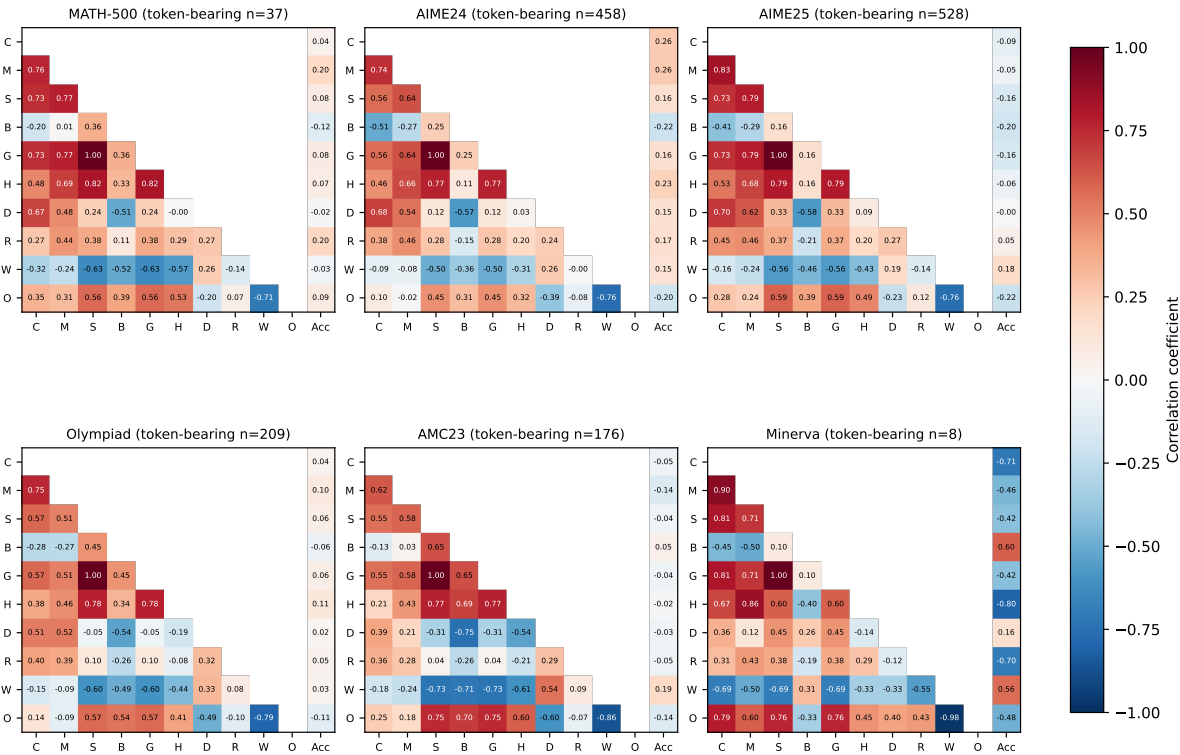


Fig. 76. GPT-OSS-20B / Position: bin 08. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 09

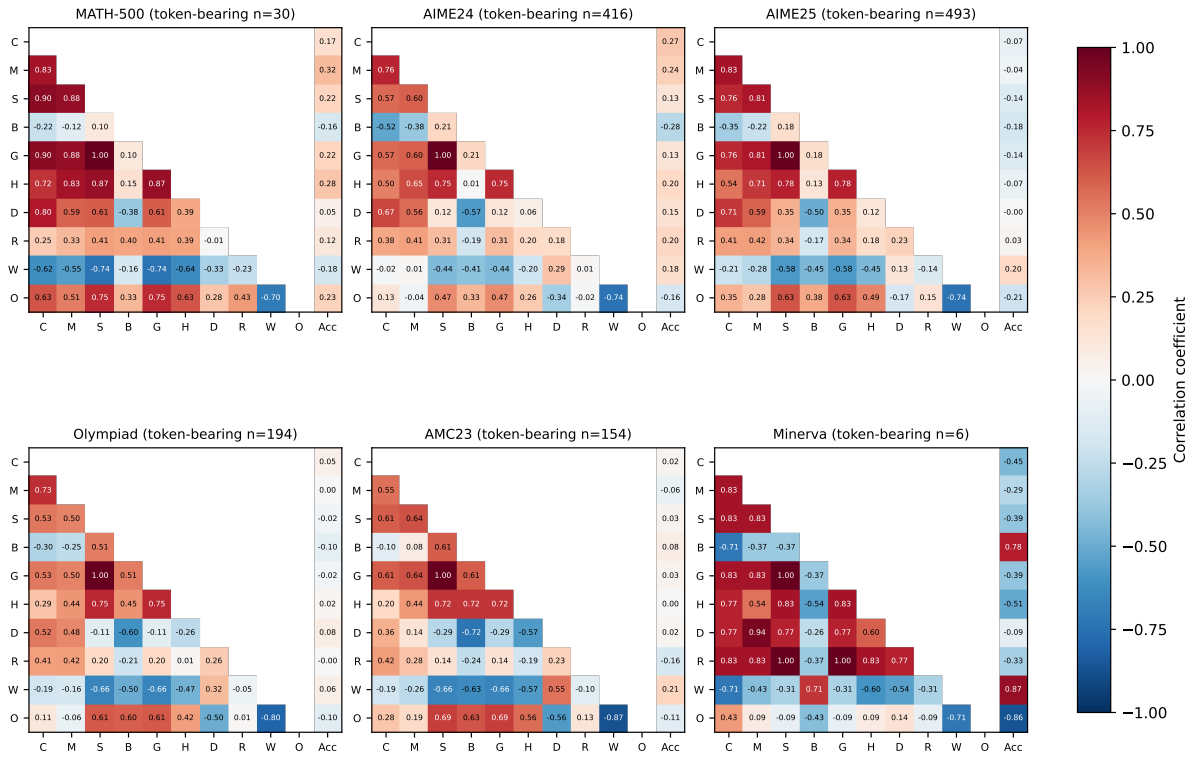


Fig. 77. GPT-OSS-20B / Position: bin 09. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: overflow

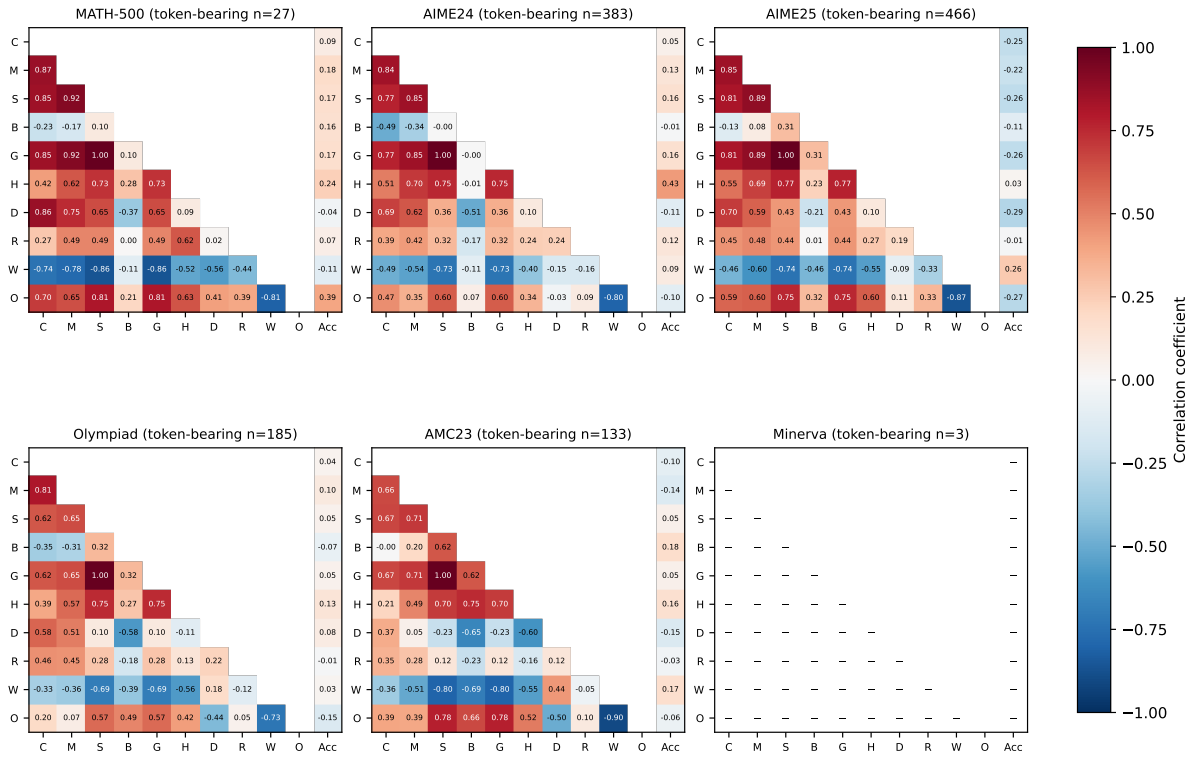


Fig. 78. GPT-OSS-20B / Position: overflow. Metric correlations and correctness; model-specific cohorts and tokenization.

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